UNIVERSITY WORKBOOK SERIES IN MATHEMATICAL SCIENCES

UNIVERSITY MATHEMATICS I

Complete Problem Workbook with Tablet Workspaces & Full Solutions

Progressive Exercises from Scratch to Mastery

Complete Step-by-Step Solutions After Every Single Exercise

Covering Logic, Set Theory, Relations, Algebraic Structures, the Real Number System, Numerical Sequences & ε-N Convergence, Real Functions & ε-δ Continuity, Rolle & Mean Value Theorems, Taylor Expansions, and Riemann Integration.

Equipped with dedicated **5 mm graphic tablet workspaces** formatted for direct digital stylus problem-solving in Xournal++, OneNote, or PDF annotation software.

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How to Use This Workbook

This problem workbook is engineered specifically for active, hands-on learning with a **graphic tablet or digital stylus** on PC. Designed to directly accompany *University Mathematics I: Comprehensive University Textbook*, it provides extensive training across all foundational concepts of higher mathematical analysis and abstract algebra.

Key Pedagogical Features

1. **Progressive Difficulty Rating:** Every exercise is categorized by difficulty:

   ★ **Foundational:** Direct application of definitions, truth tables, basic set operations, direct computations, and elementary single-step proofs.

   ★★ **Intermediate:** Multi-step proofs, ε-N and ε-δ demonstrations, equivalence classes, quotient sets, induction with inequalities, and recurring sequences.

   ★★★ **Advanced / Exam Level:** Comprehensive synthetic problems, topological characterizations, adjacent sequences, Mean Value Theorem applications, Taylor expansions with asymptotic limits, and intricate Riemann integration techniques.

2. **Dedicated 5 mm Tablet Grid Workspaces:** Each exercise is immediately followed by an authentic 5 mm Cartesian grid box calibrated for standard tablet stylus resolution. Open this PDF in your favorite annotation application (such as **Xournal++**, **Drawboard**, or **OneNote**) to solve problems by hand.

3. **Immediate, Complete Solutions:** Following each workspace, a comprehensive step-by-step solution provides every algebraic manipulation, logical justification, and formal conclusion.

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Chapter 1

Mathematical Logic, Quantifiers, Methods of Proof & Set Theory

Essential Chapter Theorems & Formulas · Core Logical Equivalences, Proof Strategies & Set Identities

1. **Basic Logical Connectives & Equivalences:**

$$P \implies Q \iff \neg P \vee Q \iff \neg Q \implies \neg P \quad \text{(Contrapositive)}.$$

$$\neg(P \wedge Q) \iff \neg P \vee \neg Q, \qquad \neg(P \vee Q) \iff \neg P \wedge \neg Q \quad \text{(De Morgan)}.$$

$$\neg(P \implies Q) \iff P \wedge \neg Q.$$

2. **Quantifiers & Negation Rules:**

$$\neg(\forall x \in E,\, P(x)) \iff \exists x \in E,\, \neg P(x), \qquad \neg(\exists x \in E,\, P(x)) \iff \forall x \in E,\, \neg P(x).$$

3. **Methods of Mathematical Proof:**

- *Direct Proof:* Assume P is true; deduce Q via logical chain.
- *Contrapositive:* Assume $\neg Q$ is true; deduce $\neg P$.
- *Proof by Contradiction:* Assume $P \wedge \neg Q$; deduce an absurd statement $(R \wedge \neg R)$.
- *Induction Principle:* Verify $P(n_0)$ (base step), prove $\forall n \geq n_0,\, P(n) \implies P(n+1)$ (inductive step).

4. **Set Operations & Indicator (Characteristic) Functions $\mathbf{1}_A$:**

$$\mathbf{1}_{A\cap B} = \mathbf{1}_A \cdot \mathbf{1}_B, \quad \mathbf{1}_{A\cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A\mathbf{1}_B, \quad \mathbf{1}_{\complement_E A} = 1 - \mathbf{1}_A.$$

$$A\Delta B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B), \quad \mathbf{1}_{A\Delta B} = (\mathbf{1}_A - \mathbf{1}_B)^2 = \mathbf{1}_A + \mathbf{1}_B - 2\mathbf{1}_A\mathbf{1}_B.$$

5. **Sentential & First-Order Compactness Theorems:**

$$\Sigma \text{ is satisfiable} \iff \text{every finite subset } \Sigma_0 \subseteq \Sigma \text{ is satisfiable.}$$

$$\Sigma \models \varphi \iff \exists \Sigma_0 \subseteq_{\text{fin}} \Sigma \text{ such that } \Sigma_0 \models \varphi.$$

6. **Soundness and Gödel's Completeness Theorem:**

$$\boldsymbol{\Sigma} \models \varphi \iff \boldsymbol{\Sigma} \vdash \varphi.$$

7. **Automorphism Invariance Criterion for Non-Definability:**

$$\pi \in \mathrm{Aut}(\mathcal{M}) \implies (\mathcal{M} \models \varphi[a] \iff \mathcal{M} \models \varphi[\pi(a)]).$$

Exercise 1.1 · Truth Tables, Tautologies, and the Law of Contraposition ★ Foundational

Let P, Q, and R be mathematical propositions.

1. Construct the complete truth table for the implication $P \implies Q$ and the contrapositive $\neg Q \implies \neg P$. Conclude that $(P \implies Q) \iff (\neg Q \implies \neg P)$ is a tautology.
2. Show via truth tables that the statement $[(P \implies Q) \wedge (Q \implies R)] \implies (P \implies R)$ (transitivity of implication / hypothetical syllogism) is a tautology.
3. Verify whether $(P \implies Q) \implies [P \implies (P \wedge Q)]$ is always true.

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Complete Step-by-Step Solution 1.1

1. Equivalence of Implication and Contrapositive: Recall that the implication $A \implies B$ is logically false if and only if A is true and B is false (i.e., equivalent to $\neg A \vee B$).

P	Q	$P \implies Q$	$\neg Q$	$\neg P$	$\neg Q \implies \neg P$	$(P \implies Q) \iff (\neg Q \implies \neg P)$
T	T	T	F	F	T	**T**
T	F	F	T	F	F	**T**
F	T	T	F	T	T	**T**
F	F	T	T	T	T	**T**

Since the final column evaluates to True (T) for all $2^2 = 4$ truth assignments, the proposition is a tautology.

2. Transitivity of Implication (Hypothetical Syllogism): Let $A = (P \implies Q) \wedge (Q \implies R)$ and $B = (P \implies R)$.

P	Q	R	$P \implies Q$	$Q \implies R$	$A = (P \implies Q) \wedge (Q \implies R)$	$B = P \implies R$	$A \implies B$
T	T	T	T	T	T	T	**T**
T	T	F	T	F	F	F	**T**
T	F	T	F	T	F	T	**T**
T	F	F	F	T	F	F	**T**
F	T	T	T	T	T	T	**T**
F	T	F	T	F	F	T	**T**
F	F	T	T	T	T	T	**T**
F	F	F	T	T	T	T	**T**

Every row results in True; hence $[(P \implies Q) \wedge (Q \implies R)] \implies (P \implies R)$ is a tautology.

3. Verification of $(P \implies Q) \implies [P \implies (P \wedge Q)]$: Assume $P \implies Q$ is True. We wish to prove $P \implies (P \wedge Q)$. Assume P is True. Since $P \implies Q$ is True and P is True, by *Modus Ponens*, Q is True. Since both P and Q are True, the conjunction $P \wedge Q$ is True. Hence $P \implies (P \wedge Q)$ holds in all cases. The overall implication is therefore a tautology.

Exercise 1.2 · Negation and Symbolic Translation of Quantified Assertions ★ Foundational

Write the formal logical negation of each of the following statements without using the negation symbol $\neg$ in front of quantifiers:

1. A_1 : $\forall x \in \mathbb{R},\ \exists y \in \mathbb{R},\ (x+y>0 \implies x^2+y^2 \geq 1)$.
2. A_2 (Boundedness of a function $f:\mathbb{R}\to\mathbb{R}$): $\exists M>0,\ \forall x \in \mathbb{R},\ |f(x)| \leq M$.
3. A_3 (Convergence of sequence (u_n) to ℓ): $\forall \varepsilon>0,\ \exists N \in \mathbb{N},\ \forall n \geq N,\ |u_n-\ell|<\varepsilon$.
4. A_4 (Strict monotonicity of f): $\forall x,y \in \mathbb{R},\ (x<y \implies f(x)<f(y))$.

State in plain words what the negation of each mathematical statement means.

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Complete Step-by-Step Solution 1.2

1. Negation of A_1: Recall that the negation of an implication $P \implies Q$ is $P \wedge \neg Q$. Applying quantifier duality:

$$\neg A_1 \iff \exists x \in \mathbb{R},\ \forall y \in \mathbb{R},\ \neg(x+y>0 \implies x^2+y^2 \geq 1) \iff \exists x \in \mathbb{R},\ \forall y \in \mathbb{R},\ (x+y>0 \wedge x^2+y^2<1).$$

Meaning: There exists a real x such that for every real y, the sum $x+y$ is strictly positive yet the sum of squares x^2+y^2 is strictly less than 1.

2. Negation of A_2 (Unboundedness):

$$\neg A_2 \iff \forall M>0,\ \exists x \in \mathbb{R},\ |f(x)|>M.$$

Meaning: The function f is unbounded on $\mathbb{R}$; no matter how large a bound $M>0$ is chosen, there exists some point $x \in \mathbb{R}$ where $|f(x)|$ exceeds M.

3. Negation of A_3 (Non-convergence to ℓ):

$$\neg A_3 \iff \exists \varepsilon>0,\ \forall N \in \mathbb{N},\ \exists n \geq N,\ |u_n-\ell| \geq \varepsilon.$$

Meaning: The sequence (u_n) does not converge to ℓ; there exists a positive error threshold $\varepsilon>0$ such that beyond any index N, there is always some term u_n that stays at least ε away from ℓ.

4. Negation of A_4 (Non-strictly increasing):

$$\neg A_4 \iff \exists x,y \in \mathbb{R},\ (x<y \wedge f(x) \geq f(y)).$$

Meaning: The function f fails to be strictly increasing; there exists at least one pair of points $x<y$ such that $f(x) \geq f(y)$.

Exercise 1.3 · Elementary Set Operations and De Morgan's Laws ★ Foundational

Let E be a universal set, and let $A, B, C \subseteq E$.

1. Prove by double containment that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.
2. Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ using logical equivalences of element membership.
3. Simplify the expression: $X = (A \cap B) \cup (A \cap \complement_E B) \cup (\complement_E A \cap B)$.

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Complete Step-by-Step Solution 1.3

1. Proof that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$: Using element membership:

$$\begin{aligned} x \in A \setminus (B \cup C) &\iff x \in A \wedge x \notin (B \cup C) \\ &\iff x \in A \wedge \neg(x \in B \vee x \in C) \\ &\iff x \in A \wedge (x \notin B \wedge x \notin C) \quad \text{(by De Morgan)} \\ &\iff (x \in A \wedge x \notin B) \wedge (x \in A \wedge x \notin C) \quad \text{(idempotence of } \wedge) \\ &\iff x \in (A \setminus B) \wedge x \in (A \setminus C) \\ &\iff x \in (A \setminus B) \cap (A \setminus C). \end{aligned}$$

Since each step is an equivalence ($\iff$), the sets are identical.

2. Distributivity $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$:

$$\begin{aligned} x \in A \cup (B \cap C) &\iff x \in A \vee (x \in B \wedge x \in C) \\ &\iff (x \in A \vee x \in B) \wedge (x \in A \vee x \in C) \quad \text{(distributivity of } \vee \text{ over } \wedge) \\ &\iff x \in (A \cup B) \wedge x \in (A \cup C) \\ &\iff x \in (A \cup B) \cap (A \cup C). \end{aligned}$$

3. Simplification of X: Notice that the first two terms can be factored using distributivity:

$$(A \cap B) \cup (A \cap \complement_E B) = A \cap (B \cup \complement_E B) = A \cap E = A.$$

Substituting this back into X:

$$X = A \cup (\complement_E A \cap B) = (A \cup \complement_E A) \cap (A \cup B) = E \cap (A \cup B) = A \cup B.$$

Hence $X = A \cup B$.

Exercise 1.4 · Direct Proofs in Number Theory and Parity ★★ Intermediate

1. Prove directly that the product of any two odd integers is odd.
2. Prove that for any integer $n \in \mathbb{Z}$, the quantity $n^3 - n$ is divisible by 6.
3. Let $a, b \in \mathbb{Z}$. Prove that if $a + b$ is odd, then one of a or b is odd and the other is even.

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Complete Step-by-Step Solution 1.4

1. Product of Two Odd Integers: Let $x, y \in \mathbb{Z}$ be odd integers. By definition, there exist $k, m \in \mathbb{Z}$ such that $x = 2k + 1$ and $y = 2m + 1$. The product is:

$$xy = (2k + 1)(2m + 1) = 4km + 2k + 2m + 1 = 2(2km + k + m) + 1.$$

Let $p = 2km + k + m \in \mathbb{Z}$. Then $xy = 2p + 1$, which is odd by definition. ■

2. Divisibility of $n^3 - n$ by 6: Factor $n^3 - n$:

$$n^3 - n = n(n^2 - 1) = (n - 1)n(n + 1).$$

This expression is the product of three consecutive integers:

- Among any two consecutive integers, at least one is even (divisible by 2). Thus $2 \mid (n^3 - n)$.
- Among any three consecutive integers, exactly one is a multiple of 3. Thus $3 \mid (n^3 - n)$.

Since $\gcd(2, 3) = 1$, by Gauss's Lemma, $2 \times 3 = 6$ divides $(n - 1)n(n + 1) = n^3 - n$. ■

3. Parity of a and b when $a + b$ is odd: Assume $a, b \in \mathbb{Z}$ such that $a + b = 2k + 1$ for some $k \in \mathbb{Z}$. Suppose for contradiction that a and b have the same parity:

- If both are even: $a = 2p$, $b = 2q \implies a + b = 2(p + q)$, which is even (contradiction).
- If both are odd: $a = 2p+1$, $b = 2q+1 \implies a+b = 2(p+q+1)$, which is even (contradiction).

Therefore, a and b cannot have the same parity; exactly one is even and the other is odd. ■

Exercise 1.5 · Proof by Contraposition vs. Direct Proof ★★ Intermediate

1. Let $n \in \mathbb{Z}$. Prove by contraposition: If n^2 is even, then n is even.
2. Let $x \in \mathbb{R}$. Prove by contraposition: If $3x^3 - x^2 + 2x - 5$ is irrational, then x is irrational.
3. Let $x, y \in \mathbb{R}$. Prove by contraposition: If $x + y > 10$, then $x > 5$ or $y > 5$.

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Complete Step-by-Step Solution 1.5

1. If n^2 is even, then n is even:

- **Statement:** $P(n) : n^2 \in 2\mathbb{Z} \implies Q(n) : n \in 2\mathbb{Z}$.
- **Contrapositive:** $\neg Q(n) \implies \neg P(n)$, i.e., "If n is odd, then n^2 is odd."
- **Proof of contrapositive:** Assume n is odd. Then $n = 2k + 1$ for some $k \in \mathbb{Z}$.

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k \in \mathbb{Z}$, n^2 is of the form $2m + 1$, hence n^2 is odd. By contraposition, if n^2 is even, then n must be even. ■

2. Irrationality of Polynomial in x:

- **Contrapositive:** "If $x \in \mathbb{Q}$, then $3x^3 - x^2 + 2x - 5 \in \mathbb{Q}$."
- **Proof:** Recall that the field of rational numbers $\mathbb{Q}$ is closed under addition, subtraction, and multiplication. If $x \in \mathbb{Q}$, then $x^2 \in \mathbb{Q}$, $x^3 \in \mathbb{Q}$, and integer multiples $3x^3, -x^2, 2x$ are rational. Consequently, the linear combination $3x^3 - x^2 + 2x - 5$ is a sum of rational numbers and is therefore in $\mathbb{Q}$. By contraposition, if the value is irrational, x must be irrational. ■

3. If $x + y > 10$, then $x > 5$ or $y > 5$:

- **Contrapositive:** $\neg(x > 5 \vee y > 5) \implies \neg(x + y > 10)$, which translates by De Morgan to:

$$(x \leq 5 \wedge y \leq 5) \implies x + y \leq 10.$$

- **Proof:** Assume $x \leq 5$ and $y \leq 5$. Adding these two real inequalities yields:

$$x + y \leq 5 + 5 = 10.$$

This establishes the contrapositive directly, hence the original statement holds. ■

Exercise 1.6 · Rigorous Proofs by Contradiction (Reductio ad Absurdum) ★★ Intermediate

1. Prove by contradiction that $\sqrt{3} \notin \mathbb{Q}$.
2. Prove by contradiction that $\log_{10}(2)$ is an irrational number.
3. Prove that there are no integers $m, n \in \mathbb{Z}$ such that $12m + 18n = 7$.

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Complete Step-by-Step Solution 1.6

1. Proof that $\sqrt{3} \notin \mathbb{Q}$: Assume for contradiction that $\sqrt{3} \in \mathbb{Q}$. Then there exist coprime integers $p \in \mathbb{Z}$ and $q \in \mathbb{N}^*$ ($\gcd(p, q) = 1$) such that:

$$\sqrt{3} = \frac{p}{q} \implies 3 = \frac{p^2}{q^2} \implies p^2 = 3q^2.$$

This implies that 3 divides p^2. Since 3 is prime, by Euclid's lemma, 3 divides p. Thus, $p = 3k$ for some integer $k \in \mathbb{Z}$. Substituting into the equation:

$$(3k)^2 = 3q^2 \implies 9k^2 = 3q^2 \implies q^2 = 3k^2.$$

This implies that 3 divides q^2, and since 3 is prime, 3 divides q. Consequently, $3 \mid p$ and $3 \mid q$, so $\gcd(p, q) \geq 3$, which contradicts the irreducible assumption $\gcd(p, q) = 1$. Therefore, $\sqrt{3}$ is irrational. ■

2. Proof that $\log_{10}(2) \notin \mathbb{Q}$: Assume for contradiction that $\log_{10}(2) \in \mathbb{Q}$. Since $10^0 = 1 < 2 < 10^1 = 10$, we have $0 < \log_{10}(2) < 1$. Thus, there exist strictly positive integers $p, q \in \mathbb{N}^*$ such that:

$$\log_{10}(2) = \frac{p}{q} \implies 10^{p/q} = 2 \implies 10^p = 2^q.$$

Decomposing 10 into its prime factors, $10 = 2 \times 5$:

$$(2 \times 5)^p = 2^q \implies 2^p \cdot 5^p = 2^q \implies 5^p = 2^{q-p}.$$

Notice that $p \geq 1$, so 5^p is a positive integer divisible by 5 (hence an odd integer ending in 5). However, the right side 2^{q-p} is either a power of 2 (an even integer if $q > p$, or 1 if $q = p$, or non-integer if $q < p$). In all cases, an odd power of 5 cannot equal a power of 2 because prime factorizations are unique (Fundamental Theorem of Arithmetic). Contradiction! Hence $\log_{10}(2) \notin \mathbb{Q}$. ■

3. Proof that $12m + 18n \neq 7$ for all $m, n \in \mathbb{Z}$: Assume for contradiction that there exist integers $m, n \in \mathbb{Z}$ such that $12m + 18n = 7$. Factor the left-hand side:

$$6(2m + 3n) = 7.$$

Since $m, n \in \mathbb{Z}$, $2m + 3n$ is an integer. Thus the left-hand side is divisible by 6. This requires $6 \mid 7$, which is absurd since 7 is prime and $6 \nmid 7$. Therefore, no such integers exist. ■

Exercise 1.7 · Standard and Shifted Mathematical Induction ★★ Intermediate

1. Prove by induction that for all $n \geq 1$:

$$\sum_{k=1}^{n} k^2 = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

2. Prove Bernoulli's Inequality: For any real number $x > -1$ and any integer $n \in \mathbb{N}$:

$$(1+x)^n \geq 1 + nx.$$

3. Prove by induction that for all $n \geq 4$, $2^n < n!$.

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Complete Step-by-Step Solution 1.7

1. Sum of Squares Formula: Let $P(n)$ be the proposition: $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

- **Base step ($n = 1$):** LHS $= 1^2 = 1$. RHS $= \frac{1(2)(3)}{6} = 1$. $P(1)$ is true.
- **Induction step:** Assume $P(n)$ is true for some $n \geq 1$. We compute:

$$\sum_{k=1}^{n+1} k^2 = \sum_{k=1}^{n} k^2 + (n+1)^2 = \frac{n(n+1)(2n+1)}{6} + (n+1)^2.$$

Factor out $(n+1)$:

$$= (n+1)\left[\frac{n(2n+1) + 6(n+1)}{6}\right] = (n+1)\left[\frac{2n^2 + 7n + 6}{6}\right].$$

Since $2n^2 + 7n + 6 = (n+2)(2n+3) = ((n+1)+1)(2(n+1)+1)$:

$$\sum_{k=1}^{n+1} k^2 = \frac{(n+1)((n+1)+1)(2(n+1)+1)}{6}.$$

Thus $P(n+1)$ is true. By the principle of mathematical induction, $P(n)$ holds for all $n \geq 1$. ■

2. Bernoulli's Inequality: Let $P(n) : (1+x)^n \geq 1 + nx$, with $x > -1$.

- **Base step ($n = 0$):** $(1+x)^0 = 1$ and $1 + 0x = 1$. $1 \geq 1$ holds.

- **Induction step:** Assume $(1+x)^n \geq 1+nx$. Since $x > -1$, we have $1+x > 0$. Multiplying both sides by $(1+x)$ preserves inequality direction:

$$(1+x)^{n+1} \geq (1+nx)(1+x) = 1+x+nx+nx^2 = 1+(n+1)x+nx^2.$$

Since $n \geq 0$ and $x^2 \geq 0$, the term $nx^2 \geq 0$. Therefore:

$$(1+x)^{n+1} \geq 1+(n+1)x+nx^2 \geq 1+(n+1)x.$$

Thus $P(n+1)$ is true. By induction, the inequality holds for all $n \in \mathbb{N}$. ■

3. Factorial Growth $2^n < n!$ for $n \geq 4$:

- **Base step ($n = 4$):** $2^4 = 16$ and $4! = 24$. Since $16 < 24$, the base case holds.
- **Induction step:** Assume $2^n < n!$ for some $n \geq 4$. Since $n \geq 4$, we have $n+1 \geq 5 > 2$. Multiplying the induction hypothesis by $n+1$:

$$(n+1)! = (n+1) \cdot n! > (n+1) \cdot 2^n > 2 \cdot 2^n = 2^{n+1}.$$

Hence $2^{n+1} < (n+1)!$, completing the induction step. ■

Exercise 1.8 · Strong Mathematical Induction & Recurrence Relations ★★ Intermediate

Consider the sequence $(u_n)_{n\in\mathbb{N}}$ defined by:

$$u_0 = 1, \quad u_1 = 3, \quad \text{and} \quad u_{n+2} = 3u_{n+1} - 2u_n \quad \text{for all } n \geq 0.$$

1. Calculate the first four terms: u_2, u_3, u_4, u_5. Conjecture a closed-form formula for u_n.
2. Prove your conjecture for all $n \in \mathbb{N}$ using strong mathematical induction.
3. Alternatively, prove the formula by solving the characteristic polynomial equation.

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Complete Step-by-Step Solution 1.8

1. Computation of Initial Terms:

- $u_2 = 3u_1 - 2u_0 = 3(3) - 2(1) = 9 - 2 = 7.$
- $u_3 = 3u_2 - 2u_1 = 3(7) - 2(3) = 21 - 6 = 15.$
- $u_4 = 3u_3 - 2u_2 = 3(15) - 2(7) = 45 - 14 = 31.$
- $u_5 = 3u_4 - 2u_3 = 3(31) - 2(15) = 93 - 30 = 63.$

Notice the pattern: $1 = 2^1 - 1$, $3 = 2^2 - 1$, $7 = 2^3 - 1$, $15 = 2^4 - 1$, $31 = 2^5 - 1$, $63 = 2^6 - 1$. We conjecture:

$$u_n = 2^{n+1} - 1 \quad \text{for all } n \geq 0.$$

2. Proof by Strong Induction: Let $P(n)$ be the statement $u_n = 2^{n+1} - 1$.

- **Base cases:** For $n = 0$: $u_0 = 2^{0+1} - 1 = 2 - 1 = 1$ (matches definition). For $n = 1$: $u_1 = 2^{1+1} - 1 = 4 - 1 = 3$ (matches definition). Both base cases are verified.
- **Induction step:** Let $k \geq 0$. Assume that $P(m)$ holds for all $0 \leq m \leq k + 1$. In particular, assume $u_k = 2^{k+1} - 1$ and $u_{k+1} = 2^{k+2} - 1$. We evaluate u_{k+2}:

$$\begin{aligned} u_{k+2} &= 3u_{k+1} - 2u_k \\ &= 3(2^{k+2} - 1) - 2(2^{k+1} - 1) \\ &= 3 \cdot 2^{k+2} - 3 - 2^{k+2} + 2 \\ &= (3 - 1) \cdot 2^{k+2} - 1 \\ &= 2 \cdot 2^{k+2} - 1 = 2^{(k+2)+1} - 1. \end{aligned}$$

Thus $P(k + 2)$ is verified. By strong mathematical induction, $u_n = 2^{n+1} - 1$ for all $n \in \mathbb{N}$. ■

3. Characteristic Equation Method: The characteristic polynomial of $r^2 - 3r + 2 = 0$ factors as $(r - 1)(r - 2) = 0$, giving distinct real roots $r_1 = 1$ and $r_2 = 2$. The general solution is $u_n = A(1)^n + B(2)^n = A + B \cdot 2^n$. Using initial conditions:

$$\begin{cases} u_0 = A + B = 1 \\ u_1 = A + 2B = 3 \end{cases} \implies B = 2, \quad A = 1 - 2 = -1.$$

Thus $u_n = -1 + 2 \cdot 2^n = 2^{n+1} - 1$, confirming the result.

Exercise 1.9 · Symmetric Difference of Sets (Δ) ★★ Intermediate

The symmetric difference of two sets A and B is defined by $A \Delta B = (A \setminus B) \cup (B \setminus A)$.

1. Prove that $A \Delta B = (A \cup B) \setminus (A \cap B)$.
2. Prove that the empty set $\emptyset$ is the neutral element for Δ, and that every set is its own inverse: $A \Delta A = \emptyset$.
3. Prove the distributivity of intersection over symmetric difference:

$$A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C).$$

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Complete Step-by-Step Solution 1.9

1. Alternative Characterization $A\Delta B = (A \cup B) \setminus (A \cap B)$:

$$\begin{aligned} x \in A\Delta B &\iff (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A) \\ &\iff [(x \in A \wedge x \notin B) \vee x \in B] \wedge [(x \in A \wedge x \notin B) \vee x \notin A] \\ &\iff [(x \in A \vee x \in B) \wedge (x \notin B \vee x \in B)] \wedge [(x \in A \vee x \notin A) \wedge (x \notin B \vee x \notin A)] \\ &\iff (x \in A \vee x \in B) \wedge \neg(x \in A \wedge x \in B) \\ &\iff x \in (A \cup B) \setminus (A \cap B). \end{aligned}$$

2. Neutral Element and Inverses:

- **Neutral element:** $A\Delta\emptyset = (A \setminus \emptyset) \cup (\emptyset \setminus A) = A \cup \emptyset = A$.
- **Self-inverse:** $A\Delta A = (A \setminus A) \cup (A \setminus A) = \emptyset \cup \emptyset = \emptyset$.

3. Distributivity of $\cap$ over Δ: Using indicator functions $\mathbf{1}_S : E \to \{0,1\}$ where operations are evaluated in $\mathbb{Z}/2\mathbb{Z}$: Recall that $\mathbf{1}_{X\Delta Y} = \mathbf{1}_X \oplus \mathbf{1}_Y = \mathbf{1}_X + \mathbf{1}_Y - 2\mathbf{1}_X\mathbf{1}_Y$ and $\mathbf{1}_{X\cap Y} = \mathbf{1}_X\mathbf{1}_Y$. For any element $x \in E$:

$$\mathbf{1}_{A\cap(B\Delta C)}(x) = \mathbf{1}_A(x) \cdot \mathbf{1}_{B\Delta C}(x) = \mathbf{1}_A(x)(\mathbf{1}_B(x) \oplus \mathbf{1}_C(x)).$$

In the ring $(\{0,1\}, \oplus, \cdot)$, multiplication distributes over addition modulo 2:

$$= (\mathbf{1}_A(x) \cdot \mathbf{1}_B(x)) \oplus (\mathbf{1}_A(x) \cdot \mathbf{1}_C(x)) = \mathbf{1}_{A\cap B}(x) \oplus \mathbf{1}_{A\cap C}(x) = \mathbf{1}_{(A\cap B)\Delta(A\cap C)}(x).$$

Since their indicator functions are identical everywhere on E, the sets are equal:

$$A \cap (B\Delta C) = (A \cap B)\Delta(A \cap C). \quad \blacksquare$$

Exercise 1.10 · Power Set $\mathcal{P}(E)$ and the Boolean Ring Structure ★★★ Advanced / Exam Level

Let E be a non-empty set and $\mathcal{P}(E)$ its power set.

1. Show that $(\mathcal{P}(E), \Delta)$ is an abelian group. Identify the identity and the inverse of any element $A \in \mathcal{P}(E)$.
2. Show that $(\mathcal{P}(E), \Delta, \cap)$ is a commutative ring with unity, and that every element is idempotent: $\forall A \in \mathcal{P}(E),\ A \cap A = A$. (Such a ring is called a *Boolean ring*).
3. Prove that if E is finite with $\text{card}(E) = n$, then $\text{card}(\mathcal{P}(E)) = 2^n$ using:
   (a) The Binomial Theorem: $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.
   (b) The bijection between $\mathcal{P}(E)$ and the set of indicator functions $\{0,1\}^E$.

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Complete Step-by-Step Solution 1.10

1. Abelian Group $(\mathcal{P}(E), \Delta)$:

- **Closure:** For any $A, B \subseteq E$, $A\Delta B \subseteq E$, so Δ is an internal binary operation.
- **Associativity:** Using indicator functions in $\mathbb{Z}/2\mathbb{Z}$, $\mathbf{1}_{(A\Delta B)\Delta C} = (\mathbf{1}_A \oplus \mathbf{1}_B) \oplus \mathbf{1}_C = \mathbf{1}_A \oplus (\mathbf{1}_B \oplus \mathbf{1}_C) = \mathbf{1}_{A\Delta(B\Delta C)}$. Thus $(A\Delta B)\Delta C = A\Delta(B\Delta C)$.
- **Identity element:** For any $A \in \mathcal{P}(E)$, $A\Delta\emptyset = A$. The identity is $\emptyset$.
- **Inverses:** $A\Delta A = \emptyset$, so every set A is its own inverse: $A^{-1} = A$.
- **Commutativity:** $A\Delta B = (A \setminus B) \cup (B \setminus A) = (B \setminus A) \cup (A \setminus B) = B\Delta A$.

Hence $(\mathcal{P}(E), \Delta)$ is an abelian group.

2. Boolean Ring Structure:

- We already proved that $(\mathcal{P}(E), \Delta)$ is an abelian group.
- Intersection $\cap$ is associative: $(A \cap B) \cap C = A \cap (B \cap C)$.
- Intersection distributes over Δ: $A \cap (B\Delta C) = (A \cap B)\Delta(A \cap C)$ (proved in Exercise 1.9).
- Intersection is commutative: $A \cap B = B \cap A$.
- Unity element: $A \cap E = A$ for all $A \subseteq E$, so E is the multiplicative identity.
- Idempotence: $A \cap A = A$ for all $A \in \mathcal{P}(E)$.

Therefore, $(\mathcal{P}(E), \Delta, \cap)$ is a commutative Boolean ring with unity.

3. Cardinality of the Power Set:

- **Method (a) Binomial expansion:** A subset of E can contain k elements, where $0 \leq k \leq n$. The number of k-element subsets is given by the binomial coefficient $\binom{n}{k}$. Summing over all possible subset sizes:

$$\operatorname{card}(\mathcal{P}(E)) = \sum_{k=0}^{n} \binom{n}{k} = \sum_{k=0}^{n} \binom{n}{k} 1^k 1^{n-k} = (1+1)^n = 2^n.$$

- **Method (b) Indicator function bijection:** Define the map $\Phi : \mathcal{P}(E) \to \{0,1\}^E$ by $\Phi(A) = \mathbf{1}_A$. For any function $f \in \{0,1\}^E$, the preimage $A_f = f^{-1}(\{1\}) = \{x \in E : f(x) = 1\}$ is a unique subset of E satisfying $\Phi(A_f) = f$. Thus Φ is a bijection. The cardinality of the function set $\{0,1\}^E$ is $|\{0,1\}|^{|E|} = 2^n$. Hence $\operatorname{card}(\mathcal{P}(E)) = 2^n$. ■

Exercise 1.11 · Arbitrary Unions, Intersections, and Interval Limits ★★★ Advanced / Exam Level

Let I be an arbitrary non-empty indexing set, and let $(A_i)_{i\in I}$ be a family of subsets of a universal set E.

1. Prove the generalized De Morgan's laws:

$$\complement_E\left(\bigcup_{i\in I} A_i\right) = \bigcap_{i\in I} \complement_E A_i \qquad \text{and} \qquad \complement_E\left(\bigcap_{i\in I} A_i\right) = \bigcup_{i\in I} \complement_E A_i.$$

2. Determine the following infinite intersections and unions of real intervals:

$$J_1 = \bigcap_{n=1}^{\infty} \left[0, \frac{1}{n}\right], \qquad J_2 = \bigcap_{n=1}^{\infty} \left(0, \frac{1}{n}\right), \qquad J_3 = \bigcup_{n=1}^{\infty} \left[\frac{1}{n}, 1\right].$$

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Complete Step-by-Step Solution 1.11

1. Generalized De Morgan's Laws: For the first identity:

$$x \in \complement_E\left(\bigcup_{i\in I} A_i\right) \iff x \notin \bigcup_{i\in I} A_i \iff \neg\,(\exists i \in I,\, x \in A_i)$$
$$\iff \forall i \in I,\, x \notin A_i \iff \forall i \in I,\, x \in \complement_E A_i \iff x \in \bigcap_{i\in I} \complement_E A_i.$$

For the second identity:

$$x \in \complement_E\left(\bigcap_{i\in I} A_i\right) \iff x \notin \bigcap_{i\in I} A_i \iff \neg\,(\forall i \in I,\, x \in A_i)$$
$$\iff \exists i \in I,\, x \notin A_i \iff \exists i \in I,\, x \in \complement_E A_i \iff x \in \bigcup_{i\in I} \complement_E A_i. \quad \blacksquare$$

2. Evaluation of Infinite Interval Families:

- **Evaluation of $J_1 = \bigcap_{n=1}^{\infty}[0, 1/n]$:** For every $n \geq 1$, $0 \leq 0 \leq 1/n$, so $0 \in [0, 1/n]$ for all n, meaning $0 \in J_1$. Suppose $x > 0$. By the Archimedean property of $\mathbb{R}$, there exists $N \in \mathbb{N}^*$ such that $1/N < x$. Then $x \notin [0, 1/N]$, so $x \notin J_1$. If $x < 0$, $x \notin [0, 1]$ so $x \notin J_1$. Therefore, $J_1 = \{0\}$.
- **Evaluation of $J_2 = \bigcap_{n=1}^{\infty}(0, 1/n)$:** Suppose there exists $x \in J_2$. Then $x \in (0, 1/n)$ for all $n \geq 1$, so $x > 0$ and $x < 1/n$ for all $n \geq 1$. However, by the Archimedean property, choosing $n > 1/x$ implies $1/n < x$, which contradicts $x < 1/n$. Hence no such x exists: $J_2 = \emptyset$.
- **Evaluation of $J_3 = \bigcup_{n=1}^{\infty}[1/n, 1]$:** Clearly for every $n \geq 1$, $[1/n, 1] \subseteq (0, 1]$, hence $J_3 \subseteq (0, 1]$. Conversely, let $x \in (0, 1]$. Since $x > 0$, by Archimedes there exists $n \in \mathbb{N}^*$ such that $1/n < x \leq 1$, meaning $x \in [1/n, 1] \subseteq J_3$. Thus $(0, 1] \subseteq J_3$. We conclude $J_3 = (0, 1]$. $\blacksquare$

Exercise 1.12 · The Pigeonhole Principle (Dirichlet Principle) and Set Partitions ★★★ Advanced / Exam Level

1. State the Dirichlet Pigeonhole Principle.
2. Let $S \subset \{1, 2, 3, \ldots, 2n\}$ be a subset containing $n+1$ distinct integers. Prove that there exist at least two distinct elements $a, b \in S$ such that a divides b ($a \mid b$).
3. Six people attend a conference. Between any two people, they are either mutual friends or mutual strangers. Prove that there must exist at least three mutual friends or three mutual strangers.

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Complete Step-by-Step Solution 1.12

1. Dirichlet Pigeonhole Principle: If m items (pigeons) are placed into k boxes (pigeonholes) with $m > k$, then at least one pigeonhole must contain two or more items. Formally, if $f : X \to Y$ is a function between finite sets with $\text{card}(X) > \text{card}(Y)$, then f is not injective.
2. Existence of Divisible Pair in $S \subset \{1, \ldots, 2n\}$ with $|S| = n+1$: Every positive integer $x \in \{1, 2, \ldots, 2n\}$ can be uniquely factored as:

$$x = 2^k \cdot q,$$

where $k \geq 0$ is a non-negative integer and q is an odd integer. Notice that the only possible odd factors q are chosen from the set of odd numbers strictly bounded by $2n$:

$$O = \{1, 3, 5, \ldots, 2n-1\}.$$

The number of elements in O is precisely n. Define the map $f : S \to O$ that assigns to each $x \in S$ its odd part q. The domain S has cardinality $\text{card}(S) = n+1$, while the codomain O has cardinality $\text{card}(O) = n$. Since $\text{card}(S) > \text{card}(O)$, by the Pigeonhole Principle, f cannot be injective: there exist two distinct elements $a, b \in S$ ($a \neq b$) that have the exact same odd part q. Thus:

$$a = 2^j \cdot q \quad \text{and} \quad b = 2^m \cdot q \quad (j, m \in \mathbb{N}).$$

Since $a \neq b$, $j \neq m$. Without loss of generality, assume $j < m$. Then:

$$\frac{b}{a} = \frac{2^m \cdot q}{2^j \cdot q} = 2^{m-j} \in \mathbb{N}^*,$$

which proves that a divides b. ■
3. Theorem on Friends and Strangers ($R(3,3) = 6$): Represent the 6 people as 6 vertices $V = \{v_1, \ldots, v_6\}$ of a complete graph K_6. Color the edge between v_i and v_j blue if they are friends, and red if they are strangers. Consider a specific person v_1. There are 5 edges incident to v_1, connecting to the remaining 5 people. By the Pigeonhole Principle (5 edges distributed

into 2 colors), at least $\lceil 5/2 \rceil = 3$ edges incident to v_1 must share the same color. Without loss of generality, assume edges $(v_1, v_2), (v_1, v_3), (v_1, v_4)$ are all **blue** (friends):

- Case A: If any edge between v_2, v_3, v_4 is blue (say (v_2, v_3) is blue), then vertices $\{v_1, v_2, v_3\}$ form a blue triangle (three mutual friends).
- Case B: If none of the edges between v_2, v_3, v_4 are blue, then all three edges $(v_2, v_3), (v_3, v_4), (v_2, v_4)$ must be **red**. Thus $\{v_2, v_3, v_4\}$ forms a red triangle (three mutual strangers).

In either case, a monochromatic triangle exists, proving that there are always at least three mutual friends or three mutual strangers. ■

Exercise 1.13 · Parenthesis Counting Lemma and Unique Readability of Formulas ★★ Intermediate

Let $\mathcal{L}_S$ be the formal language of sentential logic over sentence symbols $\mathcal{S}$ with connectives $\{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$. For any string w, let $L(w)$ and $R(w)$ denote the count of left and right parentheses in w, respectively.

1. Prove by structural induction that for every well-formed formula $\alpha \in \text{WFF}$, $L(\alpha) = R(\alpha)$.
2. Prove that for every non-empty proper initial segment (prefix) u of a well-formed formula α, $L(u) > R(u)$.
3. Deduce that no well-formed formula can be a proper initial segment of another well-formed formula.
4. Show why this guarantees that every formula $\alpha = (\beta \circ \gamma)$ has a uniquely determined principal connective $\circ$ and uniquely determined subformulas β, γ.

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Complete Step-by-Step Solution 1.13

1. Equality of Left and Right Parentheses $L(\alpha) = R(\alpha)$: We proceed by structural induction on $\alpha \in \text{WFF}$:

- *Base step:* If $\alpha = A \in \mathcal{S}$ is an atomic sentence symbol, then $L(A) = 0$ and $R(A) = 0$, so $L(\alpha) = R(\alpha) = 0$.
- *Negation step:* Let $\alpha = (\neg\beta)$ with $\beta \in \text{WFF}$. By the inductive hypothesis, $L(\beta) = R(\beta)$. Then:
$$L(\alpha) = 1 + L(\beta) = 1 + R(\beta) = R(\alpha).$$
- *Binary connective step:* Let $\alpha = (\beta \circ \gamma)$ where $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$ and $\beta, \gamma \in \text{WFF}$. By the

inductive hypothesis, $L(\beta) = R(\beta)$ and $L(\gamma) = R(\gamma)$. Then:

$$L(\alpha) = 1 + L(\beta) + L(\gamma) = 1 + R(\beta) + R(\gamma) = R(\alpha).$$

By induction, $L(\alpha) = R(\alpha)$ holds for all $\alpha \in \text{WFF}$.
2. Strict Inequality $L(u) > R(u)$ for Proper Prefixes: Let u be a non-empty proper prefix of α. Again by structural induction:

- If $\alpha = A \in \mathcal{S}$, α has length 1. Its only proper prefix is the empty string ϵ, so the statement holds vacuously.
- If $\alpha = (\neg\beta)$, a non-empty proper prefix u must take one of four forms:
  1. $u = (:$ here $L(u) = 1 > 0 = R(u)$.
  2. $u = (\neg:$ here $L(u) = 1 > 0 = R(u)$.
  3. $u = (\neg u'$ where u' is a proper prefix of β: by inductive hypothesis, $L(u') \geq R(u')$, so $L(u) = 1 + L(u') \geq 1 + R(u') > R(u') = R(u)$.
  4. $u = (\neg\beta:$ here $L(u) = 1 + L(\beta) = 1 + R(\beta) > R(\beta) = R(u)$.
- If $\alpha = (\beta \circ \gamma)$, any non-empty proper prefix u starts with the outermost left parenthesis $($. Whether u terminates inside β, at $\circ$, or inside γ, the leading $($ provides an extra left parenthesis that is not matched until the very last character of α. Hence $L(u) > R(u)$ in all cases.

3. No Formula is a Proper Prefix of Another: Suppose to the contrary that there exist $\alpha, \beta \in \text{WFF}$ such that α is a proper prefix of β. Since $\alpha \in \text{WFF}$, by Part 1 we must have $L(\alpha) = R(\alpha)$. However, because α is a non-empty proper prefix of $\beta \in \text{WFF}$, by Part 2 we must have $L(\alpha) > R(\alpha)$. This yields an immediate contradiction: $L(\alpha) = R(\alpha) < L(\alpha)$. Thus no formula can be a proper initial segment of another.
4. Unambiguous Parsing (Unique Readability): Suppose $\alpha = (\beta_1 \circ_1 \gamma_1) = (\beta_2 \circ_2 \gamma_2)$. Stripping the initial left parenthesis yields $\beta_1 \circ_1 \gamma_1) = \beta_2 \circ_2 \gamma_2)$. If $\beta_1 \neq \beta_2$, one must be a proper initial segment of the other (say β_1 is a proper prefix of β_2). By Part 3, this is impossible since both $\beta_1, \beta_2 \in \text{WFF}$. Therefore, $\beta_1 = \beta_2$. Matching the symbols immediately following β_1 forces $\circ_1 = \circ_2$, and matching the remaining suffix yields $\gamma_1 = \gamma_2$. Thus, every composite formula has a unique principal connective and unique subformulas. ■

Exercise 1.14 · Functional Completeness and the Sheffer Stroke (NAND / NOR) ★★ Intermediate

Let $P \uparrow Q \iff \neg(P \wedge Q)$ denote the Sheffer Stroke (NAND operator) and $P \downarrow Q \iff \neg(P \vee Q)$ denote the Peirce Arrow (NOR operator).

1. Construct the truth table for $P \uparrow Q$ and $P \downarrow Q$.
2. Express the conditional $P \to Q$ and biconditional $P \leftrightarrow Q$ using exclusively the Sheffer Stroke $\uparrow$.
3. Express $\neg P$, $P \wedge Q$, and $P \vee Q$ using exclusively the Peirce Arrow $\downarrow$.
4. Prove that the connective set $\{\wedge, \vee\}$ is **not** functionally complete by establishing that every formula built with $\{\wedge, \vee\}$ satisfies a monotonicity property.
5. Prove that the connective set $\{\to\}$ alone is **not** functionally complete.

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Complete Step-by-Step Solution 1.14

1. Truth Tables for NAND (↑) and NOR (↓):

P	Q	$P \wedge Q$	$P \uparrow Q = \neg(P \wedge Q)$	$P \vee Q$	$P \downarrow Q = \neg(P \vee Q)$
T	T	T	**F**	T	**F**
T	F	F	**T**	T	**F**
F	T	F	**T**	T	**F**
F	F	F	**T**	F	**T**

2. Synthesizing $P \to Q$ and $P \leftrightarrow Q$ via ↑:

- Recall that $P \to Q \iff \neg P \vee Q \iff \neg(P \wedge \neg Q) \iff P \uparrow (\neg Q)$. Since $\neg Q \iff Q \uparrow Q$, we obtain:
$$\mathbf{P \to Q \iff P \uparrow (Q \uparrow Q)}.$$
- For the biconditional: $P \leftrightarrow Q \iff (P \to Q) \wedge (Q \to P) \iff \neg((P \to Q) \uparrow (Q \to P))$. Using $A \wedge B \iff (A \uparrow B) \uparrow (A \uparrow B)$: Let $X = P \uparrow (Q \uparrow Q)$ and $Y = Q \uparrow (P \uparrow P)$. Then:
$$\mathbf{P \leftrightarrow Q \iff (X \uparrow Y) \uparrow (X \uparrow Y)}.$$

3. Synthesizing Connectives via the Peirce Arrow ↓:

- **Negation:** $P \downarrow P \iff \neg(P \vee P) \iff \neg P$.
- **Disjunction:** $P \vee Q \iff \neg\neg(P \vee Q) \iff \neg(P \downarrow Q) \iff \mathbf{(P \downarrow Q) \downarrow (P \downarrow Q)}$.
- **Conjunction:** By De Morgan: $P \wedge Q \iff \neg(\neg P \vee \neg Q) \iff (\neg P) \downarrow (\neg Q) \iff \mathbf{(P \downarrow P) \downarrow (Q \downarrow Q)}$.

4. Incompleteness of $\{\wedge, \vee\}$ (Monotonicity Barrier): Assign truth value 1 for True and 0 for False with order $0 \leq 1$. A Boolean function $F : \{0,1\}^n \to \{0,1\}$ is **monotone** if $(x_1, \ldots, x_n) \leq (y_1, \ldots, y_n)$ componentwise implies $F(x_1, \ldots, x_n) \leq F(y_1, \ldots, y_n)$. Observe that:

- The constant projections $F(x) = x_i$ are monotone.
- If F and G are monotone, then $(F \wedge G)(x) = \min(F(x), G(x))$ and $(F \vee G)(x) = \max(F(x), G(x))$ are also monotone.

By structural induction, every formula built using only $\{\wedge, \vee\}$ computes a monotone Boolean function. In particular, when all inputs are 1, the output must be 1 ($F(1, \ldots, 1) = 1$). However, the negation function $\neg x$ has $\neg(1) = 0 < 1 = \neg(0)$, so negation is strictly non-monotone. Therefore, negation cannot be expressed by $\{\wedge, \vee\}$, proving that $\{\wedge, \vee\}$ is not functionally complete.

5. Incompleteness of $\{\rightarrow\}$: Consider the truth assignment v where every sentence symbol is assigned 1 (True). Notice that $1 \rightarrow 1 = 1$. By induction on formula construction, every formula α constructed using only the connective $\rightarrow$ must evaluate to 1 when all its atomic variables are 1. Thus it is impossible to express the constant False function $\mathbf{0}$ or the negation $\neg P$, because $\neg(1) = 0 \neq 1$. Hence $\{\rightarrow\}$ alone is not functionally complete. ■

Exercise 1.15 · Topological Compactness of Sentential Logic via the Cantor Space ★★★ Advanced / Exam Level

Let $\mathcal{S} = \{A_n \mid n \in \mathbb{N}\}$ be a countable set of propositional variables. A truth assignment is a sequence $x = (x_0, x_1, x_2, \dots) \in \{0,1\}^{\mathbb{N}}$, where $x_n = v(A_n)$. Endow $X = \{0,1\}^{\mathbb{N}}$ with the product topology (the Cantor Space).

1. Describe the basic open cylinders of the product topology on X and explain why X is a compact, Hausdorff, totally disconnected topological space.
2. For any formula $\alpha \in \text{WFF}$, define the truth set $[\alpha] = \{x \in X \mid \bar{x}(\alpha) = 1\}$. Prove that $[\alpha]$ is both **closed and open** (clopen) in X.
3. Let Σ be an infinite set of sentential formulas such that every finite subset $\Sigma_0 \subseteq \Sigma$ is satisfiable. Use the **Finite Intersection Property** of compact spaces to prove that Σ is satisfiable.

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Complete Step-by-Step Solution 1.15

1. Topology of the Cantor Space $X = \{0,1\}^{\mathbb{N}}$: Give the two-point discrete space $\{0,1\}$ the discrete topology. The product space $X = \prod_{n=0}^{\infty}\{0,1\}$ has basic open sets of the form:

$$U_s = \{x \in X \mid x_0 = s_0, x_1 = s_1, \dots, x_{k-1} = s_{k-1}\},$$

for some finite binary sequence $s = (s_0, \dots, s_{k-1}) \in \{0,1\}^k$.

- **Compactness:** By **Tychonoff's Theorem**, the arbitrary product of compact spaces is compact. Since $\{0,1\}$ is finite (hence compact), X is compact.
- **Hausdorff:** For any two distinct points $x \neq y$, there is an index m where $x_m \neq y_m$. The disjoint cylinders $\{z \mid z_m = x_m\}$ and $\{z \mid z_m = y_m\}$ separate x and y.
- **Totally Disconnected:** The basic cylinder sets U_s are both open and closed (their complement is the finite union of other length-k cylinders). Thus X has a clopen base.

2. Clopen Nature of Formula Sets $[\alpha]$: Let $\alpha \in \text{WFF}$. The formula α contains only a finite set of sentence symbols $\{A_{i_1}, A_{i_2}, \dots, A_{i_m}\}$. The valuation $\bar{x}(\alpha)$ depends entirely on the coordinates $(x_{i_1}, \dots, x_{i_m}) \in \{0,1\}^m$. There are 2^m possible assignments to these m variables.

Let $S_\alpha \subseteq \{0,1\}^m$ be the subset of assignments that evaluate α to True (1). Then:

$$[\alpha] = \bigcup_{(c_1,\ldots,c_m)\in S_\alpha} \{x \in X \mid x_{i_1} = c_1, \ldots, x_{i_m} = c_m\}.$$

Since each cylinder set is clopen, and S_α is a finite set, $[\alpha]$ is a **finite union of clopen sets**. Therefore, $[\alpha]$ is both **closed and open** in X.

3. Proof of Sentential Compactness via FIP: Suppose every finite subset $\Sigma_0 \subseteq \Sigma$ is satisfiable. For each $\Sigma_0 = \{\alpha_1, \ldots, \alpha_k\}$, satisfiability means there exists an assignment $x \in X$ such that $\bar{x}(\alpha_i) = 1$ for all $1 \leq i \leq k$. In topological notation:

$$\bigcap_{\alpha\in\Sigma_0} [\alpha] = [\alpha_1] \cap [\alpha_2] \cap \cdots \cap [\alpha_k] \neq \emptyset.$$

This demonstrates that the collection of closed sets $\{[\alpha]\}_{\alpha\in\Sigma}$ possesses the **Finite Intersection Property (FIP)**. Since X is a compact topological space, every collection of closed subsets satisfying the FIP has a non-empty global intersection:

$$\bigcap_{\alpha\in\Sigma} [\alpha] \neq \emptyset.$$

Let $x^* \in \bigcap_{\alpha\in\Sigma}[\alpha]$. By definition, $x^* \in [\alpha]$ for every $\alpha \in \Sigma$, meaning $\overline{x^*}(\alpha) = 1$ for all $\alpha \in \Sigma$. Thus, x^* is a simultaneous truth assignment satisfying the entire infinite theory Σ. ■

Exercise 1.16 · The de Bruijn–Erdős Graph Coloring Theorem via Compactness ★★★ Advanced / Exam Level

Let $G = (V, E)$ be an infinite simple graph, and let $k \in \mathbb{N}_{\geq 1}$. A valid k-coloring is a function $c : V \to \{1, 2, \ldots, k\}$ such that $\{u, v\} \in E \implies c(u) \neq c(v)$.

1. Formulate the k-colorability of G as a set of sentential formulas Σ_G using sentence symbols $\{P_{v,i} \mid v \in V, 1 \leq i \leq k\}$.
2. Prove that every finite subset $\Sigma_0 \subset \Sigma_G$ involves only a finite subgraph of G.
3. Apply the Compactness Theorem to prove the **de Bruijn–Erdős Theorem**: G is k-colorable if and only if every finite subgraph $G_0 \subseteq G$ is k-colorable.
4. Deduce that if an infinite planar graph has every finite subgraph 4-colorable (by the Four Color Theorem), then the infinite planar graph is also 4-colorable.

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Complete Step-by-Step Solution 1.16

1. Formal Propositional Axiomatization Σ_G: Introduce propositional variables $P_{v,i}$ with interpretation "vertex v receives color i". The theory Σ_G consists of three axiom families:

$$\text{(Axiom 1: At least one color)} \quad \bigvee_{i=1}^{k} P_{v,i} \quad \text{for each } v \in V, \tag{1.1}$$

$$\text{(Axiom 2: At most one color)} \quad \neg(P_{v,i} \wedge P_{v,j}) \quad \text{for each } v \in V \text{ and } 1 \le i < j \le k, \tag{1.2}$$

$$\text{(Axiom 3: Adjacent vertices differ)} \quad \neg(P_{u,i} \wedge P_{v,i}) \quad \text{for each } \{u,v\} \in E \text{ and each } 1 \le i \le k. \tag{1.3}$$

2. Finite Subsets Involve Finite Subgraphs: Let $\Sigma_0 \subseteq \Sigma_G$ be an arbitrary finite subset. Since Σ_0 contains only finitely many formulas, it mentions only a finite set of vertex symbols:

$$V_0 = \{v \in V \mid P_{v,i} \text{ appears in some formula of } \Sigma_0\}.$$

Let $E_0 = E \cap (V_0 \times V_0)$ be the edges of G with both endpoints in V_0. Then $G_0 = (V_0, E_0)$ is a finite induced subgraph of G. All formulas in Σ_0 are axioms belonging to the theory Σ_{G_0} of the finite subgraph G_0.

3. Compactness Proof of the de Bruijn–Erdős Theorem:

- ($\Longrightarrow$) If G is k-colorable, restricting the coloring to any subgraph yields a valid k-coloring.
- ($\Longleftarrow$) Assume every finite subgraph of G is k-colorable. Let $\Sigma_0 \subseteq \Sigma_G$ be any finite subset, mentioning vertices V_0. The subgraph $G[V_0]$ is finite, so by hypothesis it admits a valid k-coloring $c : V_0 \to \{1, \dots, k\}$. Define a valuation v_0 on the variables in Σ_0 by:

  $$v_0(P_{u,i}) = 1 \iff c(u) = i.$$

  Because c is a valid coloring on $G[V_0]$, v_0 satisfies all clauses of Axiom 1, Axiom 2, and Axiom 3 present in Σ_0. Thus every finite subset Σ_0 is satisfiable. By the Compactness Theorem for Sentential Logic, the entire theory Σ_G is satisfiable by some global valuation v^*. Setting $c^*(u) = i \iff v^*(P_{u,i}) = 1$ yields a globally valid k-coloring of the entire infinite graph G.

4. Application to Infinite Planar Graphs: Let G be an arbitrary (possibly uncountably infinite) planar graph. Every finite subgraph $G_0 \subset G$ is a finite planar graph. By the Four Color Theorem (Appel–Haken 1976), every finite planar graph is 4-colorable. By the de Bruijn–Erdős Theorem, the full infinite planar graph G is also 4-colorable. ■

Exercise 1.17 · First-Order Structures, Terms, and Truth Evaluations ★★ Intermediate

Consider the first-order language of arithmetic $\mathcal{L} = \{0, S, +, \cdot, <\}$, where 0 is a constant, S is a unary function symbol (successor), $+$ and $\cdot$ are binary function symbols, and $<$ is a binary relation symbol. Let $\mathcal{N} = (\mathbb{N}, 0, S^{\mathbb{N}}, +^{\mathbb{N}}, \cdot^{\mathbb{N}}, <^{\mathbb{N}})$ be the standard model of natural numbers.

1. Write first-order formulas expressing:
   - "x is an even number" (Even(x)).
   - "x is a prime number" (Prime(x)).
   - "There are infinitely many twin primes" (Twin Prime Conjecture).
2. For the term $t(x, y) = S(x) \cdot (y + S(0))$, compute its valuation $\bar{s}(t)$ under the assignment $s(x) = 3, s(y) = 2$.

3. Determine the truth value in $\mathcal{N}$ of the sentence:

$$\sigma = \forall x \, \exists y \, (x < y \wedge \forall z \, (\neg(x < z \wedge z < y))).$$

What does σ express about the order structure?

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Complete Step-by-Step Solution 1.17

1. First-Order Translations:

- **Even numbers:** x is even if it is a multiple of $2 = S(S(0))$:

$$\text{Even}(x) \equiv \exists y \, (x = y \cdot S(S(0))).$$

- **Prime numbers:** x is prime if $x > 1$ and its only divisors are 1 and x:

$$\text{Prime}(x) \equiv S(0) < x \ \wedge \ \forall y \, \forall z \, (x = y \cdot z \implies (y = S(0) \vee y = x)).$$

- **Twin Prime Conjecture:** For every number n, there exist primes $p, q > n$ with $q = p + 2$:

$$\forall n \, \exists p \, \exists q \left(n < p \ \wedge \ q = p + S(S(0)) \ \wedge \ \text{Prime}(p) \ \wedge \ \text{Prime}(q) \right).$$

2. Term Valuation $\bar{s}(t)$: Given $s(x) = 3$ and $s(y) = 2$:

- $\bar{s}(0) = 0$, so $\bar{s}(S(0)) = 0 + 1 = 1$.
- $\bar{s}(y + S(0)) = s(y) + \bar{s}(S(0)) = 2 + 1 = 3$.
- $\bar{s}(S(x)) = s(x) + 1 = 3 + 1 = 4$.
- $\bar{s}(t) = \bar{s}(S(x)) \cdot \bar{s}(y + S(0)) = 4 \cdot 3 = \mathbf{12}$.

3. Truth Value and Meaning of σ: Let us analyze the subformula $\varphi(x, y) = x < y \wedge \neg \exists z \, (x < z \wedge z < y)$. This states that y is the **immediate successor** of x (there is no element strictly between x and y). The full sentence σ states: "Every element x has an immediate successor y." In $(\mathbb{N}, <)$, for any $n \in \mathbb{N}$, taking $y = n + 1$ satisfies $n < n + 1$, and there is no integer z such that $n < z < n + 1$. Therefore, σ is **True** in $\mathcal{N}$. It expresses that the order is **discrete** on the right (not dense). In contrast, in $(\mathbb{Q}, <)$ or $(\mathbb{R}, <)$, σ is False. ■

Exercise 1.18 · Definable Sets and Proofs of Non-Definability via Automorphisms ★★★ Advanced / Exam Level

Let $\mathcal{M} = (M, I)$ be an $\mathcal{L}$-structure and $\mathrm{Aut}(\mathcal{M})$ its automorphism group.

1. Prove the **Automorphism Invariance Lemma**: If $\varphi(x_1, \dots, x_k)$ is a first-order formula, $\pi \in \mathrm{Aut}(\mathcal{M})$, and $a_1, \dots, a_k \in M$, then:
$$\mathcal{M} \models \varphi[a_1, \dots, a_k] \iff \mathcal{M} \models \varphi[\pi(a_1), \dots, \pi(a_k)].$$
2. Consider the field of complex numbers $\mathcal{C} = (\mathbb{C}, +, \cdot, 0, 1)$. Show that complex conjugation $\sigma(z) = \bar{z}$ is an automorphism of $\mathcal{C}$. Deduce that the imaginary unit i is **not definable without parameters** in $\mathcal{C}$.
3. Consider the additive group of integers $\mathcal{Z} = (\mathbb{Z}, +)$. Prove that the set of positive integers $\mathbb{Z}_{>0} = \{1, 2, 3, \dots\}$ is **not definable without parameters** in $\mathcal{Z}$.
4. Is the set $\{0\}$ definable without parameters in $(\mathbb{Z}, +)$? If so, exhibit a defining formula.

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Complete Step-by-Step Solution 1.18

1. Proof of Automorphism Invariance Lemma: We proceed by induction on the structure of the formula φ:

- *Terms:* An automorphism π preserves constants ($\pi(c^{\mathcal{M}}) = c^{\mathcal{M}}$) and commutes with functions: $\pi(f^{\mathcal{M}}(a_1, \dots)) = f^{\mathcal{M}}(\pi(a_1), \dots)$. Hence for any term $t(x_1, \dots, x_k)$, $\pi(\bar{s}(t)) = \overline{(\pi \circ s)}(t)$.
- *Atomic formulas:* For equality: $t_1 = t_2 \iff \pi(t_1) = \pi(t_2)$ because π is injective. For relations: $(t_1, \dots, t_n) \in R^{\mathcal{M}} \iff (\pi(t_1), \dots, \pi(t_n)) \in R^{\mathcal{M}}$ by definition of automorphism.
- *Boolean connectives:* Trivial since $\neg$ and $\to$ commute with truth values.
- *Quantifiers:* $\mathcal{M} \models \forall x \psi[s] \iff \forall a \in M, \mathcal{M} \models \psi[s(x \mapsto a)]$. Since π is a bijection, as a ranges over M, $\pi(a)$ ranges over all of M. By induction hypothesis, $\mathcal{M} \models \psi[s(x \mapsto a)] \iff \mathcal{M} \models \psi[(\pi \circ s)(x \mapsto \pi(a))]$. Thus $\mathcal{M} \models \forall x \psi[\pi \circ s]$.

2. Complex Conjugation and Non-Definability of i:

- For all $z, w \in \mathbb{C}$: $\overline{z + w} = \bar{z} + \bar{w}$ and $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$. Also $\bar{0} = 0, \bar{1} = 1$. Since $z \mapsto \bar{z}$ is a bijection, complex conjugation is an automorphism of $\mathcal{C}$.
- Suppose i were definable by a formula $\varphi(x)$, so $\mathcal{C} \models \varphi[z] \iff z = i$. Applying the automorphism $\sigma(z) = \bar{z}$:
$$\mathcal{C} \models \varphi[i] \iff \mathcal{C} \models \varphi[\sigma(i)] \iff \mathcal{C} \models \varphi[-i].$$
This would force $-i = i \implies 2i = 0$, a contradiction! Hence, no first-order formula in the language of fields can distinguish i from $-i$ without parameters.

3. Non-Definability of $\mathbb{Z}_{>0}$ in $(\mathbb{Z},+)$: The map $\pi(x) = -x$ is an automorphism of $(\mathbb{Z},+)$ since $-(x+y) = (-x)+(-y)$ and it is bijective. If $\mathbb{Z}_{>0}$ were defined by $\psi(x)$, then for any $n \in \mathbb{Z}$:

$$n \in \mathbb{Z}_{>0} \iff \mathcal{Z} \models \psi[n] \iff \mathcal{Z} \models \psi[\pi(n)] \iff -n \in \mathbb{Z}_{>0}.$$

Taking $n = 1$: $1 \in \mathbb{Z}_{>0}$, so $-1 \in \mathbb{Z}_{>0}$, a blatant contradiction. Therefore, the positive integers are not definable in $(\mathbb{Z},+)$.

4. Definability of $\{0\}$ in $(\mathbb{Z},+)$: Yes, $\{0\}$ is definable without parameters. The element 0 is the unique additive identity of the group:

$$\varphi(x) \equiv \forall y\,(x+y=y) \quad \text{or simply} \quad x+x=x.$$

Indeed, in $\mathbb{Z}$, $x+x=x \implies 2x=x \implies x=0$. Hence $\{0\} = \{x \in \mathbb{Z} \mid (\mathbb{Z},+) \models x+x=x\}$. ■

Exercise 1.19 · Compactness and Non-Axiomatizability of Graph Connectivity ★★★ Advanced / Exam Level

Let $\mathcal{L} = \{E\}$ be the first-order language of graph theory, where E is a symmetric, irreflexive binary relation. A graph $G = (V, E)$ is **connected** if between every pair of vertices there exists a path of finite length.

1. For each $n \geq 1$, write a first-order formula $\mathrm{Path}_{\leq n}(x, y)$ asserting that there is a path of length at most n between x and y.
2. Assume for contradiction that there exists a first-order sentence σ_{conn} such that for any graph G:

   $$G \models \sigma_{\mathrm{conn}} \iff G \text{ is connected.}$$

   Expand the language by adding two new constant symbols c and d. Consider the theory:

   $$\Sigma = \{\sigma_{\mathrm{conn}}\} \cup \{\neg \mathrm{Path}_{\leq n}(c, d) \mid n \in \mathbb{N}_{\geq 1}\}.$$

   Prove that every finite subset $\Sigma_0 \subset \Sigma$ has a model.
3. Apply the First-Order Compactness Theorem to derive a contradiction. Conclude that the property of graph connectivity is **not first-order axiomatizable**.

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Complete Step-by-Step Solution 1.19

1. Path Formulas $\mathrm{Path}_{\leq n}(x, y)$**:**

$$\begin{aligned}
&\mathrm{Path}_{\leq 1}(x, y) \equiv x = y \vee E(x, y),\\
&\mathrm{Path}_{\leq 2}(x, y) \equiv \mathrm{Path}_{\leq 1}(x, y) \vee \exists z_1 \left(E(x, z_1) \wedge E(z_1, y)\right),\\
&\mathrm{Path}_{\leq n}(x, y) \equiv \exists z_1 \ldots \exists z_{n-1} \bigvee_{k=1}^{n} \left(E(x, z_1) \wedge E(z_1, z_2) \wedge \cdots \wedge E(z_{k-1}, y)\right).
\end{aligned}$$

2. Finite Satisfiability of Σ: Let $\Sigma_0 \subset \Sigma$ be an arbitrary finite subset. Σ_0 contains the connectivity sentence σ_{conn} and finitely many negation conditions:

$$\neg\mathrm{Path}_{\leq n_1}(c, d), \ldots, \neg\mathrm{Path}_{\leq n_k}(c, d).$$

Let $N = \max(n_1, \ldots, n_k)$. To satisfy Σ_0, we need a graph that is connected, yet the distance between c and d is strictly greater than N. Construct a simple cycle graph C_{2N+2} with $2N + 2$ vertices arranged in a ring.

- C_{2N+2} is connected, so $C_{2N+2} \models \sigma_{\text{conn}}$.
- Let c and d be antipodal vertices on the cycle. The shortest path between c and d has length $N + 1 > N$.
- Therefore, $C_{2N+2} \models \neg\mathrm{Path}_{\leq m}(c, d)$ for all $m \leq N$.

Thus C_{2N+2} with these constants satisfies Σ_0. Every finite subset Σ_0 has a model.

3. Derivation of Contradiction via Compactness: By the First-Order Compactness Theorem, since every finite subset Σ_0 has a model, the entire infinite theory Σ must have a model $\mathcal{G}^* = (V^*, E^*, c^*, d^*)$. Let us inspect the properties of $\mathcal{G}^*$:

1. Since $\sigma_{\text{conn}} \in \Sigma$, $\mathcal{G}^* \models \sigma_{\text{conn}}$. By our hypothesis on σ_{conn}, $\mathcal{G}^*$ must be a **connected graph**.
2. But Σ also contains $\neg\mathrm{Path}_{\leq n}(c, d)$ for **every** $n \geq 1$. This asserts that there is NO path of length 1, NO path of length 2, ..., NO path of any finite length $n \in \mathbb{N}$ connecting c^* and d^*.
3. Therefore, c^* and d^* belong to different connected components of $\mathcal{G}^*$, proving that $\mathcal{G}^*$ is **disconnected**!

This direct contradiction proves that our initial hypothesis was false: **No first-order sentence σ_{conn} can axiomatize the class of connected graphs.** ■

Exercise 1.20 · Non-Standard Models of Arithmetic via First-Order Compactness ★★★ Advanced / Exam Level

Let $\mathcal{L}_A = \{0, S, +, \cdot, <\}$ and let $\mathrm{Th}(\mathbb{N}) = \{\sigma \text{ sentence} \mid \mathbb{N} \models \sigma\}$ be True Arithmetic.

1. Add a new constant symbol c to $\mathcal{L}_A$. Consider the theory $\Sigma = \mathrm{Th}(\mathbb{N}) \cup \{c > \bar{n} \mid n \in \mathbb{N}\}$, where $\bar{n} = S^n(0)$. Prove using compactness that Σ has a model $\mathcal{M}^*$.
2. Let M^* be the universe of $\mathcal{M}^*$. Prove that the map $j : \mathbb{N} \to M^*$ given by $j(n) = \bar{n}^{\mathcal{M}^*}$ is an elementary embedding.
3. Deduce that $c^{\mathcal{M}^*}$ is an **infinite non-standard integer**.
4. Show that $\mathcal{M}^*$ also contains elements $c - 1, c + 1, 2c, c^2$ and infinitely many non-standard "galaxies" order-isomorphic to $(\mathbb{Z}, <)$.

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Complete Step-by-Step Solution 1.20

1. Satisfiability of Σ by Compactness: Let $\Sigma_0 \subset \Sigma$ be an arbitrary finite subset. Σ_0 consists of a finite set of arithmetic truths $\Phi \subset \mathrm{Th}(\mathbb{N})$ and finitely many inequalities $c > \bar{n}_1, \ldots, c > \bar{n}_k$. Let $N = \max(n_1, \ldots, n_k) + 1 \in \mathbb{N}$. In the standard structure $\mathbb{N}$, interpret the constant c as N. Then:

- All sentences in Φ hold since $\Phi \subset \mathrm{Th}(\mathbb{N})$.
- Since $N > n_i$ for all $1 \leq i \leq k$, each sentence $c > \bar{n}_i$ is true in $(\mathbb{N}, N)$.

Thus $(\mathbb{N}, N) \models \Sigma_0$. Every finite subset has a model. By the First-Order Compactness Theorem, Σ has a model $\mathcal{M}^* = (M^*, 0^*, S^*, +^*, \cdot^*, <^*, c^*)$.

2. Elementary Embedding of the Standard Numbers: Since $\mathrm{Th}(\mathbb{N}) \subseteq \Sigma$, the model $\mathcal{M}^*$ satisfies the complete first-order theory of $\mathbb{N}$:

$$\mathcal{M}^* \equiv \mathbb{N}.$$

The map $j(n) = \bar{n}^{\mathcal{M}^*}$ maps each standard natural number to the interpretation of the numeral in $\mathcal{M}^*$. Since $\mathbb{N} \models \forall x \forall y\, (x \neq y \implies S^x(0) \neq S^y(0))$, j is strictly injective. Furthermore, for any formula $\varphi(x_1, \ldots, x_k)$ and $n_1, \ldots, n_k \in \mathbb{N}$:

$$\mathbb{N} \models \varphi(n_1, \ldots, n_k) \iff \varphi(\bar{n}_1, \ldots, \bar{n}_k) \in \mathrm{Th}(\mathbb{N}) \iff \mathcal{M}^* \models \varphi(j(n_1), \ldots, j(n_k)).$$

Thus $j : \mathbb{N} \hookrightarrow \mathcal{M}^*$ is an **elementary embedding**.

3. c^* is an Infinite Non-Standard Integer: For every standard integer $n \in \mathbb{N}$, the sentence $c > \bar{n}$ belongs to Σ. Therefore:

$$\mathcal{M}^* \models c^* > j(n) \quad \text{for all } n \in \mathbb{N}.$$

The element c^* is strictly greater than every standard integer! Hence $c^* \in M^* \setminus j(\mathbb{N})$ is an **infinite (non-standard) integer**.

4. Structure of Non-Standard Elements and $\mathbb{Z}$-Galaxies: Since $\mathcal{M}^* \models \mathrm{Th}(\mathbb{N})$, every first-order theorem of Peano arithmetic holds in $\mathcal{M}^*$:

- $\mathbb{N} \models \forall x\, (x > 0 \implies \exists! y\, (S(y) = x))$. Applying this to $c^* > 0$, there exists a unique element $c^* - 1$ whose successor is c^*. Similarly, $c^* + 1 = S(c^*)$, $c^* + 2, \ldots$ all exist. The set $\{\ldots, c^* - 2, c^* - 1, c^*, c^* + 1, c^* + 2, \ldots\}$ forms a discrete chain order-isomorphic to $(\mathbb{Z}, <)$.
- Operations: $2 \cdot c^* = c^* + c^* > c^*$, and $(c^*)^2 > 2c^*$. Between the standard integers $\mathbb{N}$ and the galaxy of c^*, and between c^* and $(c^*)^2$, there exist infinitely many other $\mathbb{Z}$-galaxies. In fact, the order type of $\mathcal{M}^*$ is $\mathbb{N} + \mathbb{Z} \cdot \mathbb{Q}$ (a copy of $\mathbb{N}$ followed by a dense linear order of $\mathbb{Z}$-chains without endpoints). ■

Exercise 1.21 · The Class of Fields of Characteristic 0 is Not Strongly Elementary ★★★ Advanced / Exam Level

Let $\mathcal{L}_{\text{Ring}} = \{0, 1, +, \cdot\}$ be the language of rings. Recall that the characteristic of a field F is the smallest integer $p \geq 1$ such that $\bar{p} = \underbrace{1 + \cdots + 1}_{p \text{ times}} = 0$; if no such p exists, $\text{char}(F) = 0$.

1. Show that the class Field_0 of fields of characteristic 0 is an **elementary class** by specifying an infinite set of first-order axioms Σ_0.
2. Assume for contradiction that Field_0 is **strongly elementary**, meaning there is a single first-order sentence ψ such that:

$$F \models \psi \iff \text{char}(F) = 0.$$

   Consider the theory $\Gamma = \text{Field} \cup \{\neg\psi\} \cup \{\bar{p} \neq 0 \mid p \text{ is prime}\}$. Show that every finite subset of Γ has a model.
3. Derive a contradiction using the Compactness Theorem. Conclude that Field_0 cannot be axiomatized by any finite set of axioms.

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Complete Step-by-Step Solution 1.21

1. Field_0 is an Elementary Class: Let Σ_{Field} denote the standard finite set of field axioms (associativity, commutativity, identity elements $0 \neq 1$, inverses, distributivity). Define the infinite set of axioms:

$$\Sigma_0 = \Sigma_{\text{Field}} \cup \{\bar{p} \neq 0 \mid p \text{ is a prime number}\},$$

where $\bar{p}$ is the term $1 + 1 + \cdots + 1$ (p times). A field F satisfies $\bar{p} \neq 0$ for all primes p if and only if $\text{char}(F) = 0$. Therefore, $\text{Field}_0 = \text{Mod}(\Sigma_0)$, which proves that Field_0 is an elementary class.

2. Finite Satisfiability of Γ: Assume to the contrary that Field_0 is strongly elementary, so $F \models \psi \iff \text{char}(F) = 0$. Consequently:

$$F \models \neg\psi \iff \text{char}(F) > 0 \quad \text{(characteristic is a positive prime } p\text{)}.$$

Now consider the theory:

$$\Gamma = \Sigma_{\text{Field}} \cup \{\neg\psi\} \cup \{\bar{p} \neq 0 \mid p \text{ prime}\}.$$

Let $\Gamma_0 \subset \Gamma$ be any finite subset. Γ_0 contains Σ_{Field}, $\neg\psi$, and finitely many prime restrictions:

$$\bar{p}_1 \neq 0,\ \bar{p}_2 \neq 0,\ \ldots,\ \bar{p}_k \neq 0.$$

By Euclid's theorem, there exist infinitely many prime numbers. Choose a prime $q \in \mathbb{N}$ strictly greater than all primes in Γ_0:

$$q > \max(p_1, p_2, \ldots, p_k).$$

Consider the finite field $\mathbb{F}_q = \mathbb{Z}/q\mathbb{Z}$.

- $\mathbb{F}_q$ is a field, so $\mathbb{F}_q \models \Sigma_{\text{Field}}$.
- $\text{char}(\mathbb{F}_q) = q > 0$, which means $\mathbb{F}_q$ does not have characteristic 0. Hence $\mathbb{F}_q \models \neg\psi$.
- For any prime $p_i \leq \max(p_1, \ldots, p_k) < q$, the numeral $\bar{p}_i \not\equiv 0 \pmod{q}$. Therefore, $\mathbb{F}_q \models \bar{p}_i \neq 0$ for all $1 \leq i \leq k$.

Thus, $\mathbb{F}_q \models \Gamma_0$. Every finite subset Γ_0 is satisfiable.
3. Contradiction via Compactness: By the First-Order Compactness Theorem, since every finite subset Γ_0 has a model, the full theory Γ has a model F^*. Let us evaluate the characteristic of F^*:

- Since Γ contains $\bar{p} \neq 0$ for **every** prime p, F^* cannot have prime characteristic p. Hence $\text{char}(F^*) = 0$.
- But since $\neg\psi \in \Gamma$, $F^* \models \neg\psi$, which means $\text{char}(F^*) \neq 0$!

This contradiction proves that no such sentence ψ can exist: **The class of fields of characteristic 0 is not strongly elementary; it requires infinitely many axioms.** ■

Exercise 1.22 · Formal Deductions and the Deduction Metatheorem ★★ Intermediate

In the Hilbert deductive system with axioms (Groups 1–5) and Modus Ponens:

1. Give a formal deduction of the identity theorem $\vdash \varphi \to \varphi$ using the tautology schema:

$$A_1 \equiv \varphi \to ((\varphi \to \varphi) \to \varphi),$$
$$A_2 \equiv (\varphi \to ((\varphi \to \varphi) \to \varphi)) \to ((\varphi \to (\varphi \to \varphi)) \to (\varphi \to \varphi)),$$
$$A_3 \equiv \varphi \to (\varphi \to \varphi).$$

2. State the **Deduction Metatheorem** and explain the technical restriction required when the Generalization Rule (GEN) is applied.
3. Use the Deduction Metatheorem to prove that if $\vdash \varphi \to \psi$ and $\vdash \psi \to \theta$, then $\vdash \varphi \to \theta$.

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Complete Step-by-Step Solution 1.22

1. Formal Derivation of $\vdash \varphi \to \varphi$: We construct a 5-step formal deduction:

(1) $\varphi \to ((\varphi \to \varphi) \to \varphi)$ [Axiom Group 1 (instance of $A \to (B \to A)$)]

(2) $(\varphi \to ((\varphi \to \varphi) \to \varphi)) \to ((\varphi \to (\varphi \to \varphi)) \to (\varphi \to \varphi))$ [Axiom Group 1 (instance of $(A \to (B \to C)) \to ((A \to B) \to (A \to C))$)]

(3) $(\varphi \to (\varphi \to \varphi)) \to (\varphi \to \varphi)$ [Modus Ponens from (1) and (2)]

(4) $\varphi \to (\varphi \to \varphi)$ [Axiom Group 1 (instance of $A \to (B \to A)$)]

(5) $\varphi \to \varphi$ [Modus Ponens from (4) and (3)]

This complete 5-line sequence constitutes a formal syntactic proof of $\vdash \varphi \to \varphi$.

2. The Deduction Metatheorem and the Generalization Restriction:

- **Statement:** Let Σ be a set of formulas and φ, ψ formulas. If there is a deduction $\Sigma \cup \{\varphi\} \vdash \psi$ in which no variable x free in φ is generalized upon using GEN, then:

$$\Sigma \vdash \varphi \to \psi.$$

- **Why the Restriction is Essential:** Without the restriction, one could reason: From $\{x = 0\}$, apply Generalization to obtain $\forall x(x = 0)$. If the deduction theorem held without restriction, this would yield $\vdash x = 0 \to \forall x(x = 0)$, which is semantically false! The restriction guarantees that free variables in active hypotheses behave as constants during deduction.

3. Proof of Transitivity of Implication (Hypothetical Syllogism): We wish to show $\vdash \varphi \to \theta$ given $\vdash \varphi \to \psi$ and $\vdash \psi \to \theta$. Consider the set of premises $\Sigma = \{\varphi \to \psi,\ \psi \to \theta,\ \varphi\}$. We construct a deduction of θ from Σ:

(1) φ [Hypothesis from Σ]

(2) $\varphi \to \psi$ [Premise from Σ]

(3) ψ [Modus Ponens from (1) and (2)]

(4) $\psi \to \theta$ [Premise from Σ]

(5) θ [Modus Ponens from (3) and (4)]

We have established: $\{\varphi \to \psi,\ \psi \to \theta\} \cup \{\varphi\} \vdash \theta$. Since no Generalization was used, applying the Deduction Metatheorem with respect to hypothesis φ yields:

$$\{\varphi \to \psi,\ \psi \to \theta\} \vdash \varphi \to \theta.$$

Since the two implications are provable ($\vdash \varphi \to \psi$ and $\vdash \psi \to \theta$), we conclude:

$$\vdash \varphi \to \theta. \quad \blacksquare$$

Exercise 1.23 · The Canonical Term Model Construction in Gödel's Completeness Proof ★★★

Advanced / Exam Level

Let $\mathcal{L}$ be a language with at least one constant symbol. Let Σ be a maximal consistent theory with Henkin witnesses. Let Term_0 be the set of closed terms of $\mathcal{L}$. Define the binary relation on Term_0:

$$t_1 \sim t_2 \iff \Sigma \vdash t_1 = t_2.$$

1. Prove using the equality axioms (Group 5) that $\sim$ is an equivalence relation on Term_0.
2. Let $[t]$ denote the equivalence class of $t \in \text{Term}_0$. Prove that function interpretations $f^{\mathcal{M}}([t_1], \ldots, [t_n]) = [f(t_1, \ldots, t_n)]$ and relation interpretations $([t_1], \ldots, [t_n]) \in R^{\mathcal{M}} \iff \Sigma \vdash R(t_1, \ldots, t_n)$ are **well-defined** (independent of representatives).
3. Prove the base case of the **Truth Lemma**: For every closed atomic formula $\varphi \equiv t_1 = t_2$ and

$\varphi \equiv R(t_1, \ldots, t_n)$:

$$\mathcal{M} \models \varphi \iff \varphi \in \Sigma.$$

4. Explain how the Henkin witnessing property ensures that the inductive step holds for existential sentences $\exists x \psi(x)$.

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Complete Step-by-Step Solution 1.23

1. Equivalence Relation $\sim$:

- **Reflexivity:** By equality axiom $x = x$, $\Sigma \vdash t = t$, so $t \sim t$.
- **Symmetry:** If $t_1 \sim t_2$, then $\Sigma \vdash t_1 = t_2$. By equality axiom $x = y \to y = x$ and MP, $\Sigma \vdash t_2 = t_1$, so $t_2 \sim t_1$.
- **Transitivity:** If $t_1 \sim t_2$ and $t_2 \sim t_3$, then $\Sigma \vdash t_1 = t_2$ and $\Sigma \vdash t_2 = t_3$. By axiom $(x = y \wedge y = z) \to x = z$ and MP, $\Sigma \vdash t_1 = t_3$, so $t_1 \sim t_3$.

Thus $\sim$ is an equivalence relation on Term_0.

2. Well-Definedness of Operations and Relations:

- Suppose $t_i \sim s_i$ for all $1 \leq i \leq n$, so $\Sigma \vdash t_i = s_i$. By the function equality congruence axiom:

$$(t_1 = s_1 \wedge \cdots \wedge t_n = s_n) \implies f(t_1, \ldots, t_n) = f(s_1, \ldots, s_n).$$

  Applying Modus Ponens yields $\Sigma \vdash f(t_1, \ldots, t_n) = f(s_1, \ldots, s_n)$, which means:

$$[f(t_1, \ldots, t_n)] = [f(s_1, \ldots, s_n)].$$

  Thus $f^{\mathcal{M}}$ is well-defined.

- By the relation equality congruence axiom:

$$(t_1 = s_1 \wedge \cdots \wedge t_n = s_n) \implies \big(R(t_1, \ldots, t_n) \leftrightarrow R(s_1, \ldots, s_n)\big).$$

  Thus $\Sigma \vdash R(t_1, \ldots, t_n) \iff \Sigma \vdash R(s_1, \ldots, s_n)$, so $R^{\mathcal{M}}$ is well-defined.

3. Base Case of the Truth Lemma:

- **Equality atomic formula $(t_1 = t_2)$:**

$$\mathcal{M} \models t_1 = t_2 \iff [t_1] = [t_2] \iff t_1 \sim t_2 \iff \Sigma \vdash t_1 = t_2.$$

  Since Σ is maximal consistent and deductively closed, $\Sigma \vdash t_1 = t_2 \iff (t_1 = t_2) \in \Sigma$.

- **Relation atomic formula $(R(t_1, \ldots, t_n))$:**

$\mathcal{M} \models R(t_1, \ldots, t_n) \iff ([t_1], \ldots, [t_n]) \in R^{\mathcal{M}} \iff \Sigma \vdash R(t_1, \ldots, t_n) \iff R(t_1, \ldots, t_n) \in \Sigma.$

In both cases, $\mathcal{M} \models \varphi \iff \varphi \in \Sigma$.

4. The Henkin Witness Step: Consider an existential sentence $\varphi \equiv \exists x \psi(x)$.

- ($\Longrightarrow$) If $\mathcal{M} \models \exists x \psi(x)$, then there exists an element $m \in M$ such that $\mathcal{M} \models \psi[m]$. Since every element in the term model is the equivalence class of some closed term $t \in \text{Term}_0$, $m = [t]$. By inductive hypothesis on lower-complexity formula $\psi(t)$, $\mathcal{M} \models \psi(t) \iff \psi(t) \in \Sigma$. Since $\psi(t) \to \exists x \psi(x)$ is a logical theorem, $\exists x \psi(x) \in \Sigma$.
- ($\Longleftarrow$) Conversely, suppose $\exists x \psi(x) \in \Sigma$. Here is where the **Henkin witnessing property** is crucial: Because Σ has witnesses, there exists a constant symbol c such that:

  $$\exists x \psi(x) \to \psi(c) \in \Sigma.$$

  Since $\exists x \psi(x) \in \Sigma$ and Σ is closed under Modus Ponens, we obtain:

  $$\psi(c) \in \Sigma.$$

  By the inductive hypothesis on $\psi(c)$, $\mathcal{M} \models \psi(c)$, which means $\mathcal{M} \models \psi[[c]]$. Since $[c] \in M$, this proves that $\mathcal{M} \models \exists x \psi(x)$!

The Henkin witness constant c bridges the semantic existence in $\mathcal{M}$ with the syntactic presence of $\exists$ in Σ, completing the proof of Gödel's Completeness Theorem. ■

Chapter 2

Binary Relations, Mappings & Abstract Algebraic Structures

Essential Chapter Theorems & Formulas · Binary Relations, Mappings, Groups, Rings & Fields

1. **Equivalence Relations & Partitions:** $\mathcal{R}$ is an equivalence relation on E if it is reflexive, symmetric, and transitive. The equivalence class of x is $[x] = \{y \in E : x\mathcal{R}y\}$. The quotient set $E/\mathcal{R} = \{[x] : x \in E\}$ forms a partition of E.
2. **Order Relations (Posets):** $\mathcal{R}$ is a partial order if it is reflexive, antisymmetric ($x\mathcal{R}y \wedge y\mathcal{R}x \implies x = y$), and transitive. It is a total order if $\forall x, y \in E$, $x\mathcal{R}y \vee y\mathcal{R}x$.
3. **Mappings (Functions):**
   - *Injective:* $f(x) = f(x') \implies x = x'$.
   - *Surjective:* $\forall y \in F$, $\exists x \in E$, $f(x) = y$.
   - *Bijective:* Both injective and surjective (admits an inverse $f^{-1} : F \to E$).
4. **Subgroup Criterion:** $H \subseteq G$ is a subgroup of $(G, \cdot)$ iff $H \neq \emptyset$ and $\forall x, y \in H$, $x \cdot y^{-1} \in H$.
5. **Group Homomorphisms:** $\phi(x \cdot y) = \phi(x) * \phi(y)$. $\ker \phi = \{x \in G : \phi(x) = e_H\}$ is a subgroup of G; ϕ is injective $\iff \ker \phi = \{e_G\}$.
6. **Lagrange's Theorem:** If G is a finite group and $H \leq G$, then $\text{card}(H)$ divides $\text{card}(G)$.

Exercise 2.1 · Classification of Binary Relations ★ Foundational

For each of the following binary relations defined on a specified set, determine whether it is reflexive, symmetric, antisymmetric, or transitive:

1. On $\mathbb{Z}$: $x\mathcal{R}_1 y \iff x + y$ is even.
2. On $\mathbb{Z}$: $x\mathcal{R}_2 y \iff x \leq y + 1$.
3. On $\mathbb{R}$: $x\mathcal{R}_3 y \iff x - y \in \mathbb{Q}$.
4. On $\mathbb{N}^*$: $a \mid b$ (a divides b).

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Complete Step-by-Step Solution 2.1

1. Relation $\mathcal{R}_1$ on $\mathbb{Z}$ ($x+y$ is even):

- **Reflexive:** $\forall x \in \mathbb{Z}$, $x + x = 2x$, which is even. Yes.
- **Symmetric:** If $x + y = 2k$, then $y + x = 2k$ is even. Yes.
- **Antisymmetric:** No. Take $x = 0, y = 2$: $0 + 2 = 2$ is even and $2 + 0 = 2$ is even, but $0 \neq 2$.
- **Transitive:** If $x + y$ and $y + z$ are even, then $(x + z) = (x + y) + (y + z) - 2y$ is a sum of even integers, hence even. Yes.

$\mathcal{R}_1$ is an **equivalence relation**.

2. Relation $\mathcal{R}_2$ on $\mathbb{Z}$ ($x \leq y + 1$):

- **Reflexive:** $x \leq x + 1$ holds for all $x \in \mathbb{Z}$. Yes.
- **Symmetric:** No. $0 \leq 1 + 1$ ($0\mathcal{R}_2 1$), but $2\mathcal{R}_2 0$ means $2 \leq 1$, which is false.
- **Antisymmetric:** No. Take $x = 0, y = 1$: $0 \leq 2$ (true) and $1 \leq 1$ (true), but $0 \neq 1$.
- **Transitive:** No. $2\mathcal{R}_2 1$ ($2 \leq 2$) and $1\mathcal{R}_2 0$ ($1 \leq 1$), but $2\mathcal{R}_2 0$ requires $2 \leq 1$ (false).

$\mathcal{R}_2$ is reflexive only.

3. Relation $\mathcal{R}_3$ on $\mathbb{R}$ ($x - y \in \mathbb{Q}$):

- **Reflexive:** $x - x = 0 \in \mathbb{Q}$. Yes.
- **Symmetric:** If $x - y = q \in \mathbb{Q}$, then $y - x = -q \in \mathbb{Q}$. Yes.
- **Antisymmetric:** No. $0 - 1 = -1 \in \mathbb{Q}$ and $1 - 0 = 1 \in \mathbb{Q}$, but $0 \neq 1$.
- **Transitive:** If $x - y = q_1 \in \mathbb{Q}$ and $y - z = q_2 \in \mathbb{Q}$, then $x - z = (x - y) + (y - z) = q_1 + q_2 \in \mathbb{Q}$. Yes.

$\mathcal{R}_3$ is an **equivalence relation** (the Vitali equivalence relation on $\mathbb{R}/\mathbb{Q}$).

4. Relation of Divisibility on $\mathbb{N}^*$ ($a \mid b$):

- **Reflexive:** $a = 1 \cdot a \implies a \mid a$. Yes.
- **Antisymmetric:** If $a \mid b$ and $b \mid a$, then $b = ka$ and $a = mb$ ($k, m \in \mathbb{N}^*$). Thus $a = kma \implies km = 1 \implies k = m = 1$, so $a = b$. Yes.
- **Transitive:** If $a \mid b$ and $b \mid c$, then $b = ka$ and $c = mb \implies c = (km)a \implies a \mid c$. Yes.
- **Symmetric:** No. $2 \mid 4$ but $4 \nmid 2$.

Divisibility on $\mathbb{N}^*$ is a **partial order relation**.

Exercise 2.2 · Equivalence Classes and the Quotient Set $\mathbb{R}/\mathbb{Z}$ ★ Foundational

Let $\sim$ be the binary relation on the set of real numbers $\mathbb{R}$ defined by:

$$x \sim y \iff x - y \in \mathbb{Z}.$$

1. Prove that $\sim$ is an equivalence relation on $\mathbb{R}$.
2. Explicitly describe the equivalence class $[0]$ and $[1/2]$.
3. Show that every real number $x \in \mathbb{R}$ has a unique representative in the half-open interval $[0, 1)$.
4. Deduce that the quotient set $\mathbb{R}/\sim$ is in bijective correspondence with $[0, 1)$ (the 1-torus / circle $\mathbb{S}^1$).

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Complete Step-by-Step Solution 2.2

1. Equivalence Relation Verification:

- **Reflexivity:** $\forall x \in \mathbb{R}$, $x - x = 0 \in \mathbb{Z}$, so $x \sim x$.
- **Symmetry:** If $x \sim y$, then $x - y = k \in \mathbb{Z}$. Thus $y - x = -k \in \mathbb{Z}$, so $y \sim x$.
- **Transitivity:** If $x \sim y$ and $y \sim z$, then $x - y = k \in \mathbb{Z}$ and $y - z = m \in \mathbb{Z}$. Then $x - z = (x - y) + (y - z) = k + m \in \mathbb{Z}$, so $x \sim z$.

Thus $\sim$ is an equivalence relation on $\mathbb{R}$.

2. Description of Equivalence Classes:

- $[0] = \{y \in \mathbb{R} : y - 0 \in \mathbb{Z}\} = \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$.
- $[1/2] = \{y \in \mathbb{R} : y - 1/2 \in \mathbb{Z}\} = \{k + 1/2 : k \in \mathbb{Z}\} = \{\dots, -1/2, 1/2, 3/2, 5/2, \dots\}$.

3. Unique Representative in $[0, 1)$: For any $x \in \mathbb{R}$, let $\lfloor x \rfloor$ denote the floor function (the greatest integer less than or equal to x), and let $\{x\} = x - \lfloor x \rfloor$ denote its fractional part. By definition of the floor:

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1 \implies 0 \leq x - \lfloor x \rfloor < 1 \implies \{x\} \in [0, 1).$$

Furthermore, $x - \{x\} = \lfloor x \rfloor \in \mathbb{Z}$, which proves $x \sim \{x\}$. To show uniqueness: suppose $r_1, r_2 \in [0, 1)$ such that $r_1 \sim r_2$. Then $r_1 - r_2 \in \mathbb{Z}$. But since $0 \leq r_1 < 1$ and $-1 < -r_2 \leq 0$, we have:

$$-1 < r_1 - r_2 < 1.$$

The only integer in the open interval $(-1, 1)$ is 0. Therefore $r_1 - r_2 = 0 \implies r_1 = r_2$.

4. Quotient Set Bijection: The map $\Psi : \mathbb{R}/\sim \to [0, 1)$ defined by $\Psi([x]) = \{x\}$ is well-defined and bijective. Geometrically, identifying $0 \sim 1$ wraps the interval $[0, 1)$ into the unit circle $\mathbb{S}^1 \subset \mathbb{C}$ via $x \mapsto e^{2\pi i x}$. ■

Exercise 2.3 · Injective, Surjective, and Bijective Mappings ★ Foundational

Consider the rational function:

$$f : \mathbb{R} \setminus \{1\} \to \mathbb{R} \setminus \{2\}, \qquad f(x) = \frac{2x+1}{x-1}.$$

1. Prove that f is well-defined (i.e., that $f(x) \neq 2$ for all $x \neq 1$).
2. Prove that f is injective directly from the definition.
3. Prove that f is surjective.
4. Determine the explicit formula for the inverse bijection $f^{-1}(y)$.

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Complete Step-by-Step Solution 2.3

1. Well-definedness: Suppose for contradiction that there exists $x \in \mathbb{R} \setminus \{1\}$ such that $f(x) = 2$.

$$\frac{2x+1}{x-1} = 2 \implies 2x+1 = 2(x-1) = 2x-2 \implies 1 = -2,$$

which is impossible. Thus $f(x) \neq 2$ for all $x \in \mathbb{R} \setminus \{1\}$, so the codomain is indeed $\mathbb{R} \setminus \{2\}$.

2. Injectivity of f: Let $x_1, x_2 \in \mathbb{R} \setminus \{1\}$ such that $f(x_1) = f(x_2)$.

$$\begin{aligned} \frac{2x_1+1}{x_1-1} = \frac{2x_2+1}{x_2-1} &\implies (2x_1+1)(x_2-1) = (2x_2+1)(x_1-1) \\ &\implies 2x_1x_2 - 2x_1 + x_2 - 1 = 2x_1x_2 - 2x_2 + x_1 - 1 \\ &\implies -2x_1 + x_2 = -2x_2 + x_1 \\ &\implies 3x_2 = 3x_1 \implies x_1 = x_2. \end{aligned}$$

Hence f is injective.

3. Surjectivity of f: Let $y \in \mathbb{R} \setminus \{2\}$. We seek $x \in \mathbb{R} \setminus \{1\}$ such that $f(x) = y$:

$$y = \frac{2x+1}{x-1} \implies y(x-1) = 2x+1 \implies yx - y = 2x+1 \implies x(y-2) = y+1.$$

Since $y \neq 2$, we can divide by $y-2$:

$$x = \frac{y+1}{y-2}.$$

Notice that if x were equal to 1:

$$\frac{y+1}{y-2} = 1 \implies y+1 = y-2 \implies 1 = -2 \quad \text{(absurd)}.$$

Thus $x \neq 1$, so $x \in \mathbb{R} \setminus \{1\}$. Therefore, every $y \in \mathbb{R} \setminus \{2\}$ possesses a preimage under f. f is surjective.

4. Inverse Bijection: Since f is both injective and surjective, it is bijective, and its inverse is:

$$f^{-1} : \mathbb{R} \setminus \{2\} \to \mathbb{R} \setminus \{1\}, \qquad f^{-1}(y) = \frac{y+1}{y-2}. \quad \blacksquare$$

Exercise 2.4 · Posets and Extremal Elements ★★ Intermediate

Let $E = \{1, 2, 3, 4, 6, 12\}$ ordered by the divisibility relation $a \mid b$.

1. Verify that $(\mathcal{E}, |)$ is a partially ordered set (poset). Is it totally ordered?
2. Determine the minimal, maximal, least (minimum), and greatest (maximum) elements of E.
3. Consider the subset $A = \{2, 3\}$. Determine the set of lower bounds, upper bounds, infimum ($\inf A = \gcd(2, 3)$), and supremum ($\sup A = \operatorname{lcm}(2, 3)$) of A in E.

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Complete Step-by-Step Solution 2.4

1. Poset Properties:

- Reflexivity: $\forall x \in E$, $x \mid x$ ($x = 1 \cdot x$).
- Antisymmetry: If $x \mid y$ and $y \mid x$, then $x = y$ for positive integers.
- Transitivity: If $x \mid y$ and $y \mid z$, then $x \mid z$.

$(E, |)$ is a poset. It is **not totally ordered** because, for instance, elements 2 and 3 are incomparable: $2 \nmid 3$ and $3 \nmid 2$.

2. Extremal Elements of $(E, |)$:

- **Least element (Minimum):** 1 is the minimum, since $1 \mid x$ for all $x \in E$.
- **Minimal elements:** The only minimal element is 1.
- **Greatest element (Maximum):** 12 is the maximum, since $x \mid 12$ for all $x \in E$.
- **Maximal elements:** The only maximal element is 12.

3. Bounds of $A = \{2, 3\}$ in $(E, |)$:

- **Lower bounds of A:** Elements $d \in E$ such that $d \mid 2$ and $d \mid 3$. The common divisors of 2 and 3 in E are $\{1\}$.
- **Infimum ($\inf A$):** The greatest lower bound is $\gcd(2, 3) = 1$.
- **Upper bounds of A:** Elements $m \in E$ such that $2 \mid m$ and $3 \mid m$ (common multiples in E). The multiples of 6 in E are $\{6, 12\}$.
- **Supremum ($\sup A$):** The least upper bound under divisibility is the element dividing all upper bounds: $\operatorname{lcm}(2, 3) = 6$. $\blacksquare$

Exercise 2.5 · Direct and Inverse Images of Sets ★★ Intermediate

Let $f : E \to F$ be a mapping, $A, B \subseteq E$, and $C, D \subseteq F$.

1. Prove that $f(A \cap B) \subseteq f(A) \cap f(B)$. Provide a counterexample showing that equality does not hold in general.
2. Prove that $f(A \cap B) = f(A) \cap f(B)$ for all $A, B \subseteq E$ if and only if f is injective.
3. Prove that the inverse image always preserves set operations:

$$f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D) \qquad \text{and} \qquad f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D).$$

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Complete Step-by-Step Solution 2.5

1. Proof of $f(A \cap B) \subseteq f(A) \cap f(B)$ & Counterexample: Let $y \in f(A \cap B)$. By definition, $\exists x \in A \cap B$ such that $f(x) = y$. Since $x \in A$, $y = f(x) \in f(A)$. Since $x \in B$, $y = f(x) \in f(B)$. Thus $y \in f(A) \cap f(B)$. *Counterexample to equality:* Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = x^2$. Let $A = [-2, 0]$ and $B = [0, 2]$. Then $A \cap B = \{0\}$, so $f(A \cap B) = \{0\}$. However, $f(A) = [0, 4]$ and $f(B) = [0, 4]$, so $f(A) \cap f(B) = [0, 4] \neq \{0\}$.

2. Equality Holds for all $A, B \iff f$ is Injective:

- ($\implies$) Assume $f(A \cap B) = f(A) \cap f(B)$ for all subsets. Let $x_1, x_2 \in E$ with $f(x_1) = f(x_2) = y$. Choose singletons $A = \{x_1\}$ and $B = \{x_2\}$. Then $f(A) \cap f(B) = \{y\} \cap \{y\} = \{y\} \neq \emptyset$. By our hypothesis, $f(A \cap B) = \{y\} \neq \emptyset \implies A \cap B \neq \emptyset$. Since A and B are singletons, $\{x_1\} \cap \{x_2\} \neq \emptyset \implies x_1 = x_2$. Thus f is injective.
- ($\impliedby$) Assume f is injective. We already have $f(A \cap B) \subseteq f(A) \cap f(B)$. Now take $y \in f(A) \cap f(B)$. Then $\exists x_A \in A$ and $x_B \in B$ such that $f(x_A) = y = f(x_B)$. By injectivity, $x_A = x_B = x$. Thus $x \in A \cap B$, meaning $y = f(x) \in f(A \cap B)$. Hence $f(A) \cap f(B) \subseteq f(A \cap B)$, proving equality.

3. Inverse Image Preserves Operations:

$$\begin{aligned} x \in f^{-1}(C \cap D) &\iff f(x) \in C \cap D \iff f(x) \in C \wedge f(x) \in D \\ &\iff x \in f^{-1}(C) \wedge x \in f^{-1}(D) \iff x \in f^{-1}(C) \cap f^{-1}(D). \end{aligned}$$

Similarly:

$$\begin{aligned} x \in f^{-1}(C \cup D) &\iff f(x) \in C \cup D \iff f(x) \in C \vee f(x) \in D \\ &\iff x \in f^{-1}(C) \vee x \in f^{-1}(D) \iff x \in f^{-1}(C) \cup f^{-1}(D). \quad \blacksquare \end{aligned}$$

Exercise 2.6 · Composition of Mappings and Invertibility ★★ Intermediate

Let $f : E \to F$ and $g : F \to G$ be two mappings.

1. Prove that if $g \circ f$ is injective, then f is injective.
2. Prove that if $g \circ f$ is surjective, then g is surjective.
3. Deduce that if $g \circ f$ is bijective, then f is injective and g is surjective.
4. Give an explicit example where $g \circ f$ is bijective, but f is not surjective and g is not injective.

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Complete Step-by-Step Solution 2.6

1. $g \circ f$ injective $\implies$ f injective: Let $x_1, x_2 \in E$ such that $f(x_1) = f(x_2)$. Applying g to both sides:

$$g(f(x_1)) = g(f(x_2)) \iff (g \circ f)(x_1) = (g \circ f)(x_2).$$

Since $g \circ f$ is injective, this implies $x_1 = x_2$. Hence f is injective. ■

2. $g \circ f$ surjective $\implies$ g surjective: Let $z \in G$. Since $g \circ f$ is surjective, there exists $x \in E$ such that:

$$(g \circ f)(x) = z \iff g(f(x)) = z.$$

Setting $y = f(x) \in F$, we have found an element $y \in F$ such that $g(y) = z$. Thus g is surjective. ■

3. Bijectivity of $g \circ f$: Since $g \circ f$ is bijective, it is both injective and surjective. By part 1, f is injective. By part 2, g is surjective.

4. Counterexample (f not surjective, g not injective): Let $E = \{1\}$, $F = \{a, b\}$, and $G = \{1\}$. Define:

$$f : \{1\} \to \{a, b\} \quad \text{by } f(1) = a, \qquad g : \{a, b\} \to \{1\} \quad \text{by } g(a) = 1,\ g(b) = 1.$$

The composite is $(g \circ f)(1) = g(a) = 1$, which is a bijection from $\{1\}$ to $\{1\}$. However:

- f is not surjective because $b \in F$ has no preimage ($f(\{1\}) = \{a\} \neq F$).
- g is not injective because $g(a) = g(b) = 1$ while $a \neq b$. ■

Exercise 2.7 · Subgroups and the Structure of Subgroups of $(\mathbb{Z}, +)$ ★★ Intermediate

1. State the One-Step Subgroup Criterion.
2. Prove that for any $n \in \mathbb{Z}$, the set $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ is a subgroup of $(\mathbb{Z}, +)$.
3. Prove that every subgroup H of $(\mathbb{Z}, +)$ is of the form $H = n\mathbb{Z}$ for some unique non-negative integer $n \in \mathbb{N}$.

4. For two subgroups $a\mathbb{Z}$ and $b\mathbb{Z}$, identify $a\mathbb{Z} \cap b\mathbb{Z}$ and $a\mathbb{Z} + b\mathbb{Z}$.

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Complete Step-by-Step Solution 2.7

1. Subgroup Criterion: A non-empty subset $H \subseteq G$ of a group $(G, \cdot)$ is a subgroup ($H \leq G$) if and only if $\forall x, y \in H$, $x \cdot y^{-1} \in H$. In additive notation: $\forall x, y \in H$, $x - y \in H$.
2. $n\mathbb{Z} \leq (\mathbb{Z}, +)$**:** $0 = n \cdot 0 \in n\mathbb{Z}$, so $n\mathbb{Z} \neq \emptyset$. For any $x, y \in n\mathbb{Z}$, $x = nk_1$ and $y = nk_2$. Then $x - y = nk_1 - nk_2 = n(k_1 - k_2) \in n\mathbb{Z}$. Thus $n\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$.
3. Every Subgroup of $\mathbb{Z}$ is of the form $n\mathbb{Z}$: Let $H \leq (\mathbb{Z}, +)$.

- Case 1: If $H = \{0\}$, then $H = 0\mathbb{Z}$ (with $n = 0$).
- Case 2: If $H \neq \{0\}$, then H contains a non-zero element x. Since H is a subgroup, $-x \in H$. Thus H contains at least one strictly positive integer. Let $S = H \cap \mathbb{N}^* = \{x \in H : x > 0\}$. Since S is a non-empty subset of $\mathbb{N}^*$, by the Well-Ordering Principle of $\mathbb{N}$, S possesses a least element, which we call $n = \min(S) > 0$.
  - We first show $n\mathbb{Z} \subseteq H$: Since $n \in H$ and H is closed under addition and negation, $nk \in H$ for all $k \in \mathbb{Z}$.
  - Next, we show $H \subseteq n\mathbb{Z}$: Let $m \in H$. By the Euclidean Division Theorem in $\mathbb{Z}$, there exist unique integers q, r such that:

    $$m = nq + r \quad \text{with } 0 \leq r < n.$$

    Rearranging gives $r = m - nq$. Since $m \in H$ and $nq \in H$, we have $r \in H$. If $r > 0$, then $r \in S$ and $r < n$, contradicting the minimality of $n = \min(S)$. Therefore, the remainder must be $r = 0$, which implies $m = nq \in n\mathbb{Z}$.

Hence $H = n\mathbb{Z}$.
4. Intersection and Sum of Subgroups:

- $x \in a\mathbb{Z} \cap b\mathbb{Z} \iff a \mid x \wedge b \mid x \iff \operatorname{lcm}(a, b) \mid x$. Thus $a\mathbb{Z} \cap b\mathbb{Z} = \operatorname{lcm}(a, b)\mathbb{Z}$.
- $a\mathbb{Z} + b\mathbb{Z} = \{au + bv : u, v \in \mathbb{Z}\}$. By Bézout's Identity, this set consists of all multiples of $\gcd(a, b)$: $a\mathbb{Z} + b\mathbb{Z} = \gcd(a, b)\mathbb{Z}$. ■

Exercise 2.8 · The Klein Four-Group V_4 and Cayley Tables ★★ Intermediate

Let $V_4 = \{e, a, b, c\}$ be a set of four elements equipped with a binary operation $*$ whose Cayley table is:

$*$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

1. Verify from the table that $(V_4, *)$ is an abelian group. What is the identity element?
2. What is the order of each element in V_4?
3. Show that V_4 is not isomorphic to the cyclic group $(\mathbb{Z}/4\mathbb{Z}, +)$.
4. Prove that $V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.8

1. Verification of Group Axioms:

- **Identity:** The row and column for e reproduce the header elements: $e * x = x * e = x$. Thus e is the identity element.
- **Inverses:** The diagonal elements are all e: $e^2 = a^2 = b^2 = c^2 = e$. Thus every element is its own inverse ($x^{-1} = x$).
- **Commutativity:** The table is strictly symmetric with respect to the main diagonal. Thus $x * y = y * x$ for all $x, y \in V_4$.
- **Associativity:** Verified case-by-case (or by realization as symmetries of a rectangle).

Thus $(V_4, *)$ is an abelian group.

2. Order of Elements:

- $\text{ord}(e) = 1$.
- $\text{ord}(a) = 2$ (since $a \neq e$ and $a^2 = e$).
- $\text{ord}(b) = 2$ (since $b \neq e$ and $b^2 = e$).
- $\text{ord}(c) = 2$ (since $c \neq e$ and $c^2 = e$).

3. Non-isomorphism with $\mathbb{Z}/4\mathbb{Z}$: In $(\mathbb{Z}/4\mathbb{Z}, +)$, the element $\bar{1}$ has order 4 (since $1+1 = 2, 1+1+1 = 3, 1+1+1+1 = 0$). Thus $\mathbb{Z}/4\mathbb{Z}$ is cyclic of order 4. In V_4, the maximum element order is 2. Since group isomorphisms preserve element orders, V_4 cannot be isomorphic to $\mathbb{Z}/4\mathbb{Z}$.

4. Isomorphism $V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$: The direct product $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ contains 4 elements: $(0,0), (1,0), (0,1), (1,1)$. Define the map $\phi : V_4 \to \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ by:

$$\phi(e) = (0,0), \quad \phi(a) = (1,0), \quad \phi(b) = (0,1), \quad \phi(c) = (1,1).$$

It is directly verified that $\phi(x * y) = \phi(x) + \phi(y)$ (addition modulo 2 in each component) and ϕ is bijective. Thus $V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. ■

Exercise 2.9 · Group Homomorphisms, Kernels, and Images ★★ Intermediate

Let $(G, \cdot)$ and $(H, *)$ be two groups, and let $\phi : G \to H$ be a group homomorphism.

1. Prove that $\phi(e_G) = e_H$ and $\phi(x^{-1}) = (\phi(x))^{-1}$ for all $x \in G$.
2. Prove that $\ker \phi = \{x \in G : \phi(x) = e_H\}$ is a subgroup of G.
3. Prove that ϕ is injective if and only if $\ker \phi = \{e_G\}$.
4. Prove that the image $\operatorname{Im} \phi = \{\phi(x) : x \in G\}$ is a subgroup of H.

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Complete Step-by-Step Solution 2.9

1. Preservation of Identity and Inverses:

- **Identity:** Since $e_G \cdot e_G = e_G$, applying ϕ gives $\phi(e_G) * \phi(e_G) = \phi(e_G) = \phi(e_G) * e_H$. Multiplying both sides by $(\phi(e_G))^{-1}$ yields $\phi(e_G) = e_H$.
- **Inverses:** For any $x \in G$, $x \cdot x^{-1} = e_G$. Applying ϕ:

$$\phi(x) * \phi(x^{-1}) = \phi(x \cdot x^{-1}) = \phi(e_G) = e_H.$$

By uniqueness of inverses in H, $\phi(x^{-1}) = (\phi(x))^{-1}$.

2. $\ker \phi \leq G$**:**

- Since $\phi(e_G) = e_H$, $e_G \in \ker \phi$, so $\ker \phi \neq \emptyset$.
- Let $x, y \in \ker \phi$. Then $\phi(x) = e_H$ and $\phi(y) = e_H$.

$$\phi(x \cdot y^{-1}) = \phi(x) * \phi(y^{-1}) = \phi(x) * (\phi(y))^{-1} = e_H * e_H^{-1} = e_H.$$

Thus $x \cdot y^{-1} \in \ker \phi$. By the subgroup criterion, $\ker \phi \leq G$.

3. Injectivity Criterion (ϕ **injective** $\iff$ $\ker \phi = \{e_G\}$**):**

- ($\implies$) Assume ϕ is injective. If $x \in \ker \phi$, then $\phi(x) = e_H = \phi(e_G)$. By injectivity, $x = e_G$. Thus $\ker \phi = \{e_G\}$.
- ($\impliedby$) Assume $\ker \phi = \{e_G\}$. Let $x_1, x_2 \in G$ such that $\phi(x_1) = \phi(x_2)$. Multiplying by $\phi(x_2)^{-1}$:

$$\phi(x_1) * (\phi(x_2))^{-1} = e_H \implies \phi(x_1 \cdot x_2^{-1}) = e_H \implies x_1 \cdot x_2^{-1} \in \ker \phi.$$

Since $\ker \phi = \{e_G\}$, $x_1 \cdot x_2^{-1} = e_G \implies x_1 = x_2$. Hence ϕ is injective.

4. $\operatorname{Im} \phi \leq H$**:** $e_H = \phi(e_G) \in \operatorname{Im} \phi \implies \operatorname{Im} \phi \neq \emptyset$. Let $u, v \in \operatorname{Im} \phi$. Then $\exists x, y \in G$ such that $u = \phi(x)$ and $v = \phi(y)$. Then $u * v^{-1} = \phi(x) * \phi(y)^{-1} = \phi(x \cdot y^{-1}) \in \operatorname{Im} \phi$ since $x \cdot y^{-1} \in G$. Thus $\operatorname{Im} \phi$ is a subgroup of H. ■

Exercise 2.10 · Lagrange's Theorem and Applications in Number Theory ★★★ Advanced / Exam Level

1. State and prove Lagrange's Theorem for a finite group G and subgroup $H \leq G$.
2. Deduce that the order of any element $g \in G$ divides the order of the group $|G|$.
3. Prove Fermat's Little Theorem using group theory: For any prime p and integer a with $\gcd(a, p) = 1$,
$$a^{p-1} \equiv 1 \pmod{p}.$$
4. Let G be a group of prime order p. Prove that G is cyclic and generated by any non-identity element.

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Complete Step-by-Step Solution 2.10

1. Proof of Lagrange's Theorem: Let G be a finite group and $H \leq G$. For any $x \in G$, define the left coset of H containing x as $xH = \{xh : h \in H\}$.

- The map $\psi_x : H \to xH$ given by $\psi_x(h) = xh$ is a bijection (injective by left cancellation: $xh_1 = xh_2 \implies h_1 = h_2$, and surjective by definition of xH). Thus every left coset has the exact same cardinality:
$$|xH| = |H|.$$
- The relation $x \sim y \iff x^{-1}y \in H$ is an equivalence relation whose equivalence classes are precisely the left cosets xH. Since equivalence classes partition G, the cosets are mutually disjoint and their union is G:
$$G = x_1H \cup x_2H \cup \cdots \cup x_mH,$$
where $m = [G : H]$ is the number of distinct cosets (the index of H in G). Summing cardinalities:
$$|G| = \sum_{i=1}^{m} |x_iH| = m \cdot |H| = [G : H] \cdot |H|.$$
Therefore, $|H|$ divides $|G|$. ■

2. Element Order Divides Group Order: For any $g \in G$, consider the cyclic subgroup generated by g: $\langle g \rangle = \{g^k : k \in \mathbb{Z}\}$. The cardinality of this subgroup is precisely the order of the element: $|\langle g \rangle| = \text{ord}(g)$. By Lagrange's Theorem, $|\langle g \rangle|$ divides $|G|$, hence $\text{ord}(g)$ divides $|G|$.

3. Proof of Fermat's Little Theorem: Consider the set of non-zero congruence classes modulo p:
$$G = (\mathbb{Z}/p\mathbb{Z})^* = \{\bar{1}, \bar{2}, \ldots, \overline{p-1}\}.$$

Since p is prime, every non-zero element has a multiplicative inverse by Bézout's identity. Thus $(G, \cdot)$ is a group under multiplication modulo p, with order $|G| = p-1$. For any integer a coprime to p, the class $\bar{a} \in G$. By part 2, the order $k = \text{ord}(\bar{a})$ divides $|G| = p-1$. Thus $p-1 = kq$ for some integer q, which implies:

$$\bar{a}^{p-1} = (\bar{a}^k)^q = \bar{1}^q = \bar{1} \iff a^{p-1} \equiv 1 \pmod{p}. \quad \blacksquare$$

4. Groups of Prime Order are Cyclic: Let $|G| = p$ where p is prime. Take any element $x \in G$ with $x \neq e$. Consider the subgroup $H = \langle x \rangle$. By Lagrange's Theorem, $|H|$ divides $|G| = p$. The only positive divisors of the prime p are 1 and p. Since $x \neq e$, $|H| \geq 2$, which forces $|H| = p$. Since $H \subseteq G$ and $|H| = |G|$, $H = G$, meaning $G = \langle x \rangle$ is cyclic. $\blacksquare$

Exercise 2.11 · The Ring $\mathbb{Z}/n\mathbb{Z}$ and Field Invertibility Criterion ★★★ Advanced / Exam Level

Consider the quotient ring $(\mathbb{Z}/n\mathbb{Z}, +, \cdot)$ where $n \geq 2$.

1. Prove that an element $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$ is invertible (a unit) if and only if $\gcd(a, n) = 1$.
2. Deduce that $(\mathbb{Z}/n\mathbb{Z}, +, \cdot)$ is a field if and only if n is prime.
3. In the field $\mathbb{Z}/13\mathbb{Z}$, determine the multiplicative inverse of $\bar{5}$ using the Extended Euclidean Algorithm.
4. Solve the linear congruence in $\mathbb{Z}$: $5x \equiv 9 \pmod{13}$.

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Complete Step-by-Step Solution 2.11

1. Invertibility Criterion in $\mathbb{Z}/n\mathbb{Z}$: $\bar{a}$ is invertible in $\mathbb{Z}/n\mathbb{Z} \iff \exists \bar{u} \in \mathbb{Z}/n\mathbb{Z}$ such that $\bar{a} \cdot \bar{u} = \bar{1}$.

$$\bar{a} \cdot \bar{u} = \bar{1} \iff au \equiv 1 \pmod{n} \iff \exists v \in \mathbb{Z},\ au - 1 = vn \iff au - vn = 1.$$

By Bézout's Identity, integers u, v satisfying $au + n(-v) = 1$ exist if and only if $\gcd(a, n) = 1$.
2. $\mathbb{Z}/n\mathbb{Z}$ is a Field $\iff$ n is Prime:

- ($\implies$) Assume $\mathbb{Z}/n\mathbb{Z}$ is a field. Then every non-zero element is invertible. If n were composite, $n = ab$ with $1 < a, b < n$. Then $\bar{a} \neq \bar{0}$ and $\bar{b} \neq \bar{0}$, but $\bar{a} \cdot \bar{b} = \bar{n} = \bar{0}$, making $\bar{a}$ a zero divisor, which cannot happen in a field. Contradiction.
- ($\impliedby$) Assume $n = p$ is prime. For any $\bar{a} \neq \bar{0}$, $1 \leq a \leq p-1$. Since p is prime, $\gcd(a, p) = 1$. By part 1, $\bar{a}$ is invertible. Since $(\mathbb{Z}/p\mathbb{Z}, +, \cdot)$ is a non-trivial commutative ring with unity where every non-zero element is invertible, it is a field.

3. Inverse of $\bar{5}$ in $\mathbb{Z}/13\mathbb{Z}$ via Extended Euclidean Algorithm: Compute $\gcd(13, 5)$:

$$\begin{aligned} 13 &= 2 \times 5 + 3 \implies 3 = 13 - 2 \times 5 \\ 5 &= 1 \times 3 + 2 \implies 2 = 5 - 3 = 5 - (13 - 2 \times 5) = 3 \times 5 - 13 \\ 3 &= 1 \times 2 + 1 \implies 1 = 3 - 2 = (13 - 2 \times 5) - (3 \times 5 - 13) = 2 \times 13 - 5 \times 5. \end{aligned}$$

Thus:

$$-5 \times 5 + 2 \times 13 = 1 \implies -5 \times 5 \equiv 1 \pmod{13}.$$

Modulo 13: $-5 \equiv 13 - 5 = 8 \pmod{13}$. Check: $5 \times 8 = 40 = 3 \times 13 + 1 \equiv 1 \pmod{13}$. Thus $\bar{5}^{-1} = \bar{8}$ in $\mathbb{Z}/13\mathbb{Z}$.

4. Solving $5x \equiv 9 \pmod{13}$: Multiply both sides of the congruence by the modular inverse $\bar{5}^{-1} = \bar{8}$:

$$x \equiv 8 \times 9 \pmod{13} \implies x \equiv 72 \pmod{13}.$$

Dividing 72 by 13: $72 = 5 \times 13 + 7$. Hence $x \equiv 7 \pmod{13}$. Verification: $5 \times 7 = 35 = 2 \times 13 + 9 \equiv 9 \pmod{13}$ (correct). The general solution in $\mathbb{Z}$ is $x \in \{7 + 13k : k \in \mathbb{Z}\}$. ■

Exercise 2.12 · Canonical Decomposition and the First Isomorphism Theorem for Sets ★★★

Advanced / Exam Level

Let $f : E \to F$ be any mapping between sets. Define a binary relation $\mathcal{R}_f$ on E by:

$$x \mathcal{R}_f y \iff f(x) = f(y).$$

1. Prove that $\mathcal{R}_f$ is an equivalence relation on E (the *fiber relation* or kernel equivalence).
2. Let $\pi : E \to E/\mathcal{R}_f$ be the canonical projection $\pi(x) = [x]$. Show that π is surjective.
3. Define $\tilde{f} : E/\mathcal{R}_f \to \mathrm{Im}(f)$ by $\tilde{f}([x]) = f(x)$. Prove that $\tilde{f}$ is well-defined and bijective.
4. Deduce the canonical commutative diagram factorization: $f = i \circ \tilde{f} \circ \pi$, where $i : \mathrm{Im}(f) \hookrightarrow F$ is the inclusion injection.

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Complete Step-by-Step Solution 2.12

1. Equivalence Relation $\mathcal{R}_f$:

- **Reflexivity:** $\forall x \in E,\ f(x) = f(x) \implies x \mathcal{R}_f x$.
- **Symmetry:** $x \mathcal{R}_f y \implies f(x) = f(y) \implies f(y) = f(x) \implies y \mathcal{R}_f x$.
- **Transitivity:** $x \mathcal{R}_f y \wedge y \mathcal{R}_f z \implies f(x) = f(y) \wedge f(y) = f(z) \implies f(x) = f(z) \implies x \mathcal{R}_f z$.

Thus $\mathcal{R}_f$ is an equivalence relation on E.

2. Canonical Projection π is Surjective: By definition of the quotient set $E/\mathcal{R}_f$, any element in the quotient set is an equivalence class $[x]$ for some representative $x \in E$. Thus for every $[x] \in E/\mathcal{R}_f$, we have $\pi(x) = [x]$, which demonstrates that π is surjective.

3. $\tilde{f}$ is Well-Defined and Bijective:

- **Well-defined:** Suppose $[x] = [x']$. Then $x\mathcal{R}_f x' \implies f(x) = f(x')$. Thus the definition $\tilde{f}([x]) = f(x)$ is independent of the choice of representative.
- **Injectivity:** Suppose $\tilde{f}([x_1]) = \tilde{f}([x_2])$. Then $f(x_1) = f(x_2) \implies x_1\mathcal{R}_f x_2 \implies [x_1] = [x_2]$. Thus $\tilde{f}$ is injective.
- **Surjectivity:** Let $y \in \mathrm{Im}(f)$. Then $\exists x \in E$ such that $f(x) = y$. Thus $\tilde{f}([x]) = f(x) = y$. Hence $\tilde{f}$ is surjective onto $\mathrm{Im}(f)$.

Therefore, $\tilde{f} : E/\mathcal{R}_f \to \mathrm{Im}(f)$ is a bijection.

4. Canonical Factorization: For any $x \in E$:

$$(i \circ \tilde{f} \circ \pi)(x) = i(\tilde{f}(\pi(x))) = i(\tilde{f}([x])) = i(f(x)) = f(x).$$

Hence $f = i \circ \tilde{f} \circ \pi$. This expresses every mapping as the composition of a canonical surjection (π), a bijection ($\tilde{f}$), and a canonical injection (i). ■

Chapter 3

The Real Number System and Completeness of $\mathbb{R}$

Essential Chapter Theorems & Formulas · Bounds, Completeness, Density & the Floor Function

1. **ε-Characterization of Supremum and Infimum:** Let $A \subset \mathbb{R}$ be non-empty and bounded above. $M = \sup A \iff$:

$$\begin{cases} \forall x \in A,\ x \leq M & (M \text{ is an upper bound}) \\ \forall \varepsilon > 0,\ \exists x \in A,\ x > M - \varepsilon & (\text{least upper bound}). \end{cases}$$

Let $A \subset \mathbb{R}$ be non-empty and bounded below. $m = \inf A \iff$:

$$\begin{cases} \forall x \in A,\ x \geq m & (m \text{ is a lower bound}) \\ \forall \varepsilon > 0,\ \exists x \in A,\ x < m + \varepsilon & (\text{greatest lower bound}). \end{cases}$$

2. **Axiom of Completeness (Least Upper Bound Property):** Every non-empty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.
3. **Archimedean Property of $\mathbb{R}$:** $\forall x \in \mathbb{R},\ \exists n \in \mathbb{N},\ n > x \iff \forall \varepsilon > 0,\ \exists n \in \mathbb{N}^*,\ \frac{1}{n} < \varepsilon$.
4. **Floor Function $\lfloor x \rfloor$:** $\forall x \in \mathbb{R}$, $\lfloor x \rfloor$ is the unique integer satisfying:

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1 \iff x - 1 < \lfloor x \rfloor \leq x.$$

5. **Density of $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ in $\mathbb{R}$:** For any real numbers $a < b$:

$$\exists q \in \mathbb{Q} \quad \text{such that} \quad a < q < b, \qquad \exists r \in \mathbb{R} \setminus \mathbb{Q} \quad \text{such that} \quad a < r < b.$$

Exercise 3.1 · Determination of Bounds, Extremities, Sup, and Inf ★ Foundational

For each of the following subsets of $\mathbb{R}$, determine whether it is bounded above, bounded below, or bounded. Find its supremum, infimum, and determine whether a maximum or minimum exists:

1. $A_1 = \left\{1 - \frac{1}{n} : n \in \mathbb{N}^*\right\}$.
2. $A_2 = \left\{(-1)^n + \frac{1}{n} : n \in \mathbb{N}^*\right\}$.
3. $A_3 = \{x \in \mathbb{R} : x^2 - 3x + 2 < 0\}$.
4. $A_4 = \left\{\frac{n-1}{n+1} : n \in \mathbb{N}^*\right\}$.

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Complete Step-by-Step Solution 3.1

1. Set $A_1 = \{1 - 1/n : n \geq 1\}$: The elements are $0, 1/2, 2/3, 3/4, \ldots$

- For all $n \geq 1$, $1/n > 0 \implies 1 - 1/n < 1$. Thus 1 is an upper bound. Since $\lim_{n\to\infty}(1-1/n) = 1$, $\sup A_1 = 1$. Because $1 \notin A_1$, $\max A_1$ does not exist.
- For $n = 1$, $x_1 = 0$. Since $1/n \leq 1$ for all $n \geq 1$, $1 - 1/n \geq 0$. Thus $\inf A_1 = 0$. Since $0 \in A_1$ (for $n = 1$), $\min A_1 = 0$.

2. Set $A_2 = \{(-1)^n + 1/n : n \geq 1\}$:

- For even $n = 2k$: $x_{2k} = 1 + \frac{1}{2k}$. The maximum occurs at $k = 1$ ($n = 2$): $x_2 = 1 + 1/2 = 3/2$. For all other even n, $1 < 1 + \frac{1}{2k} < 3/2$. For odd $n = 2k + 1$: $x_{2k+1} = -1 + \frac{1}{2k+1} \leq 0 < 3/2$. Thus $\sup A_2 = \max A_2 = 3/2$.
- For odd n: terms approach -1 from above. $-1 + \frac{1}{2k+1} > -1$. As $k \to \infty$, the terms approach -1. Even terms are all > 1. Thus $\inf A_2 = -1$. Since $(-1)^n + 1/n = -1$ has no solution in $\mathbb{N}^*$, $-1 \notin A_2$, so $\min A_2$ does not exist.

3. Set $A_3 = \{x \in \mathbb{R} : x^2 - 3x + 2 < 0\}$: Factoring the quadratic: $(x-1)(x-2) < 0 \iff x \in (1, 2)$. A_3 is the open interval $(1, 2)$. Thus $\sup A_3 = 2$, $\inf A_3 = 1$. Neither maximum nor minimum exists because the endpoints are excluded.

4. Set $A_4 = \{(n-1)/(n+1) : n \in \mathbb{N}^*\}$: Rewrite: $\frac{n-1}{n+1} = 1 - \frac{2}{n+1}$. For $n = 1$, $x_1 = 0$. As n increases, $\frac{2}{n+1}$ decreases strictly to 0, so x_n increases strictly towards 1. Thus $\min A_4 = \inf A_4 = 0$ (achieved at $n = 1$). $\sup A_4 = 1$. Since $x_n < 1$ for all n, $1 \notin A_4$, so $\max A_4$ does not exist. ■

Exercise 3.2 · The ε-Characterization of Supremum and Infimum ★ Foundational

Consider the set:

$$A = \left\{ \frac{2n+1}{n+2} : n \in \mathbb{N} \right\}.$$

1. Compute the first three elements of A ($n = 0, 1, 2$).
2. Show that $\min A = 1/2$.
3. Conjecture the value of $\sup A$.
4. Using the formal ε-characterization of the supremum, prove rigorously that $\sup A = 2$.

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Complete Step-by-Step Solution 3.2

1. First Elements of A:

- For $n = 0$: $x_0 = \frac{2(0)+1}{0+2} = \frac{1}{2}$.
- For $n = 1$: $x_1 = \frac{2(1)+1}{1+2} = \frac{3}{3} = 1$.
- For $n = 2$: $x_2 = \frac{2(2)+1}{2+2} = \frac{5}{4} = 1.25$.

2. Minimum of A: Rewrite x_n:

$$x_n = \frac{2n+1}{n+2} = \frac{2(n+2)-3}{n+2} = 2 - \frac{3}{n+2}.$$

As n increases on $\mathbb{N}$, the denominator $n+2$ increases, so $\frac{3}{n+2}$ decreases strictly, which implies that $x_n = 2 - \frac{3}{n+2}$ is strictly increasing. Therefore, the smallest value occurs at the minimal index $n = 0$:

$$\forall n \in \mathbb{N}, \quad x_n \geq x_0 = \frac{1}{2}.$$

Since $1/2 \in A$, we have $\min A = \inf A = 1/2$.

3. Conjecture for Supremum: As $n \to \infty$, $x_n \to 2$. We conjecture $\sup A = 2$.

4. Rigorous ε-Proof that $\sup A = 2$: We must verify two conditions:

1. **2 is an Upper Bound:** For all $n \in \mathbb{N}$:

$$x_n = 2 - \frac{3}{n+2} < 2 \quad \left(\text{since } \frac{3}{n+2} > 0\right).$$

   Thus $x \leq 2$ for all $x \in A$.

2. **Least Upper Bound Condition:** Let $\varepsilon > 0$ be arbitrary. We must find $n \in \mathbb{N}$ such that $x_n > 2 - \varepsilon$.

$$2 - \frac{3}{n+2} > 2 - \varepsilon \iff \frac{3}{n+2} < \varepsilon \iff n+2 > \frac{3}{\varepsilon} \iff n > \frac{3}{\varepsilon} - 2.$$

   By the Archimedean property of $\mathbb{R}$, there exists an integer $N \in \mathbb{N}$ such that $N > \frac{3}{\varepsilon} - 2$. Choosing $n = \max\left(0, \lfloor \frac{3}{\varepsilon} - 2 \rfloor + 1\right) \in \mathbb{N}$, we obtain:

$$x_n > 2 - \varepsilon.$$

Both conditions are satisfied. Therefore, $\sup A = 2$. ■

Exercise 3.3 · Properties of the Floor Function $\lfloor x \rfloor$ ★ Foundational

1. State the defining inequality of the floor function $\lfloor x \rfloor$ for any $x \in \mathbb{R}$.
2. Solve the following equation for $x \in \mathbb{R}$:

$$\lfloor 2x \rfloor = 3x - 1.$$

3. Prove Hermite's identity for $n = 2$: For all $x \in \mathbb{R}$,

$$\lfloor x \rfloor + \left\lfloor x + \frac{1}{2} \right\rfloor = \lfloor 2x \rfloor.$$

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Complete Step-by-Step Solution 3.3

1. Defining Inequality: For any $x \in \mathbb{R}$, $\lfloor x \rfloor \in \mathbb{Z}$ is the unique integer satisfying:

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1 \iff x - 1 < \lfloor x \rfloor \leq x.$$

2. Solving $\lfloor 2x \rfloor = 3x - 1$: Let $k = \lfloor 2x \rfloor \in \mathbb{Z}$. Then $k = 3x - 1$, which gives:

$$x = \frac{k+1}{3}.$$

Applying the definition of the floor to $2x$:

$$k \leq 2x < k + 1.$$

Substitute $x = \frac{k+1}{3}$:

$$k \leq 2\left(\frac{k+1}{3}\right) < k + 1 \iff 3k \leq 2k + 2 < 3k + 3.$$

This splits into two inequalities:

- $3k \leq 2k + 2 \implies k \leq 2$.
- $2k + 2 < 3k + 3 \implies k > -1 \implies k \geq 0$.

Since $k \in \mathbb{Z}$, the possible values are $k \in \{0, 1, 2\}$.

- For $k = 0$: $x = 1/3$. Check: $\lfloor 2/3 \rfloor = 0$, $3(1/3) - 1 = 0$ (Valid).
- For $k = 1$: $x = 2/3$. Check: $\lfloor 4/3 \rfloor = 1$, $3(2/3) - 1 = 1$ (Valid).
- For $k = 2$: $x = 3/3 = 1$. Check: $\lfloor 2 \rfloor = 2$, $3(1) - 1 = 2$ (Valid).

The solution set is $S = \{1/3, 2/3, 1\}$.
3. Proof of Hermite's Identity ($n = 2$): Every real number can be written as $x = m + \alpha$ where $m = \lfloor x \rfloor \in \mathbb{Z}$ and $\alpha \in [0, 1)$. We analyze two cases based on the position of α:

- **Case 1:** $0 \leq \alpha < 1/2$. Then $x + 1/2 = m + \alpha + 1/2$ where $1/2 \leq \alpha + 1/2 < 1$. Thus $\lfloor x + 1/2 \rfloor = m$. Also $2x = 2m + 2\alpha$ where $0 \leq 2\alpha < 1$, so $\lfloor 2x \rfloor = 2m$. LHS $= \lfloor x \rfloor + \lfloor x + 1/2 \rfloor = m + m = 2m = \lfloor 2x \rfloor =$ RHS.
- **Case 2:** $1/2 \leq \alpha < 1$. Then $x+1/2 = m+\alpha+1/2 = m+1+(\alpha-1/2)$ where $0 \leq \alpha-1/2 < 1/2$. Thus $\lfloor x + 1/2 \rfloor = m + 1$. Also $2x = 2m + 2\alpha = 2m + 1 + (2\alpha - 1)$ where $0 \leq 2\alpha - 1 < 1$. Thus $\lfloor 2x \rfloor = 2m + 1$. LHS $= \lfloor x \rfloor + \lfloor x + 1/2 \rfloor = m + (m + 1) = 2m + 1 = \lfloor 2x \rfloor =$ RHS.

In both cases, the equality holds identically. ■

Exercise 3.4 · Supremum of Unions and Minkowski Sums of Sets ★★ Intermediate

Let A and B be two non-empty, bounded subsets of $\mathbb{R}$.

1. Prove that $A \cup B$ is bounded and:
$$\sup(A \cup B) = \max(\sup A, \sup B) \qquad \text{and} \qquad \inf(A \cup B) = \min(\inf A, \inf B).$$
2. Define the Minkowski sum $A + B = \{a + b : a \in A,\, b \in B\}$. Prove that $A + B$ is bounded above and:
$$\sup(A + B) = \sup A + \sup B.$$
3. State and prove the corresponding result for $\inf(A + B)$.

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Complete Step-by-Step Solution 3.4

1. Supremum of the Union $A \cup B$: Let $M_A = \sup A$ and $M_B = \sup B$, and let $M = \max(M_A, M_B)$.

- **M is an upper bound of $A \cup B$:** Let $x \in A \cup B$. If $x \in A$, then $x \leq M_A \leq M$. If $x \in B$, then $x \leq M_B \leq M$. In both cases, $x \leq M$.
- **M is the least upper bound:** Let $\varepsilon > 0$. Assume without loss of generality that $M = M_A$. Since $M_A = \sup A$, by the ε-characterization, there exists $a \in A$ such that $a > M_A - \varepsilon = M - \varepsilon$. Since $a \in A \subseteq A \cup B$, we have found an element in $A \cup B$ strictly greater than $M - \varepsilon$.

Thus $\sup(A \cup B) = \max(\sup A, \sup B)$. By an identical argument, $\inf(A \cup B) = \min(\inf A, \inf B)$.
2. Supremum of the Minkowski Sum $A + B$: Let $M_A = \sup A$ and $M_B = \sup B$.

- $M_A + M_B$ **is an upper bound:** For any $z \in A + B$, $z = a + b$ with $a \in A, b \in B$. Since $a \le M_A$ and $b \le M_B$, adding inequalities gives:

$$a + b \le M_A + M_B.$$

Thus $M_A + M_B$ is an upper bound.

- **Least upper bound property:** Let $\varepsilon > 0$. By the $\varepsilon/2$-characterization of $\sup A$ and $\sup B$:

$$\exists a \in A \quad \text{such that} \quad a > M_A - \frac{\varepsilon}{2},$$

$$\exists b \in B \quad \text{such that} \quad b > M_B - \frac{\varepsilon}{2}.$$

Adding these two inequalities:

$$a + b > \left(M_A - \frac{\varepsilon}{2}\right) + \left(M_B - \frac{\varepsilon}{2}\right) = M_A + M_B - \varepsilon.$$

Since $a + b \in A + B$, for every $\varepsilon > 0$, there exists an element of $A + B$ strictly greater than $(M_A + M_B) - \varepsilon$.

Therefore, $\sup(A + B) = \sup A + \sup B$. ■

3. Infimum of $A + B$: By replacing $\varepsilon/2$ characterizations for lower bounds: $a \ge m_A$ and $b \ge m_B \implies a + b \ge m_A + m_B$. For any $\varepsilon > 0$, $\exists a \in A, b \in B$ with $a < m_A + \varepsilon/2$ and $b < m_B + \varepsilon/2$, so $a + b < m_A + m_B + \varepsilon$. Hence $\inf(A + B) = \inf A + \inf B$. ■

Exercise 3.5 · Duality and Products of Bounded Sets ★★ Intermediate

Let $A \subset \mathbb{R}$ be a non-empty, bounded set. Define $-A = \{-x : x \in A\}$.

1. Prove that $\sup(-A) = -\inf A$ and $\inf(-A) = -\sup A$.
2. Let $A, B \subset (0, +\infty)$ be bounded subsets of positive real numbers. Define $A \cdot B = \{ab : a \in A,\ b \in B\}$. Prove that:
$$\sup(A \cdot B) = (\sup A)(\sup B).$$
3. Show by counterexample that the product rule fails if elements can take negative values.

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Complete Step-by-Step Solution 3.5

1. Proof that $\sup(-A) = -\inf A$: Let $m = \inf A$.

- For all $x \in A$, $x \ge m \iff -x \le -m$. Thus $-m$ is an upper bound of $-A$.

- Let $\varepsilon > 0$. Since $m = \inf A$, $\exists x \in A$ such that $x < m + \varepsilon$. Multiplying by -1 reverses the inequality: $-x > -m - \varepsilon$. Since $-x \in -A$, this satisfies the ε-characterization of the supremum.

Hence $\sup(-A) = -m = -\inf A$. Applying this to $-(-A) = A$ gives $\sup(A) = -\inf(-A) \iff \inf(-A) = -\sup A$.

2. Supremum of Product of Positive Sets: Let $M_A = \sup A > 0$ and $M_B = \sup B > 0$.

- For all $a \in A, b \in B$, since $a > 0$ and $b > 0$, $a \le M_A$ and $b \le M_B \implies ab \le M_A M_B$. Thus $M_A M_B$ is an upper bound of $A \cdot B$.
- Let $\varepsilon > 0$. Choose $\eta > 0$ such that $\eta < \min(M_A, M_B)$ and $\eta(M_A + M_B) < \varepsilon$. By ε-characterization, $\exists a \in A, b \in B$ such that $a > M_A - \eta$ and $b > M_B - \eta$. Multiplying these positive inequalities:

$$ab > (M_A - \eta)(M_B - \eta) = M_A M_B - \eta(M_A + M_B) + \eta^2 > M_A M_B - \varepsilon.$$

  Thus $\sup(A \cdot B) = M_A M_B = (\sup A)(\sup B)$.

3. Counterexample for General Signs: Let $A = [-3, 1]$ and $B = [-2, 1]$. $\sup A = 1$ and $\sup B = 1 \implies (\sup A)(\sup B) = 1$. However, $(-3) \times (-2) = 6 \in A \cdot B$, so $\sup(A \cdot B) \ge 6 \neq 1$. ■

Exercise 3.6 · The Archimedean Property and Infinitesimals ★★ Intermediate

1. Prove that the set of natural numbers $\mathbb{N}$ is not bounded above in $\mathbb{R}$ using the Completeness Axiom.
2. Deduce the Archimedean Property: $\forall x \in \mathbb{R},\ \exists n \in \mathbb{N},\ n > x$.
3. Prove that $\inf\left\{\frac{1}{n} : n \in \mathbb{N}^*\right\} = 0$.
4. Prove that if $x \in \mathbb{R}$ satisfies $\forall \varepsilon > 0,\ |x| < \varepsilon$, then $x = 0$.

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Complete Step-by-Step Solution 3.6

1. $\mathbb{N}$ is Unbounded Above in $\mathbb{R}$: Suppose for contradiction that $\mathbb{N}$ is bounded above. Since $\mathbb{N} \neq \emptyset$ and $\mathbb{N}$ is bounded above, by the Completeness Axiom of $\mathbb{R}$, $\mathbb{N}$ admits a supremum $M = \sup \mathbb{N} \in \mathbb{R}$. By the ε-characterization with $\varepsilon = 1 > 0$, there exists an element $n_0 \in \mathbb{N}$ such that:

$$n_0 > M - 1.$$

Adding 1 to both sides yields:

$$n_0 + 1 > M.$$

Since $n_0 \in \mathbb{N}$, $n_0 + 1$ is also a natural number ($n_0 + 1 \in \mathbb{N}$). Thus $n_0 + 1 \in \mathbb{N}$ exceeds the upper bound M, which contradicts $M = \sup \mathbb{N}$. Therefore, $\mathbb{N}$ is not bounded above. ■

2. Deduction of the Archimedean Property: Let $x \in \mathbb{R}$. If $\mathbb{N}$ were bounded above by x, then x would be an upper bound of $\mathbb{N}$, which is impossible by part 1. Therefore, x cannot be an upper bound of $\mathbb{N}$, which means $\exists n \in \mathbb{N}$ such that $n > x$.

3. $\inf\{1/n : n \in \mathbb{N}^*\} = 0$**:** Let $S = \{1/n : n \geq 1\}$.

- For all $n \geq 1$, $n > 0 \implies 1/n > 0$. Thus 0 is a lower bound.
- Let $\varepsilon > 0$. By the Archimedean property, there exists $n \in \mathbb{N}^*$ such that $n > 1/\varepsilon$. Taking reciprocals yields $\frac{1}{n} < \varepsilon = 0 + \varepsilon$.

Hence $\inf S = 0$.

4. Proof that $|x| < \varepsilon$ **for all** $\varepsilon > 0 \implies x = 0$**:** Suppose for contradiction that $x \neq 0$. Then $|x| > 0$. Choose $\varepsilon_0 = \frac{|x|}{2} > 0$. The hypothesis implies $|x| < \varepsilon_0 = \frac{|x|}{2}$, which simplifies to $1 < 1/2$, a clear contradiction. Therefore, $x = 0$. ■

Exercise 3.7 · Rigorous Proof of the Density of $\mathbb{Q}$ in $\mathbb{R}$ ★★ Intermediate

A subset $D \subset \mathbb{R}$ is said to be dense in $\mathbb{R}$ if for every pair of real numbers $a, b \in \mathbb{R}$ with $a < b$, there exists $d \in D$ such that $a < d < b$.

1. Let $a, b \in \mathbb{R}$ with $a < b$. Prove that there exists $q \in \mathbb{N}^*$ such that $q(b - a) > 1$.
2. Let $p = \lfloor qa \rfloor + 1 \in \mathbb{Z}$. Prove that $qa < p \leq qa + 1$.
3. Deduce that $a < \frac{p}{q} < b$, establishing the density of the rational numbers $\mathbb{Q}$ in $\mathbb{R}$.

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Complete Step-by-Step Solution 3.7

1. Existence of Denominator $q \in \mathbb{N}^*$**:** Since $a < b$, the difference $b - a > 0$ is strictly positive. Thus $\frac{1}{b-a} \in \mathbb{R}$. By the Archimedean property of $\mathbb{R}$, there exists an integer $q \in \mathbb{N}^*$ such that:

$$q > \frac{1}{b - a}.$$

Multiplying both sides by the positive quantity $b - a$:

$$q(b - a) > 1 \iff qb - qa > 1 \iff qa + 1 < qb.$$

2. Bounds on $p = \lfloor qa \rfloor + 1$**:** By definition of the floor function applied to $qa \in \mathbb{R}$:

$$\lfloor qa \rfloor \leq qa < \lfloor qa \rfloor + 1.$$

Substituting $p = \lfloor qa \rfloor + 1$:

$$p - 1 \leq qa < p.$$

From the right inequality, $qa < p$. From the left inequality, $p - 1 \leq qa \implies p \leq qa + 1$. Thus:

$$qa < p \leq qa + 1.$$

3. Deduction that $a < p/q < b$: Combining the results:

$$qa < p \leq qa + 1 < qb.$$

Therefore:

$$qa < p < qb.$$

Dividing through by the strictly positive integer $q > 0$:

$$a < \frac{p}{q} < b.$$

Since $p \in \mathbb{Z}$ and $q \in \mathbb{N}^*$, the ratio $\frac{p}{q}$ is a rational number ($\frac{p}{q} \in \mathbb{Q}$). Since a and b were arbitrary real numbers with $a < b$, this completes the proof that $\mathbb{Q}$ is dense in $\mathbb{R}$. ■

Exercise 3.8 · Density of the Irrational Numbers $\mathbb{R} \setminus \mathbb{Q}$ ★★ Intermediate

1. Show that if $q \in \mathbb{Q}$ and $r \in \mathbb{R} \setminus \mathbb{Q}$, then $q + r \in \mathbb{R} \setminus \mathbb{Q}$.
2. Let $a, b \in \mathbb{R}$ with $a < b$. Consider the shifted interval $\left(a - \sqrt{2}, b - \sqrt{2}\right)$. Using the density of $\mathbb{Q}$, prove that there exists an irrational number $\xi \in \mathbb{R} \setminus \mathbb{Q}$ such that:

$$a < \xi < b.$$

3. Deduce that between any two distinct real numbers, there are infinitely many rational numbers and infinitely many irrational numbers.

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Complete Step-by-Step Solution 3.8

1. Sum of Rational and Irrational is Irrational: Let $q \in \mathbb{Q}$ and $r \notin \mathbb{Q}$. Suppose for contradiction that $s = q + r \in \mathbb{Q}$. Then $r = s - q$. Since $\mathbb{Q}$ is a field, the difference of two rational numbers is rational ($s - q \in \mathbb{Q}$), which implies $r \in \mathbb{Q}$, contradicting $r \notin \mathbb{Q}$. Thus $q + r$ is irrational.
2. Proof of Density of $\mathbb{R} \setminus \mathbb{Q}$: Let $a, b \in \mathbb{R}$ with $a < b$. Subtracting $\sqrt{2}$ yields the real numbers $a' = a - \sqrt{2}$ and $b' = b - \sqrt{2}$, with $a' < b'$. By the density of $\mathbb{Q}$ in $\mathbb{R}$ (proved in Exercise 3.7),

there exists a rational number $q \in \mathbb{Q}$ such that:

$$a - \sqrt{2} < q < b - \sqrt{2}.$$

Adding $\sqrt{2}$ across the inequalities:

$$a < q + \sqrt{2} < b.$$

Let $\xi = q + \sqrt{2}$. Since $q \in \mathbb{Q}$ and $\sqrt{2} \notin \mathbb{Q}$, by part 1, $\xi \in \mathbb{R} \setminus \mathbb{Q}$ is irrational. Hence ξ is an irrational number in the open interval (a, b).

3. Infinitely Many Rationals and Irrationals: Suppose there were only finitely many rational numbers in (a, b), say $\{q_1, \ldots, q_k\}$. Let $q_{\min} = \min_{1 \le i \le k}\{q_i : q_i > a\}$. Then the non-empty interval $(a, q_{\min})$ would contain no rational numbers, which directly contradicts the density of $\mathbb{Q}$. The exact same contradiction applies to irrationals. Thus every non-empty interval (a, b) contains infinitely many rational and infinitely many irrational numbers. ■

Exercise 3.9 · Cantor's Nested Interval Theorem (Théorème des Segments Emboîtés) ★★

Intermediate

Let $([a_n, b_n])_{n \in \mathbb{N}}$ be a sequence of closed, bounded intervals in $\mathbb{R}$ that are nested:

$$[a_0, b_0] \supseteq [a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots \supseteq [a_n, b_n] \supseteq \ldots$$

1. Prove that the sequence (a_n) is increasing, (b_n) is decreasing, and $\forall n, p \in \mathbb{N}$, $a_n \le b_p$.
2. Let $A = \{a_n : n \in \mathbb{N}\}$. Prove that A is bounded above and that $\sup A \le b_p$ for all $p \in \mathbb{N}$.
3. Conclude that the intersection $\bigcap_{n=0}^{\infty}[a_n, b_n]$ is non-empty.
4. If in addition $\lim_{n \to \infty}(b_n - a_n) = 0$, prove that the intersection contains exactly one point.

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Complete Step-by-Step Solution 3.9

1. Monotonicity and Cross-Inequality: The containment $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]$ means:

$$a_n \le a_{n+1} \le b_{n+1} \le b_n.$$

Thus (a_n) is increasing and (b_n) is decreasing. Let $n, p \in \mathbb{N}$. If $n \le p$, then $a_n \le a_p \le b_p$. If $n > p$, then $a_n \le b_n \le b_p$. In all cases, $a_n \le b_p$.

2. Supremum of A: Every b_p is an upper bound of the set $A = \{a_n : n \in \mathbb{N}\}$. Since $A \neq \emptyset$ and A is bounded above (for instance by b_0), by the Completeness Axiom of $\mathbb{R}$, $c = \sup A$ exists in $\mathbb{R}$. By definition of the supremum (least upper bound), since every b_p is an upper bound of A:

$$c = \sup A \le b_p \quad \text{for all } p \in \mathbb{N}.$$

Furthermore, since c is an upper bound of A, $a_n \leq c$ for all $n \in \mathbb{N}$.
3. Non-Empty Intersection: Combining the inequalities: for all $n \in \mathbb{N}$:

$$a_n \leq c \leq b_n \iff c \in [a_n, b_n].$$

Since $c \in [a_n, b_n]$ for every $n \in \mathbb{N}$, we have $c \in \bigcap_{n=0}^{\infty}[a_n, b_n]$, so the intersection is non-empty.
4. Uniqueness when $\lim(b_n - a_n) = 0$: Suppose $c_1, c_2 \in \bigcap_{n=0}^{\infty}[a_n, b_n]$ with $c_1 \leq c_2$. Then for all $n \in \mathbb{N}$, $a_n \leq c_1 \leq c_2 \leq b_n$. This implies:

$$0 \leq c_2 - c_1 \leq b_n - a_n.$$

Taking the limit as $n \to \infty$:

$$0 \leq c_2 - c_1 \leq \lim_{n\to\infty}(b_n - a_n) = 0 \implies c_2 - c_1 = 0 \implies c_1 = c_2.$$

Therefore, the intersection consists of the single point $\{c\}$. ■

Exercise 3.10 · Incompleteness of $\mathbb{Q}$: Absence of Supremum in $\mathbb{Q}$ ★★★ Advanced / Exam Level

Consider the subset of rational numbers:

$$S = \{q \in \mathbb{Q} : q > 0 \text{ and } q^2 < 2\}.$$

1. Prove that S is non-empty and bounded above in $\mathbb{Q}$.
2. For any $q \in S$, define:
$$q' = q - \frac{q^2 - 2}{q + 2} = \frac{2q + 2}{q + 2}.$$
Show that $q' \in \mathbb{Q}$, $q' > q$, and $(q')^2 < 2$, proving that S has no maximum in $\mathbb{Q}$.
3. Let $M \in \mathbb{Q}$ be any upper bound of S. Show that there exists a smaller upper bound $M' \in \mathbb{Q}$ with $M' < M$.
4. Conclude that the ordered field $\mathbb{Q}$ does not satisfy the Least Upper Bound Axiom (Completeness).

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Complete Step-by-Step Solution 3.10

1. S is Non-Empty and Bounded Above in $\mathbb{Q}$: $1 \in \mathbb{Q}$, $1 > 0$, and $1^2 = 1 < 2 \implies 1 \in S \neq \emptyset$. If $q \geq 2$, then $q^2 \geq 4 > 2$, so $q \notin S$. Thus 2 (or any rational number ≥ 2) is an upper bound of S in $\mathbb{Q}$.

2. Construction of a Strictly Larger Element q' in S: Let $q \in S$, so $q > 0$ and $q^2 < 2$.

$$q' = \frac{2q+2}{q+2}.$$

Clearly $q' \in \mathbb{Q}$ and $q' > 0$. Compute $q' - q$:

$$q' - q = \frac{2q+2-q(q+2)}{q+2} = \frac{2-q^2}{q+2}.$$

Since $q^2 < 2$, the numerator $2 - q^2 > 0$, so $q' - q > 0 \implies q' > q$. Now compute $(q')^2 - 2$:

$$(q')^2 - 2 = \left(\frac{2q+2}{q+2}\right)^2 - 2 = \frac{4q^2+8q+4-2(q^2+4q+4)}{(q+2)^2} = \frac{2(q^2-2)}{(q+2)^2}.$$

Since $q^2 - 2 < 0$, we have $(q')^2 - 2 < 0 \implies (q')^2 < 2$. Thus $q' \in S$ and $q' > q$. This proves that no element in S can be the maximum of S.

3. Absence of a Least Upper Bound in $\mathbb{Q}$: Let $M \in \mathbb{Q}$ be an upper bound of S. Since $\sqrt{2} \notin \mathbb{Q}$, we cannot have $M^2 = 2$. If $M^2 < 2$, then by part 2, there exists $M' \in S$ with $M' > M$, contradicting that M is an upper bound. Thus we must have $M^2 > 2$. Now consider:

$$M' = \frac{2M+2}{M+2}.$$

Then $M - M' = \frac{M^2-2}{M+2} > 0 \implies M' < M$. Furthermore:

$$(M')^2 - 2 = \frac{2(M^2-2)}{(M+2)^2} > 0 \implies (M')^2 > 2.$$

This implies that M' is still an upper bound of S in $\mathbb{Q}$, but $M' < M$. Hence no rational upper bound can be the *least* upper bound.

4. Conclusion: The set $S \subset \mathbb{Q}$ is non-empty and bounded above, yet it possesses no supremum in $\mathbb{Q}$. Therefore, the rational number system $\mathbb{Q}$ is incomplete. ■

Exercise 3.11 · Supremum and Infimum of Doubly-Indexed Sets ★★★ Advanced / Exam Level

Consider the set:

$$A = \left\{ \frac{1}{n} - \frac{1}{m} : n, m \in \mathbb{N}^* \right\}.$$

1. Determine upper and lower bounds for A.
2. Prove that $\sup A = 1$ and $\inf A = -1$.
3. Does A possess a maximum or a minimum? Justify carefully.
4. Prove that 0 is an accumulation point of A.

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Complete Step-by-Step Solution 3.11

1. Bounds of A: For all $n, m \in \mathbb{N}^*$, $0 < \frac{1}{n} \leq 1$ and $0 < \frac{1}{m} \leq 1$. Thus:

$$-1 < -\frac{1}{m} \leq \frac{1}{n} - \frac{1}{m} < \frac{1}{n} \leq 1.$$

Hence $-1 < x < 1$ for all $x \in A$. A is bounded: bounded below by -1 and bounded above by 1.

2. Proof that $\sup A = 1$ and $\inf A = -1$:

- Upper bound: $\forall x \in A, x = \frac{1}{n} - \frac{1}{m} < \frac{1}{n} \leq 1$, so 1 is an upper bound.
- Let $\varepsilon > 0$. Fix $n = 1$. Then $x_m = 1 - \frac{1}{m}$. By Archimedes, $\exists m \in \mathbb{N}^*$ such that $\frac{1}{m} < \varepsilon$. Then $x_m = 1 - \frac{1}{m} > 1 - \varepsilon$. Since $x_m \in A$, this satisfies the ε-characterization of the supremum. Thus $\sup A = 1$.
- For the infimum: Fix $m = 1$. Then $x_n = \frac{1}{n} - 1$. By Archimedes, $\exists n \in \mathbb{N}^*$ such that $\frac{1}{n} < \varepsilon$. Then $x_n = \frac{1}{n} - 1 < -1 + \varepsilon$. Thus $\inf A = -1$.

3. Non-existence of Maximum and Minimum: Suppose $\max A$ exists. Then there would exist $n, m \in \mathbb{N}^*$ such that $\frac{1}{n} - \frac{1}{m} = 1$. This requires $\frac{1}{n} = 1 + \frac{1}{m} > 1$, which is impossible since $n \geq 1 \implies \frac{1}{n} \leq 1$. Thus $\max A$ does not exist. By symmetry, $\frac{1}{n} - \frac{1}{m} = -1 \implies \frac{1}{m} = 1 + \frac{1}{n} > 1$, which is impossible. Thus $\min A$ does not exist.

4. 0 is an Accumulation Point of A: Taking $n = m$ gives $\frac{1}{n} - \frac{1}{n} = 0 \in A$. Taking $m = n + 1$ gives $y_n = \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)} \in A$. For any $\varepsilon > 0$, choosing $n > 1/\sqrt{\varepsilon}$ gives $0 < y_n < \varepsilon$. Thus every neighborhood of 0 contains points of $A \setminus \{0\}$, proving 0 is an accumulation point. ■

Exercise 3.12 · Uncountability of $\mathbb{R}$ and Cantor's Diagonal Argument ★★★ Advanced / Exam Level

1. Define what it means for a set E to be countable.
2. State the Nested Interval Property of $\mathbb{R}$ (proved in Exercise 3.9).
3. Prove by contradiction that the open interval $(0, 1)$ is uncountable using Cantor's diagonal argument: Assume $(0, 1) = \{x_1, x_2, x_3, \dots\}$ is an enumeration, where each x_n has a decimal expansion:
$$x_n = 0.d_{n,1}d_{n,2}d_{n,3}\dots \quad (d_{n,k} \in \{0, 1, \dots, 9\}).$$
Construct an explicit real number $y \in (0, 1)$ that cannot belong to the list.
4. Deduce that the set of irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ is uncountable.

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Complete Step-by-Step Solution 3.12

1. Definition of Countability: A set E is countable if there exists an injective map $f : E \to \mathbb{N}$. If E is infinite and there exists a bijection $f : E \to \mathbb{N}$, E is countably infinite (or denumerable).
2. Nested Interval Property: Every sequence of nested non-empty closed bounded intervals $[a_n, b_n]$ has a non-empty intersection $\bigcap_{n=0}^{\infty}[a_n, b_n] \neq \emptyset$.
3. Cantor's Diagonal Argument: Assume for contradiction that the interval $(0, 1)$ is countable. Then there exists a bijection $f : \mathbb{N}^* \to (0, 1)$, listing all numbers in $(0, 1)$ as a sequence $(x_n)_{n\geq 1}$:

$$\begin{aligned} x_1 &= 0.d_{1,1}d_{1,2}d_{1,3}d_{1,4}\dots \\ x_2 &= 0.d_{2,1}d_{2,2}d_{2,3}d_{2,4}\dots \\ x_3 &= 0.d_{3,1}d_{3,2}d_{3,3}d_{3,4}\dots \\ &\vdots \end{aligned}$$

(To avoid ambiguous representations like $0.1999\cdots = 0.2000\dots$, we prohibit representations ending in infinitely repeating 9s). Now construct a number $y = 0.c_1c_2c_3\cdots \in (0, 1)$ by defining its n-th decimal digit c_n along the diagonal:

$$c_n = \begin{cases} 4 & \text{if } d_{n,n} \neq 4 \\ 5 & \text{if } d_{n,n} = 4. \end{cases}$$

Notice that:

- All digits of y are either 4 or 5, so $0 < y < 1$, and y does not end in trailing 0s or 9s.
- For every $n \geq 1$, the n-th digit of y differs from the n-th digit of x_n:

$$c_n \neq d_{n,n} \implies y \neq x_n.$$

Thus y cannot equal x_n for any $n \in \mathbb{N}^*$. This means $y \in (0, 1)$ is not in the enumeration, which directly contradicts the assumption that (x_n) enumerated all elements of $(0, 1)$. Therefore, $(0, 1)$ is uncountable, and consequently $\mathbb{R}$ is uncountable. ■
4. Uncountability of $\mathbb{R} \setminus \mathbb{Q}$: We know that $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$. The set of rational numbers $\mathbb{Q}$ is countable. If $\mathbb{R} \setminus \mathbb{Q}$ were also countable, then $\mathbb{R}$, being the union of two countable sets, would be countable. This contradicts the fact that $\mathbb{R}$ is uncountable. Therefore, the set of irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ is uncountable. ■

Chapter 4

Numerical Sequences & Limits in $\mathbb{R}$

Essential Chapter Theorems & Formulas · Convergence Criteria, Monotonicity, Cauchy & Recurrence

1. **ε-N Definition of Sequence Limit:**

$$\lim_{n\to\infty} u_n = \ell \iff \forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall n \geq N,\ |u_n - \ell| < \varepsilon.$$

2. **Squeeze Theorem (Théorème des Gendarmes):** If $v_n \leq u_n \leq w_n$ for all $n \geq N_0$ and $\lim v_n = \lim w_n = \ell$, then $\lim u_n = \ell$.

3. **Monotone Convergence Theorem:**

   - If (u_n) is increasing and bounded above, then (u_n) converges and $\lim u_n = \sup\{u_n : n \in \mathbb{N}\}$.
   - If (u_n) is decreasing and bounded below, then (u_n) converges and $\lim u_n = \inf\{u_n : n \in \mathbb{N}\}$.

4. **Bolzano-Weierstrass Theorem:** Every bounded real sequence $(u_n)_{n\in\mathbb{N}}$ possesses at least one convergent subsequence $(u_{\phi(n)})_{n\in\mathbb{N}}$.

5. **Cauchy Criterion:** A sequence (u_n) in $\mathbb{R}$ converges if and only if it is a Cauchy sequence:

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall p, q \geq N,\ |u_p - u_q| < \varepsilon.$$

6. **Adjacent Sequences (Suites Adjacentes):** Two sequences (u_n) and (v_n) are adjacent if (u_n) is increasing, (v_n) is decreasing, and $\lim(v_n - u_n) = 0$. They necessarily converge to the exact same finite limit ℓ, with $u_n \leq \ell \leq v_n$ for all n.

7. **Recurrence Relations $u_{n+1} = f(u_n)$:** If $f : I \to I$ is continuous and (u_n) converges to $\ell \in I$, then ℓ is a fixed point: $f(\ell) = \ell$.

Exercise 4.1 · Direct ε-N Proofs of Sequence Convergence ★ Foundational

Using the formal ε-N definition of limit:

1. Prove that $\lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$.
2. Prove that $\lim_{n\to\infty} \frac{3n-1}{2n+5} = \frac{3}{2}$.
3. For $\varepsilon = 10^{-3}$, calculate an explicit integer threshold N for the sequence in part 2.

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Complete Step-by-Step Solution 4.1

1. Proof that $\lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$: Let $\varepsilon > 0$ be given. We must find $N \in \mathbb{N}$ such that:

$$\forall n \geq N, \quad \left|\frac{1}{\sqrt{n}} - 0\right| < \varepsilon \iff \frac{1}{\sqrt{n}} < \varepsilon \iff \sqrt{n} > \frac{1}{\varepsilon} \iff n > \frac{1}{\varepsilon^2}.$$

By the Archimedean property of $\mathbb{R}$, there exists $N \in \mathbb{N}$ such that $N > \frac{1}{\varepsilon^2}$ (for instance, $N = \lfloor 1/\varepsilon^2 \rfloor + 1$). Then for all $n \geq N$:

$$n \geq N > \frac{1}{\varepsilon^2} \implies \frac{1}{\sqrt{n}} < \varepsilon.$$

Thus $\lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$. ■

2. Proof that $\lim_{n\to\infty} \frac{3n-1}{2n+5} = \frac{3}{2}$: Compute the difference $|u_n - \ell|$:

$$\left|\frac{3n-1}{2n+5} - \frac{3}{2}\right| = \left|\frac{2(3n-1) - 3(2n+5)}{2(2n+5)}\right| = \left|\frac{6n - 2 - 6n - 15}{4n+10}\right| = \frac{17}{4n+10}.$$

Notice that for all $n \geq 1$, $4n + 10 > 4n$, so:

$$\frac{17}{4n+10} < \frac{17}{4n}.$$

To ensure $|u_n - 3/2| < \varepsilon$, it suffices to enforce $\frac{17}{4n} < \varepsilon$:

$$\frac{17}{4n} < \varepsilon \iff n > \frac{17}{4\varepsilon}.$$

Let $\varepsilon > 0$. Choose $N = \left\lfloor \frac{17}{4\varepsilon} \right\rfloor + 1 \in \mathbb{N}$. Then for all $n \geq N$:

$$\left|\frac{3n-1}{2n+5} - \frac{3}{2}\right| = \frac{17}{4n+10} < \frac{17}{4n} \leq \frac{17}{4N} < \varepsilon.$$

This proves that $\lim_{n\to\infty} u_n = 3/2$. ■

3. Threshold for $\varepsilon = 10^{-3}$:

$$N = \left\lfloor \frac{17}{4 \times 10^{-3}} \right\rfloor + 1 = \left\lfloor \frac{17000}{4} \right\rfloor + 1 = 4250 + 1 = 4251.$$

For any index $n \geq 4251$, the term u_n is guaranteed to be within 0.001 of 1.5.

Exercise 4.2 · Computation of Limits and Resolution of Indeterminate Forms ★ Foundational

Calculate the limit of each of the following sequences as $n \to \infty$:

1. $u_n = \sqrt{n^2 + 4n + 1} - n$ (Indeterminate form $\infty - \infty$).
2. $v_n = \frac{2^n + 3^n}{2^{n+1} + 3^{n+1}}$ (Dominant geometric base).
3. $w_n = \left(1 + \frac{2}{n}\right)^n$ (Euler exponential form 1^∞).
4. $z_n = \frac{n^3 + 5n - 2}{4n^3 - 7n^2 + 1}$.

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Complete Step-by-Step Solution 4.2

1. Limit of $u_n = \sqrt{n^2 + 4n + 1} - n$: Multiply and divide by the conjugate expression:

$$\begin{aligned} u_n &= \frac{(\sqrt{n^2 + 4n + 1} - n)(\sqrt{n^2 + 4n + 1} + n)}{\sqrt{n^2 + 4n + 1} + n} \\ &= \frac{(n^2 + 4n + 1) - n^2}{\sqrt{n^2(1 + 4/n + 1/n^2)} + n} \\ &= \frac{4n + 1}{n\sqrt{1 + 4/n + 1/n^2} + n} = \frac{n(4 + 1/n)}{n(\sqrt{1 + 4/n + 1/n^2} + 1)} \\ &= \frac{4 + 1/n}{\sqrt{1 + 4/n + 1/n^2} + 1}. \end{aligned}$$

As $n \to \infty$, $1/n \to 0$ and $1/n^2 \to 0$:

$$\lim_{n\to\infty} u_n = \frac{4 + 0}{\sqrt{1 + 0 + 0} + 1} = \frac{4}{2} = 2.$$

2. Limit of v_n: Factor the dominant base 3^n from numerator and denominator:

$$v_n = \frac{3^n\left[(2/3)^n + 1\right]}{3^{n+1}\left[(2/3)^{n+1} + 1\right]} = \frac{1}{3} \cdot \frac{(2/3)^n + 1}{(2/3)^{n+1} + 1}.$$

Since $|2/3| < 1$, $\lim_{n\to\infty}(2/3)^n = 0$:

$$\lim_{n\to\infty} v_n = \frac{1}{3} \cdot \frac{0 + 1}{0 + 1} = \frac{1}{3}.$$

3. Limit of $w_n = (1 + 2/n)^n$: Use the logarithmic transformation:

$$\ln(w_n) = n \ln\left(1 + \frac{2}{n}\right).$$

As $n \to \infty$, $2/n \to 0$. Using the standard equivalence $\ln(1+u) \sim u$ as $u \to 0$:

$$\ln\left(1+\frac{2}{n}\right) \sim \frac{2}{n} \implies \ln(w_n) \sim n \cdot \frac{2}{n} = 2.$$

By continuity of the exponential function:

$$\lim_{n\to\infty} w_n = e^2.$$

4. Limit of z_n: Factor the highest power n^3:

$$z_n = \frac{n^3(1+5/n^2-2/n^3)}{n^3(4-7/n+1/n^3)} = \frac{1+5/n^2-2/n^3}{4-7/n+1/n^3} \xrightarrow{n\to\infty} \frac{1+0-0}{4-0+0} = \frac{1}{4}. \quad \blacksquare$$

Exercise 4.3 · The Squeeze (Sandwich) Theorem ★ Foundational

1. State the Squeeze Theorem for numerical sequences.
2. Determine $\lim_{n\to\infty} \frac{\cos(n^2)+(-1)^n \sin(n)}{n}$.
3. Determine the limit of the sum:

$$S_n = \sum_{k=1}^{n} \frac{1}{\sqrt{n^2+k}} = \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \cdots + \frac{1}{\sqrt{n^2+n}}.$$

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Complete Step-by-Step Solution 4.3

1. Statement of Squeeze Theorem: Let $(u_n), (v_n), (w_n)$ be real sequences. If there exists $N_0 \in \mathbb{N}$ such that $v_n \le u_n \le w_n$ for all $n \ge N_0$, and $\lim_{n\to\infty} v_n = \lim_{n\to\infty} w_n = \ell \in \mathbb{R}$, then (u_n) converges and $\lim_{n\to\infty} u_n = \ell$.

2. Limit of $x_n = \frac{\cos(n^2)+(-1)^n \sin(n)}{n}$: Using the triangle inequality on the numerator:

$$|\cos(n^2) + (-1)^n \sin(n)| \le |\cos(n^2)| + |\sin(n)| \le 1 + 1 = 2.$$

Dividing by $n > 0$:

$$-\frac{2}{n} \le x_n \le \frac{2}{n}.$$

Since $\lim_{n\to\infty}(-2/n) = 0$ and $\lim_{n\to\infty}(2/n) = 0$, by the Squeeze Theorem:

$$\lim_{n\to\infty} x_n = 0.$$

3. Limit of $S_n = \sum_{k=1}^{n} \frac{1}{\sqrt{n^2+k}}$: For each $k \in \{1, 2, \ldots, n\}$, we have $1 \le k \le n$, which gives:

$$n^2 + 1 \le n^2 + k \le n^2 + n.$$

Taking square roots and reciprocals reverses the inequality:

$$\frac{1}{\sqrt{n^2+n}} \leq \frac{1}{\sqrt{n^2+k}} \leq \frac{1}{\sqrt{n^2+1}}.$$

Summing this inequality for all k from 1 to n (n identical terms):

$$\sum_{k=1}^{n} \frac{1}{\sqrt{n^2+n}} \leq S_n \leq \sum_{k=1}^{n} \frac{1}{\sqrt{n^2+1}} \iff \frac{n}{\sqrt{n^2+n}} \leq S_n \leq \frac{n}{\sqrt{n^2+1}}.$$

Compute the limits of the bounding sequences:

- $\frac{n}{\sqrt{n^2+n}} = \frac{n}{n\sqrt{1+1/n}} = \frac{1}{\sqrt{1+1/n}} \xrightarrow{n\to\infty} 1.$
- $\frac{n}{\sqrt{n^2+1}} = \frac{n}{n\sqrt{1+1/n^2}} = \frac{1}{\sqrt{1+1/n^2}} \xrightarrow{n\to\infty} 1.$

Since both bounds converge to 1, by the Squeeze Theorem:

$$\lim_{n\to\infty} S_n = 1. \quad \blacksquare$$

Exercise 4.4 · Monotone Convergence: The Basel Partial Sums ★★ Intermediate

Consider the sequence of partial sums:

$$u_n = \sum_{k=1}^{n} \frac{1}{k^2} = 1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} \quad (n \geq 1).$$

1. Prove that (u_n) is strictly increasing.
2. Prove that for all $k \geq 2$, $\frac{1}{k^2} < \frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}$.
3. Deduce by telescoping summation that for all $n \geq 1$, $u_n < 2$.
4. Conclude that (u_n) converges to a finite limit $\ell \in (1, 2]$ using the Monotone Convergence Theorem.

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Complete Step-by-Step Solution 4.4

1. Monotonicity: Compute the difference between consecutive terms:

$$u_{n+1} - u_n = \left(\sum_{k=1}^{n+1} \frac{1}{k^2}\right) - \left(\sum_{k=1}^{n} \frac{1}{k^2}\right) = \frac{1}{(n+1)^2} > 0.$$

Since $u_{n+1} - u_n > 0$ for all $n \geq 1$, the sequence (u_n) is strictly increasing.

2. Partial Fraction Bound: For any integer $k \geq 2$, $k^2 > k(k-1)$. Taking reciprocals:

$$\frac{1}{k^2} < \frac{1}{k(k-1)}.$$

Decomposing into partial fractions:

$$\frac{1}{k-1} - \frac{1}{k} = \frac{k-(k-1)}{k(k-1)} = \frac{1}{k(k-1)}.$$

Thus $\frac{1}{k^2} < \frac{1}{k-1} - \frac{1}{k}$.

3. Telescoping Upper Bound: Isolate the first term ($k = 1$):

$$u_n = 1 + \sum_{k=2}^{n} \frac{1}{k^2} < 1 + \sum_{k=2}^{n} \left(\frac{1}{k-1} - \frac{1}{k} \right).$$

Expand the telescoping sum:

$$\sum_{k=2}^{n} \left(\frac{1}{k-1} - \frac{1}{k} \right) = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n}\right) = 1 - \frac{1}{n}.$$

Therefore:

$$u_n < 1 + \left(1 - \frac{1}{n}\right) = 2 - \frac{1}{n} < 2.$$

Thus $u_n < 2$ for all $n \geq 1$.

4. Convergence: The sequence (u_n) is strictly increasing and bounded above by 2. By the Monotone Convergence Theorem, (u_n) converges to a finite limit $\ell = \sup\{u_n : n \geq 1\}$. Since $u_1 = 1 < u_n < 2$, we have $1 < \ell \leq 2$. (Euler famously proved in 1734 that $\ell = \frac{\pi^2}{6} \approx 1.64493$). ■

Exercise 4.5 · Divergence to Infinity and the Harmonic Series ★★ Intermediate

1. State the formal definition of $\lim_{n\to\infty} u_n = +\infty$.
2. Prove directly using the definition that $\lim_{n\to\infty} \frac{n^2-4}{n+3} = +\infty$.
3. Consider the harmonic sequence $H_n = \sum_{k=1}^{n} \frac{1}{k}$. Prove Oresme's inequality:

$$H_{2^p} \geq 1 + \frac{p}{2} \quad \text{for all } p \in \mathbb{N}.$$

4. Deduce that $\lim_{n\to\infty} H_n = +\infty$ (the harmonic series diverges).

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Complete Step-by-Step Solution 4.5

1. Definition of Divergence to $+\infty$:

$$\lim_{n\to\infty} u_n = +\infty \iff \forall A > 0,\ \exists N \in \mathbb{N},\ \forall n \geq N,\ u_n > A.$$

2. Limit of $u_n = \frac{n^2-4}{n+3}$: For $n \geq 3$, $n^2 - 4 \geq n^2 - \frac{1}{2}n^2 = \frac{1}{2}n^2$. Also for $n \geq 3$, $n+3 \leq n+n = 2n$. Thus:

$$u_n = \frac{n^2-4}{n+3} \geq \frac{n^2/2}{2n} = \frac{n}{4}.$$

Let $A > 0$ be arbitrary. We require $\frac{n}{4} > A \iff n > 4A$. Choose $N = \max(3, \lfloor 4A \rfloor + 1)$. Then for all $n \geq N$:

$$u_n \geq \frac{n}{4} \geq \frac{N}{4} > A.$$

Hence $\lim_{n\to\infty} u_n = +\infty$.

3. Oresme's Inequality for H_{2^p}: We group the terms in dyadic blocks:

$$H_{2^p} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^{p-1}+1} + \cdots + \frac{1}{2^p}\right).$$

In each block $B_j = \sum_{k=2^{j-1}+1}^{2^j} \frac{1}{k}$ (for $j = 1, \ldots, p$): There are $2^j - 2^{j-1} = 2^{j-1}$ terms, and the smallest term is the last one, $\frac{1}{2^j}$. Thus:

$$B_j \geq 2^{j-1} \times \frac{1}{2^j} = \frac{1}{2}.$$

Summing over all p blocks:

$$H_{2^p} = 1 + \sum_{j=1}^{p} B_j \geq 1 + \sum_{j=1}^{p} \frac{1}{2} = 1 + \frac{p}{2}.$$

4. Divergence of H_n: The sequence (H_n) is strictly increasing since $H_{n+1} - H_n = \frac{1}{n+1} > 0$. The subsequence $H_{2^p} \geq 1 + p/2$ is unbounded as $p \to \infty$. An increasing sequence that possesses an unbounded subsequence must diverge to $+\infty$. Therefore, $\lim_{n\to\infty} H_n = +\infty$. ■

Exercise 4.6 · Subsequences and the Bolzano-Weierstrass Theorem ★★ Intermediate

Consider the sequence:

$$u_n = \sin\left(\frac{n\pi}{2}\right) + \frac{1}{n} \quad (n \geq 1).$$

1. Prove that (u_n) is bounded.
2. Extract four convergent subsequences corresponding to indices $n = 4k, 4k+1, 4k+2, 4k+3$. Determine their respective limits.
3. Deduce that (u_n) does not converge.
4. State the Bolzano-Weierstrass Theorem and explain how this sequence illustrates it.

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Complete Step-by-Step Solution 4.6

1. Boundedness: Using the triangle inequality:

$$|u_n| = \left|\sin\left(\frac{n\pi}{2}\right) + \frac{1}{n}\right| \leq \left|\sin\left(\frac{n\pi}{2}\right)\right| + \frac{1}{n} \leq 1 + 1 = 2.$$

Thus $-2 \leq u_n \leq 2$ for all $n \geq 1$. (u_n) is bounded.

2. Four Subsequences:

- For $n = 4k$: $\sin(4k\pi/2) = \sin(2k\pi) = 0$. $u_{4k} = 0 + \frac{1}{4k} \xrightarrow{k\to\infty} 0$.
- For $n = 4k+1$: $\sin((4k+1)\pi/2) = \sin(2k\pi + \pi/2) = 1$. $u_{4k+1} = 1 + \frac{1}{4k+1} \xrightarrow{k\to\infty} 1$.
- For $n = 4k+2$: $\sin((4k+2)\pi/2) = \sin(2k\pi + \pi) = 0$. $u_{4k+2} = 0 + \frac{1}{4k+2} \xrightarrow{k\to\infty} 0$.
- For $n = 4k+3$: $\sin((4k+3)\pi/2) = \sin(2k\pi + 3\pi/2) = -1$. $u_{4k+3} = -1 + \frac{1}{4k+3} \xrightarrow{k\to\infty} -1$.

3. Non-convergence of (u_n): Recall that if a sequence (u_n) converges to a limit ℓ, then *all* its subsequences must converge to the same limit ℓ. Here, we have found two subsequences converging to distinct limits:

$$\lim_{k\to\infty} u_{4k+1} = 1 \neq -1 = \lim_{k\to\infty} u_{4k+3}.$$

Therefore, (u_n) diverges (it oscillates among the accumulation values $\{-1, 0, 1\}$).

4. Bolzano-Weierstrass Illustration: The Bolzano-Weierstrass Theorem states that every bounded real sequence has at least one convergent subsequence. Here, (u_n) is bounded, and indeed we extracted multiple convergent subsequences (converging to $-1, 0$, and 1). The theorem does not guarantee that the whole sequence converges, only that convergent subsequences exist.

■

Exercise 4.7 · The Cauchy Criterion for Convergence ★★ Intermediate

1. State the definition of a Cauchy sequence in $\mathbb{R}$.
2. Prove that every convergent sequence in $\mathbb{R}$ is a Cauchy sequence.
3. Consider the sequence:

$$u_n = \sum_{k=1}^{n} \frac{\cos(k)}{2^k} \quad (n \geq 1).$$

Prove that (u_n) is a Cauchy sequence, and deduce that (u_n) converges.

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Complete Step-by-Step Solution 4.7

1. Definition of Cauchy Sequence: A sequence $(u_n)_{n\in\mathbb{N}}$ in $\mathbb{R}$ is a Cauchy sequence if:

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall p, q \geq N,\ |u_p - u_q| < \varepsilon.$$

2. Convergent $\implies$ Cauchy: Assume $\lim_{n\to\infty} u_n = \ell \in \mathbb{R}$. Let $\varepsilon > 0$. By definition of convergence with threshold $\varepsilon/2$:

$$\exists N \in \mathbb{N},\ \forall n \geq N, \quad |u_n - \ell| < \frac{\varepsilon}{2}.$$

Let $p, q \geq N$. Using the triangle inequality:

$$|u_p - u_q| = |(u_p - \ell) - (u_q - \ell)| \leq |u_p - \ell| + |u_q - \ell| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus (u_n) is a Cauchy sequence.

3. Proof that $u_n = \sum_{k=1}^{n} \frac{\cos(k)}{2^k}$ is Cauchy: Let $p, q \in \mathbb{N}^*$ with $q > p \geq N$.

$$\begin{aligned} |u_q - u_p| &= \left| \sum_{k=p+1}^{q} \frac{\cos(k)}{2^k} \right| \leq \sum_{k=p+1}^{q} \frac{|\cos(k)|}{2^k} \leq \sum_{k=p+1}^{q} \frac{1}{2^k} \\ &= \frac{1}{2^{p+1}} \sum_{j=0}^{q-p-1} \left(\frac{1}{2}\right)^j = \frac{1}{2^{p+1}} \cdot \frac{1 - (1/2)^{q-p}}{1 - 1/2} \\ &< \frac{1}{2^{p+1}} \cdot \frac{1}{1/2} = \frac{1}{2^p}. \end{aligned}$$

Let $\varepsilon > 0$. Since $\lim_{p\to\infty} \frac{1}{2^p} = 0$, by the Archimedean property there exists $N \in \mathbb{N}$ such that:

$$\frac{1}{2^N} < \varepsilon.$$

Then for all $q > p \geq N$:

$$|u_q - u_p| < \frac{1}{2^p} \leq \frac{1}{2^N} < \varepsilon.$$

Thus (u_n) is a Cauchy sequence. Since $\mathbb{R}$ is complete (every Cauchy sequence in $\mathbb{R}$ converges), (u_n) converges to a finite limit $\ell \in \mathbb{R}$. ■

Exercise 4.8 · Adjacent Sequences and the Irrationality of e ★★ Intermediate

Consider the two sequences defined for $n \geq 1$ by:

$$u_n = \sum_{k=0}^{n} \frac{1}{k!} = 1 + 1 + \frac{1}{2} + \cdots + \frac{1}{n!} \qquad \text{and} \qquad v_n = u_n + \frac{1}{n \cdot n!}.$$

1. Prove that (u_n) is strictly increasing.
2. Prove that (v_n) is strictly decreasing.
3. Prove that $\lim_{n\to\infty}(v_n - u_n) = 0$.
4. Deduce that (u_n) and (v_n) are adjacent and converge to the same limit e.
5. Prove by contradiction that e is irrational.

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Complete Step-by-Step Solution 4.8

1. Monotonicity of (u_n):

$$u_{n+1} - u_n = \frac{1}{(n+1)!} > 0.$$

Thus (u_n) is strictly increasing.

2. Monotonicity of (v_n):

$$\begin{aligned} v_{n+1} - v_n &= \left(u_{n+1} + \frac{1}{(n+1)(n+1)!}\right) - \left(u_n + \frac{1}{n \cdot n!}\right) \\ &= \frac{1}{(n+1)!} + \frac{1}{(n+1)(n+1)!} - \frac{1}{n \cdot n!} \\ &= \frac{1}{(n+1)!}\left[1 + \frac{1}{n+1} - \frac{n+1}{n}\right] \\ &= \frac{1}{(n+1)!}\left[\frac{n+2}{n+1} - \frac{n+1}{n}\right] = \frac{1}{(n+1)!}\left[\frac{n(n+2) - (n+1)^2}{n(n+1)}\right] \\ &= \frac{1}{(n+1)!}\left[\frac{n^2 + 2n - (n^2 + 2n + 1)}{n(n+1)}\right] = -\frac{1}{n(n+1)(n+1)!} < 0. \end{aligned}$$

Thus (v_n) is strictly decreasing.

3. Gap Limit:

$$v_n - u_n = \frac{1}{n \cdot n!} \xrightarrow{n\to\infty} 0.$$

4. Adjacency and Limit e: Since (u_n) is increasing, (v_n) is decreasing, and $v_n - u_n \to 0$, the sequences are adjacent. By the Adjacent Sequences Theorem, they converge to the exact same

limit $\ell = e \approx 2.71828$, and:

$$u_n < e < v_n \iff u_n < e < u_n + \frac{1}{n \cdot n!} \quad \text{for all } n \geq 1.$$

5. Proof that $e \notin \mathbb{Q}$: Assume for contradiction that $e \in \mathbb{Q}$, so $e = \frac{p}{q}$ for some $p, q \in \mathbb{N}^*$. Choose $n = q \geq 1$. From the strict inequality $u_q < e < v_q$:

$$u_q < \frac{p}{q} < u_q + \frac{1}{q \cdot q!}.$$

Multiply all sides by $q!$:

$$q! \cdot u_q < p \cdot (q-1)! < q! \cdot u_q + \frac{1}{q}.$$

Notice that $q! \cdot u_q = q! \sum_{k=0}^{q} \frac{1}{k!} = \sum_{k=0}^{q} \frac{q!}{k!} \in \mathbb{Z}$, because for all $k \leq q$, $k!$ divides $q!$. Let $M = q! \cdot u_q \in \mathbb{Z}$. Also $p(q-1)! = K \in \mathbb{Z}$ is an integer. The inequality becomes:

$$M < K < M + \frac{1}{q}.$$

Since $q \geq 1$, $\frac{1}{q} \leq 1$, meaning $M < K < M+1$. This requires the integer K to lie strictly between two consecutive integers M and $M+1$, which is impossible! Contradiction. Therefore, e is irrational. ■

Exercise 4.9 · Arithmetic-Geometric Recurrence Sequences ★★ Intermediate

Consider the sequence $(u_n)_{n \in \mathbb{N}}$ defined by:

$$u_0 = 5 \qquad \text{and} \qquad u_{n+1} = \frac{1}{3}u_n + 4 \quad \text{for all } n \geq 0.$$

1. Determine the fixed point α of the recurrence relation: $\alpha = \frac{1}{3}\alpha + 4$.
2. Define the auxiliary sequence $v_n = u_n - \alpha$. Prove that (v_n) is a geometric sequence.
3. Express v_n and u_n explicitly as functions of n.
4. Compute $\lim_{n \to \infty} u_n$ and the sum $S_n = \sum_{k=0}^{n} u_k$.

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Complete Step-by-Step Solution 4.9

1. Fixed Point α:

$$\alpha = \frac{1}{3}\alpha + 4 \iff \alpha - \frac{1}{3}\alpha = 4 \iff \frac{2}{3}\alpha = 4 \iff \alpha = 4 \times \frac{3}{2} = 6.$$

2. Auxiliary Sequence $v_n = u_n - 6$:

$$v_{n+1} = u_{n+1} - 6 = \left(\frac{1}{3}u_n + 4\right) - 6 = \frac{1}{3}u_n - 2 = \frac{1}{3}(u_n - 6) = \frac{1}{3}v_n.$$

Thus (v_n) is a geometric sequence with common ratio $q = 1/3$. Initial term: $v_0 = u_0 - 6 = 5 - 6 = -1$.

3. Explicit Expressions:

$$v_n = v_0 \cdot q^n = -1 \cdot \left(\frac{1}{3}\right)^n = -\frac{1}{3^n}.$$

Since $u_n = v_n + 6$:

$$u_n = 6 - \frac{1}{3^n} = 6 - 3^{-n}.$$

4. Limit and Sum: Since $|1/3| < 1$, $\lim_{n\to\infty} 3^{-n} = 0$:

$$\lim_{n\to\infty} u_n = 6 - 0 = 6.$$

Sum $S_n = \sum_{k=0}^{n} u_k$:

$$S_n = \sum_{k=0}^{n}(6-3^{-k}) = 6(n+1) - \sum_{k=0}^{n}\left(\frac{1}{3}\right)^k = 6(n+1) - \frac{1-(1/3)^{n+1}}{1-1/3} = 6n+6-\frac{3}{2}\left(1-\frac{1}{3^{n+1}}\right). \quad \blacksquare$$

Exercise 4.10 · **Non-Linear Recurrence: $u_{n+1} = \sqrt{2+u_n}$** ★★★ Advanced / Exam Level

Consider the sequence $(u_n)_{n\in\mathbb{N}}$ defined by:

$$u_0 = 0 \qquad \text{and} \qquad u_{n+1} = \sqrt{2+u_n} \quad \text{for } n \geq 0.$$

1. Let $f(x) = \sqrt{2+x}$. Show that $f([0,2]) \subseteq [0,2]$.
2. Prove by induction that $0 \leq u_n \leq u_{n+1} \leq 2$ for all $n \in \mathbb{N}$.
3. Deduce that (u_n) converges and calculate its limit ℓ.
4. Establish the error estimate: $|u_{n+1} - 2| \leq \frac{1}{2}|u_n - 2|$, and deduce that $|u_n - 2| \leq 2 \cdot 2^{-n}$.

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Complete Step-by-Step Solution 4.10

1. Stability of $I = [0, 2]$: $f(x) = \sqrt{2 + x}$ is strictly increasing on $[-2, +\infty)$. For $x \in [0, 2]$:

$$f(0) = \sqrt{2} \approx 1.414 \geq 0, \qquad f(2) = \sqrt{2 + 2} = \sqrt{4} = 2.$$

Thus $f([0, 2]) = [\sqrt{2}, 2] \subseteq [0, 2]$.
2. Induction: $0 \leq u_n \leq u_{n+1} \leq 2$: Let $P(n) : 0 \leq u_n \leq u_{n+1} \leq 2$.

- Base step ($n = 0$): $u_0 = 0$, $u_1 = \sqrt{2}$. Since $0 \leq 0 \leq \sqrt{2} \leq 2$, $P(0)$ holds.
- Induction step: Assume $0 \leq u_n \leq u_{n+1} \leq 2$. Since f is strictly increasing on $[0, 2]$, applying f preserves the order:

$$f(0) \leq f(u_n) \leq f(u_{n+1}) \leq f(2) \implies \sqrt{2} \leq u_{n+1} \leq u_{n+2} \leq 2.$$

  Since $0 \leq \sqrt{2}$, we obtain $0 \leq u_{n+1} \leq u_{n+2} \leq 2$, which is $P(n + 1)$.

By induction, (u_n) is increasing and bounded above by 2.
3. Convergence and Limit: By the Monotone Convergence Theorem, (u_n) converges to a finite limit $\ell \in [0, 2]$. Since f is continuous on $[0, 2]$, passing to the limit in $u_{n+1} = f(u_n)$:

$$\ell = f(\ell) \iff \ell = \sqrt{2 + \ell} \iff \ell^2 = 2 + \ell \iff \ell^2 - \ell - 2 = 0.$$

Factoring: $(\ell - 2)(\ell + 1) = 0 \implies \ell = 2$ or $\ell = -1$. Since $u_n \geq 0$ for all n, $\ell \geq 0$, discarding $\ell = -1$. Thus $\lim_{n\to\infty} u_n = 2$.
4. Error Bound: Multiply by conjugate:

$$|u_{n+1} - 2| = |\sqrt{2 + u_n} - 2| = \frac{|(2 + u_n) - 4|}{\sqrt{2 + u_n} + 2} = \frac{|u_n - 2|}{\sqrt{2 + u_n} + 2}.$$

Since $u_n \geq 0$, the denominator satisfies $\sqrt{2 + u_n} + 2 \geq \sqrt{2} + 2 > 2$. Therefore:

$$|u_{n+1} - 2| \leq \frac{1}{2}|u_n - 2|.$$

By induction:

$$|u_n - 2| \leq \left(\frac{1}{2}\right)^n |u_0 - 2| = 2 \cdot 2^{-n} = 2^{1-n}. \quad \blacksquare$$

Exercise 4.11 · The Cesàro Mean Theorem and Applications ★★★ Advanced / Exam Level

Let $(u_n)_{n\geq 1}$ be a real sequence. Define its sequence of arithmetic means (Cesàro means):

$$c_n = \frac{1}{n}\sum_{k=1}^{n} u_k = \frac{u_1 + u_2 + \cdots + u_n}{n}.$$

1. Prove the Cesàro Theorem: If $\lim_{n\to\infty} u_n = \ell \in \mathbb{R}$, then $\lim_{n\to\infty} c_n = \ell$.
2. Show that the converse is false by analyzing $u_n = (-1)^n$.
3. Application: Let $(x_n)_{n\geq 1}$ be a sequence of strictly positive numbers. Prove that if $\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = \ell > 0$, then $\lim_{n\to\infty} \sqrt[n]{x_n} = \ell$.
4. Calculate: $\lim_{n\to\infty} \frac{\sqrt[n]{n!}}{n}$.

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Complete Step-by-Step Solution 4.11

1. Proof of the Cesàro Theorem: Assume $\lim_{n\to\infty} u_n = \ell$. Let $\varepsilon > 0$. There exists $N_0 \in \mathbb{N}^*$ such that for all $k \geq N_0$, $|u_k - \ell| < \varepsilon/2$. For $n > N_0$, split the Cesàro sum:

$$|c_n - \ell| = \left|\frac{1}{n}\sum_{k=1}^{n}(u_k - \ell)\right| \leq \frac{1}{n}\sum_{k=1}^{N_0}|u_k - \ell| + \frac{1}{n}\sum_{k=N_0+1}^{n}|u_k - \ell|$$
$$< \frac{1}{n}\sum_{k=1}^{N_0}|u_k - \ell| + \frac{n - N_0}{n}\frac{\varepsilon}{2} < \frac{M}{n} + \frac{\varepsilon}{2},$$

where $M = \sum_{k=1}^{N_0}|u_k - \ell|$ is a fixed constant independent of n. Since $\lim_{n\to\infty}\frac{M}{n} = 0$, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $\frac{M}{n} < \frac{\varepsilon}{2}$. Setting $N = \max(N_0, N_1)$, for all $n \geq N$:

$$|c_n - \ell| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus $\lim_{n\to\infty} c_n = \ell$. ■

2. Converse is False (Counterexample): Let $u_n = (-1)^n$. The sequence alternates $-1, 1, -1, 1, \ldots$ and diverges. However:

$$c_n = \frac{1}{n}\sum_{k=1}^{n}(-1)^k = \begin{cases} 0 & \text{if } n \text{ is even} \\ -1/n & \text{if } n \text{ is odd} \end{cases} \implies \lim_{n\to\infty} c_n = 0.$$

The Cesàro mean converges to 0 even though (u_n) diverges.

3. Root Limit from Ratio Limit: Let $y_n = \ln(x_n)$. Then $\sqrt[n]{x_n} = \exp\left(\frac{1}{n}\ln(x_n)\right)$. Notice that $\ln(x_n) = \ln(x_1) + \sum_{k=1}^{n-1}\ln\left(\frac{x_{k+1}}{x_k}\right)$. Let $w_k = \ln\left(\frac{x_{k+1}}{x_k}\right)$. By hypothesis, $w_k \to \ln(\ell)$. Applying Cesàro to (w_k):

$$\frac{\ln(x_n)}{n} \xrightarrow{n\to\infty} \ln(\ell).$$

By continuity of the exponential function, $\lim_{n\to\infty}\sqrt[n]{x_n} = e^{\ln(\ell)} = \ell$.

4. Computation of $\lim \frac{\sqrt[n]{n!}}{n}$: Rewrite $\frac{\sqrt[n]{n!}}{n} = \sqrt[n]{\frac{n!}{n^n}}$. Let $x_n = \frac{n!}{n^n}$. Compute the ratio $\frac{x_{n+1}}{x_n}$:

$$\frac{x_{n+1}}{x_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{(n+1)n! \cdot n^n}{(n+1)(n+1)^n n!} = \left(\frac{n}{n+1}\right)^n = \frac{1}{(1+1/n)^n}.$$

As $n \to \infty$, $(1+1/n)^n \to e$, so $\lim_{n\to\infty}\frac{x_{n+1}}{x_n} = \frac{1}{e}$. By part 3:

$$\lim_{n\to\infty}\frac{\sqrt[n]{n!}}{n} = \frac{1}{e}.$$ ■

Exercise 4.12 · Gauss's Arithmetic-Geometric Mean (AGM) ★★★ Advanced / Exam Level

Let $0 < b_0 < a_0$. Define the coupled sequences for $n \geq 0$ by:

$$a_{n+1} = \frac{a_n + b_n}{2} \qquad \text{and} \qquad b_{n+1} = \sqrt{a_n b_n}.$$

1. Prove that for all $n \geq 0$, $b_n \leq b_{n+1} \leq a_{n+1} \leq a_n$ using the AM-GM inequality.
2. Prove that (a_n) is decreasing, (b_n) is increasing, and both converge.
3. Show that $a_{n+1} - b_{n+1} = \frac{(a_n - b_n)^2}{2(\sqrt{a_n} + \sqrt{b_n})^2} \leq \frac{(a_n - b_n)^2}{8b_0}$.
4. Deduce that $\lim_{n\to\infty}(a_n - b_n) = 0$, so (a_n) and (b_n) are adjacent and converge to a common value $M(a_0, b_0)$ (Gauss's AGM).

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Complete Step-by-Step Solution 4.12

1. AM-GM Inequality and Ordering: By the Arithmetic-Geometric Mean inequality, for any positive numbers $x, y > 0$:

$$\sqrt{xy} \leq \frac{x+y}{2}, \quad \text{with equality iff } x = y.$$

Applied to a_n, b_n:

$$b_{n+1} = \sqrt{a_n b_n} \leq \frac{a_n + b_n}{2} = a_{n+1}.$$

Since $b_n < a_n$, we also have:

$$a_{n+1} = \frac{a_n + b_n}{2} < \frac{a_n + a_n}{2} = a_n,$$

$$b_{n+1} = \sqrt{a_n b_n} > \sqrt{b_n b_n} = b_n.$$

Thus for all $n \geq 0$:

$$b_0 \leq b_n < b_{n+1} \leq a_{n+1} < a_n \leq a_0.$$

2. Convergence of Both Sequences:

- (a_n) is strictly decreasing and bounded below by b_0. By the Monotone Convergence Theorem, $a_n \to \ell_a$.
- (b_n) is strictly increasing and bounded above by a_0. By the Monotone Convergence Theorem, $b_n \to \ell_b$.

3. Quadratic Difference Formula:

$$a_{n+1} - b_{n+1} = \frac{a_n + b_n}{2} - \sqrt{a_n b_n} = \frac{a_n - 2\sqrt{a_n b_n} + b_n}{2} = \frac{(\sqrt{a_n} - \sqrt{b_n})^2}{2}$$
$$= \frac{1}{2}\left(\frac{a_n - b_n}{\sqrt{a_n} + \sqrt{b_n}}\right)^2 = \frac{(a_n - b_n)^2}{2(\sqrt{a_n} + \sqrt{b_n})^2}.$$

Since $a_n \geq b_n \geq b_0$, we have $\sqrt{a_n} + \sqrt{b_n} \geq 2\sqrt{b_0}$, so $(\sqrt{a_n} + \sqrt{b_n})^2 \geq 4b_0$. Therefore:

$$a_{n+1} - b_{n+1} \leq \frac{(a_n - b_n)^2}{8b_0}.$$

4. Adjacency and Gauss's AGM: From the recursive relation $a_{n+1} = \frac{a_n + b_n}{2}$, taking limits on both sides:

$$\ell_a = \frac{\ell_a + \ell_b}{2} \implies 2\ell_a = \ell_a + \ell_b \implies \ell_a = \ell_b.$$

Thus $\lim_{n\to\infty}(a_n - b_n) = 0$. The sequences (a_n) and (b_n) are adjacent, converging to the exact same value:

$$M(a_0, b_0) = \lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n.$$

This common limit is the Arithmetic-Geometric Mean of Gauss, famously used to rapidly compute elliptic integrals and digits of π. ■

Chapter 5

Real Functions of a Real Variable, Continuity & Derivatives

Essential Chapter Theorems & Formulas · Limits, Continuity, Extreme Values, Rolle, MVT & L'Hôpital

1. ε-δ **Definition of Function Limit:**

$$\lim_{x \to x_0} f(x) = \ell \iff \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x \in D,\ (0 < |x - x_0| < \delta \implies |f(x) - \ell| < \varepsilon).$$

2. **Sequential Characterization (Heine's Criterion):** $\lim_{x \to x_0} f(x) = \ell \iff$ for every sequence (x_n) in $D \setminus \{x_0\}$ converging to x_0, $\lim f(x_n) = \ell$.

3. **Intermediate Value Theorem (Bolzano / TVI):** If $f : [a, b] \to \mathbb{R}$ is continuous and y is strictly between $f(a)$ and $f(b)$, then $\exists c \in (a, b)$ such that $f(c) = y$. In particular, if $f(a)f(b) < 0$, then $\exists c \in (a, b)$ such that $f(c) = 0$.

4. **Weierstrass Extreme Value Theorem:** A continuous function on a closed bounded interval $[a, b]$ is bounded and attains its absolute minimum and maximum: $\exists x_{\min}, x_{\max} \in [a, b]$ such that $f(x_{\min}) \leq f(x) \leq f(x_{\max})$ for all $x \in [a, b]$.

5. **Rolle's Theorem:** If $f \in C([a, b])$ and f is differentiable on (a, b) with $f(a) = f(b)$, then $\exists c \in (a, b)$ such that $f'(c) = 0$.

6. **Mean Value Theorem (Lagrange):** If $f \in C([a, b])$ and f is differentiable on (a, b), then $\exists c \in (a, b)$ such that:

$$f(b) - f(a) = f'(c)(b - a).$$

7. **Cauchy's Mean Value Theorem:** $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$ for some $c \in (a, b)$.

Exercise 5.1 · Direct ε-δ Proofs of Function Limits ★ Foundational

1. Using the formal ε-δ definition, prove that $\lim_{x \to 3}(4x - 5) = 7$.
2. Prove using the ε-δ definition that $\lim_{x \to 2} x^2 = 4$.
3. Prove using the ε-δ definition that $\lim_{x \to 2} \frac{1}{x} = \frac{1}{2}$.

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Complete Step-by-Step Solution 5.1

1. Proof that $\lim_{x\to 3}(4x-5)=7$: We must find $\delta>0$ such that $0<|x-3|<\delta \implies |(4x-5)-7|<\varepsilon$. Compute the error:

$$|(4x-5)-7| = |4x-12| = 4|x-3|.$$

To have $4|x-3|<\varepsilon$, it suffices to choose $|x-3|<\frac{\varepsilon}{4}$. Let $\varepsilon>0$. Choose $\delta=\frac{\varepsilon}{4}>0$. Then if $0<|x-3|<\delta$:

$$|(4x-5)-7| = 4|x-3| < 4\delta = 4\left(\frac{\varepsilon}{4}\right) = \varepsilon.$$

Thus $\lim_{x\to 3}(4x-5)=7$. ■

2. Proof that $\lim_{x\to 2} x^2 = 4$: Compute $|x^2-4| = |x-2|\cdot|x+2|$. To control $|x+2|$, restrict x to a neighborhood of 2, say $|x-2|<1 \implies 1<x<3$. Then $3<x+2<5 \implies |x+2|<5$. Thus, provided $|x-2|<1$:

$$|x^2-4| = |x-2|\cdot|x+2| < 5|x-2|.$$

To have $5|x-2|<\varepsilon$, we require $|x-2|<\frac{\varepsilon}{5}$. Let $\varepsilon>0$. Choose $\delta=\min\left(1,\frac{\varepsilon}{5}\right)>0$. Then if $0<|x-2|<\delta$:

$$|x^2-4| < 5|x-2| < 5\left(\frac{\varepsilon}{5}\right) = \varepsilon.$$

Thus $\lim_{x\to 2} x^2 = 4$. ■

3. Proof that $\lim_{x\to 2}\frac{1}{x}=\frac{1}{2}$: Compute:

$$\left|\frac{1}{x}-\frac{1}{2}\right| = \left|\frac{2-x}{2x}\right| = \frac{|x-2|}{2|x|}.$$

Restrict $|x-2|<1 \implies 1<x<3 \implies |x|>1 \implies \frac{1}{|x|}<1$. Then:

$$\left|\frac{1}{x}-\frac{1}{2}\right| = \frac{|x-2|}{2|x|} < \frac{|x-2|}{2}.$$

Let $\varepsilon>0$. Choose $\delta=\min(1,2\varepsilon)>0$. Then for $0<|x-2|<\delta$:

$$\left|\frac{1}{x}-\frac{1}{2}\right| < \frac{|x-2|}{2} < \frac{2\varepsilon}{2} = \varepsilon.$$

Thus $\lim_{x\to 2}\frac{1}{x}=\frac{1}{2}$. ■

Exercise 5.2 · Sequential Criterion (Heine) and Non-Existence of Limits ★ Foundational

1. State Heine's Sequential Criterion for the limit of a function at x_0.
2. Consider the function $f(x) = \sin(1/x)$ defined on $\mathbb{R}^*$. Consider the two sequences $x_n = \frac{1}{2n\pi}$ and $y_n = \frac{1}{2n\pi+\pi/2}$. Show that $x_n \to 0$ and $y_n \to 0$, but $\lim f(x_n) \neq \lim f(y_n)$.
3. Deduce that $\lim_{x\to 0} \sin(1/x)$ does not exist.
4. Determine whether $\lim_{x\to 0} x\sin(1/x)$ exists.

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Complete Step-by-Step Solution 5.2

1. Heine's Sequential Criterion: $\lim_{x\to x_0} f(x) = \ell$ if and only if for every sequence $(x_n)_{n\in\mathbb{N}}$ in $D_f \setminus \{x_0\}$ such that $\lim_{n\to\infty} x_n = x_0$, the image sequence converges to ℓ: $\lim_{n\to\infty} f(x_n) = \ell$.
2. Subsequences for $f(x) = \sin(1/x)$: Both sequences converge to 0:

$$x_n = \frac{1}{2n\pi} \xrightarrow{n\to\infty} 0, \qquad y_n = \frac{1}{2n\pi + \pi/2} \xrightarrow{n\to\infty} 0.$$

Evaluating f on these sequences:

$$f(x_n) = \sin\left(\frac{1}{1/(2n\pi)}\right) = \sin(2n\pi) = 0 \xrightarrow{n\to\infty} 0,$$

$$f(y_n) = \sin\left(\frac{1}{1/(2n\pi+\pi/2)}\right) = \sin\left(2n\pi + \frac{\pi}{2}\right) = 1 \xrightarrow{n\to\infty} 1.$$

We observe that $\lim_{n\to\infty} f(x_n) = 0 \neq 1 = \lim_{n\to\infty} f(y_n)$.
3. Non-Existence of Limit: Since two sequences converging to 0 produce distinct image limits ($0 \neq 1$), by Heine's Theorem, the limit $\lim_{x\to 0} \sin(1/x)$ does not exist.
4. Limit of $g(x) = x\sin(1/x)$: Since $|\sin(1/x)| \leq 1$ for all $x \neq 0$:

$$|g(x)| = |x\sin(1/x)| = |x| \cdot |\sin(1/x)| \leq |x| \cdot 1 = |x|.$$

Thus $-|x| \leq g(x) \leq |x|$. Since $\lim_{x\to 0}(-|x|) = 0$ and $\lim_{x\to 0}|x| = 0$, by the Squeeze Theorem:

$$\lim_{x\to 0} x\sin\left(\frac{1}{x}\right) = 0. \quad \blacksquare$$

Exercise 5.3 · Continuous Extensions (Prolongement par Continuité) ★ Foundational

Determine whether each of the following functions can be extended by continuity to the specified point x_0. If so, state the value $\tilde{f}(x_0)$:

1. $f_1(x) = \frac{\sin(3x)}{x}$ at $x_0 = 0$.
2. $f_2(x) = \frac{\sqrt{1+x}-1}{x}$ at $x_0 = 0$.
3. $f_3(x) = \frac{x^2-4}{|x-2|}$ at $x_0 = 2$.
4. $f_4(x) = x\ln(x)$ at $x_0 = 0$.

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Complete Step-by-Step Solution 5.3

A function f admits a continuous extension at $x_0 \notin D_f$ if and only if $\lim_{x\to x_0} f(x) = \ell \in \mathbb{R}$ exists and is finite. The extension is defined by $\tilde{f}(x) = f(x)$ for $x \neq x_0$ and $\tilde{f}(x_0) = \ell$.
1. $f_1(x) = \frac{\sin(3x)}{x}$ **at** $x_0 = 0$**:** Rewrite: $f_1(x) = 3 \cdot \frac{\sin(3x)}{3x}$. Let $u = 3x$. As $x \to 0$, $u \to 0$. Using the fundamental limit $\lim_{u\to 0} \frac{\sin u}{u} = 1$:

$$\lim_{x\to 0} f_1(x) = 3 \times 1 = 3.$$

Yes, extendable with $\tilde{f}_1(0) = 3$.
2. $f_2(x) = \frac{\sqrt{1+x}-1}{x}$ **at** $x_0 = 0$**:** Multiply numerator and denominator by the conjugate $\sqrt{1+x}+1$:

$$f_2(x) = \frac{(\sqrt{1+x}-1)(\sqrt{1+x}+1)}{x(\sqrt{1+x}+1)} = \frac{(1+x)-1}{x(\sqrt{1+x}+1)} = \frac{x}{x(\sqrt{1+x}+1)} = \frac{1}{\sqrt{1+x}+1}.$$

Taking the limit as $x \to 0$:

$$\lim_{x\to 0} f_2(x) = \frac{1}{\sqrt{1+0}+1} = \frac{1}{2}.$$

Yes, extendable with $\tilde{f}_2(0) = 1/2$.
3. $f_3(x) = \frac{x^2-4}{|x-2|}$ **at** $x_0 = 2$**:** Factor $x^2 - 4 = (x-2)(x+2)$.

- Right limit ($x \to 2^+$): $|x-2| = x-2$, so $f_3(x) = \frac{(x-2)(x+2)}{x-2} = x+2 \to 4$.
- Left limit ($x \to 2^-$): $|x-2| = -(x-2)$, so $f_3(x) = \frac{(x-2)(x+2)}{-(x-2)} = -(x+2) \to -4$.

Since the left and right limits disagree ($4 \neq -4$), $\lim_{x\to 2} f_3(x)$ does not exist. f_3 cannot be extended by continuity at $x = 2$.
4. $f_4(x) = x\ln(x)$ **at** $x_0 = 0$**:** By standard logarithmic growth hierarchy:

$$\lim_{x\to 0^+} x\ln(x) = 0.$$

Yes, extendable by continuity from the right with $\tilde{f}_4(0) = 0$. ■

Exercise 5.4 · The Intermediate Value Theorem and Fixed Points ★★ Intermediate

1. Prove that the polynomial equation $x^5 - 3x - 1 = 0$ has at least one real root in the interval $(1, 2)$.
2. Prove that any continuous function $f : [0, 1] \to [0, 1]$ has at least one fixed point, i.e., $\exists c \in [0, 1]$ such that $f(c) = c$.
3. A hiker starts climbing a mountain trail at 8:00 AM and reaches the summit at 5:00 PM. The next day, she departs the summit at 8:00 AM along the same trail and arrives at the base at 5:00 PM. Prove that there exists a point on the trail that she crossed at the exact same time on both days.

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Complete Step-by-Step Solution 5.4

1. Root of $P(x) = x^5 - 3x - 1$ in $(1, 2)$: P is a polynomial function, hence continuous on $[1, 2]$. Compute the values at endpoints:

$$P(1) = 1^5 - 3(1) - 1 = 1 - 3 - 1 = -3 < 0,$$

$$P(2) = 2^5 - 3(2) - 1 = 32 - 6 - 1 = 25 > 0.$$

Since $P(1) < 0 < P(2)$ and P is continuous, by Bolzano's Intermediate Value Theorem, there exists at least one $c \in (1, 2)$ such that $P(c) = 0$.

2. Fixed Point of $f : [0, 1] \to [0, 1]$: Define the auxiliary function $g : [0, 1] \to \mathbb{R}$ by:

$$g(x) = f(x) - x.$$

Since f and $x \mapsto x$ are continuous, g is continuous on $[0, 1]$. Evaluate g at endpoints:

- $g(0) = f(0) - 0 = f(0) \geq 0$ (since $f([0, 1]) \subseteq [0, 1]$).
- $g(1) = f(1) - 1 \leq 0$ (since $f(1) \leq 1$).

If $g(0) = 0$, then $c = 0$ is a fixed point. If $g(1) = 0$, then $c = 1$ is a fixed point. Otherwise, $g(0) > 0$ and $g(1) < 0$. By the Intermediate Value Theorem, there exists $c \in (0, 1)$ such that $g(c) = 0 \iff f(c) = c$.

3. The Mountain Climber Problem: Let $s(t)$ denote the hiker's distance from the base along the trail on Day 1, where $t \in [8, 17]$ (hours). Then $s(8) = 0$ and $s(17) = L$ (summit distance). Let $r(t)$ denote the hiker's distance from the base on Day 2. Then $r(8) = L$ and $r(17) = 0$. Both $s(t)$ and $r(t)$ are continuous functions of time. Define $h(t) = s(t) - r(t)$ for $t \in [8, 17]$.

- $h(8) = s(8) - r(8) = 0 - L = -L < 0$.
- $h(17) = s(17) - r(17) = L - 0 = L > 0$.

Since h is continuous and changes sign, by the Intermediate Value Theorem, there exists $t^* \in (8, 17)$ such that:

$$h(t^*) = 0 \iff s(t^*) = r(t^*).$$

At time t^*, the hiker was at the exact same location on the trail on both days. ■

Exercise 5.5 · Differentiability vs. Continuity and Class C^1 ★★ Intermediate

1. Prove that if a function f is differentiable at x_0, then f is continuous at x_0.
2. Consider $f(x) = |x|$. Show that f is continuous at 0, but not differentiable at 0.
3. Consider the function:

$$g(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

   (a) Prove that g is differentiable at $x = 0$ and calculate $g'(0)$.
   (b) Calculate $g'(x)$ for $x \neq 0$.
   (c) Prove that g' is not continuous at $x = 0$, so $g \in C^0(\mathbb{R}) \cap D^1(\mathbb{R})$ but $g \notin C^1(\mathbb{R})$.

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Complete Step-by-Step Solution 5.5

1. Differentiability $\implies$ Continuity: Assume $f'(x_0) = \lim_{x \to x_0} \frac{f(x)-f(x_0)}{x-x_0}$ exists. For $x \neq x_0$, write:

$$f(x) - f(x_0) = \frac{f(x) - f(x_0)}{x - x_0} \cdot (x - x_0).$$

Taking limits as $x \to x_0$:

$$\lim_{x \to x_0} [f(x) - f(x_0)] = \lim_{x \to x_0} \left[\frac{f(x) - f(x_0)}{x - x_0}\right] \cdot \lim_{x \to x_0} (x - x_0) = f'(x_0) \cdot 0 = 0.$$

Thus $\lim_{x \to x_0} f(x) = f(x_0)$, which proves that f is continuous at x_0.
2. Counterexample $f(x) = |x|$: $|x| \to 0 = f(0)$ as $x \to 0$, so f is continuous at 0. Compute the difference quotient at 0:

$$\frac{f(h) - f(0)}{h} = \frac{|h|}{h} = \begin{cases} 1 & \text{if } h > 0 \\ -1 & \text{if } h < 0. \end{cases}$$

Right derivative $f'_+(0) = 1$, left derivative $f'_-(0) = -1$. Since $f'_+(0) \neq f'_-(0)$, $f'(0)$ does not exist.
3. Pathological Function $g(x)$:

(a) **Derivative at 0:**

$$\lim_{h\to 0}\frac{g(h)-g(0)}{h}=\lim_{h\to 0}\frac{h^2\sin(1/h)}{h}=\lim_{h\to 0}h\sin\left(\frac{1}{h}\right).$$

Since $|h\sin(1/h)|\le |h|\to 0$, the limit is 0. Thus g is differentiable at 0 and $g'(0)=0$.

(b) **Derivative for $x\neq 0$:** Using the product and chain rules:

$$g'(x)=\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2\sin(1/x)\right)=2x\sin\left(\frac{1}{x}\right)+x^2\cos\left(\frac{1}{x}\right)\left(-\frac{1}{x^2}\right)=2x\sin\left(\frac{1}{x}\right)-\cos\left(\frac{1}{x}\right).$$

(c) **Discontinuity of g' at 0:** As $x\to 0$, $2x\sin(1/x)\to 0$. However, $\cos(1/x)$ oscillates between -1 and 1 and has no limit. Thus $\lim_{x\to 0}g'(x)$ does not exist. Since $\lim_{x\to 0}g'(x)\neq g'(0)=0$, g' is discontinuous at 0. Hence g is differentiable everywhere, but not continuously differentiable ($g\notin C^1(\mathbb{R})$). ■

Exercise 5.6 · Applications of Rolle's Theorem ★★ Intermediate

1. State Rolle's Theorem precisely, listing all hypotheses.
2. Let $P(x)=x^3-3x+k$. Prove that $P(x)=0$ cannot have more than one real root in $[-1,1]$ regardless of the value of k.
3. Let $f:[a,b]\to\mathbb{R}$ be twice differentiable on (a,b). Suppose $f(a)=f(b)=0$ and there exists $x_0\in(a,b)$ such that $f(x_0)>0$. Prove that there exists $c\in(a,b)$ such that $f''(c)<0$.

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Complete Step-by-Step Solution 5.6

1. Statement of Rolle's Theorem: Let $f:[a,b]\to\mathbb{R}$. If:

1. f is continuous on the closed interval $[a,b]$,
2. f is differentiable on the open interval (a,b),
3. $f(a)=f(b)$,

then there exists at least one $c\in(a,b)$ such that $f'(c)=0$.

2. At Most One Root of $P(x)=x^3-3x+k$ in $[-1,1]$: Suppose for contradiction that $P(x)=0$ has two or more distinct roots in $[-1,1]$, say $x_1<x_2\in[-1,1]$. Then $P(x_1)=P(x_2)=0$. Since P is a polynomial, it is continuous on $[x_1,x_2]$ and differentiable on (x_1,x_2). By Rolle's Theorem, there must exist $c\in(x_1,x_2)\subset(-1,1)$ such that $P'(c)=0$. Compute $P'(x)$:

$$P'(x)=3x^2-3=3(x^2-1).$$

For $c \in (-1, 1)$, we have $c^2 < 1$, which gives:

$$P'(c) = 3(c^2 - 1) < 0.$$

Thus $P'(c)$ cannot equal zero for any $c \in (-1, 1)$. Contradiction! Therefore, $P(x)$ has at most one real root in $[-1, 1]$.

3. Existence of c such that $f''(c) < 0$: Since f is continuous on the compact interval $[a, b]$, by Weierstrass's Extreme Value Theorem, f attains its absolute maximum at some point $x_{\max} \in [a, b]$. Since $f(x_0) > 0$ and $f(a) = f(b) = 0$, the maximum must be strictly positive:

$$f(x_{\max}) \geq f(x_0) > 0.$$

Thus $x_{\max} \neq a$ and $x_{\max} \neq b$, meaning $x_{\max} \in (a, b)$. Since f is differentiable on (a, b), by Fermat's Theorem on local extrema:

$$f'(x_{\max}) = 0.$$

Now apply Rolle's Theorem to f on $[a, x_{\max}]$: there exists $c_1 \in (a, x_{\max})$ such that $f'(c_1) = \frac{f(x_{\max}) - f(a)}{x_{\max} - a} > 0$. Apply the Mean Value Theorem to f' on $[c_1, x_{\max}]$: There exists $c \in (c_1, x_{\max}) \subset (a, b)$ such that:

$$f''(c) = \frac{f'(x_{\max}) - f'(c_1)}{x_{\max} - c_1} = \frac{0 - f'(c_1)}{x_{\max} - c_1} < 0. \quad \blacksquare$$

Exercise 5.7 · The Mean Value Theorem and Fundamental Inequalities ★★ Intermediate

1. State the Mean Value Theorem (Lagrange).
2. Applying the MVT to $f(t) = \ln(t)$ on $[1, 1 + x]$ for $x > 0$, prove the fundamental inequality:

$$\frac{x}{1 + x} < \ln(1 + x) < x \quad \text{for all } x > 0.$$

3. Prove that for all $x, y \in \mathbb{R}$:

$$|\sin(x) - \sin(y)| \leq |x - y| \qquad \text{and} \qquad |\arctan(x) - \arctan(y)| \leq |x - y|.$$

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Complete Step-by-Step Solution 5.7

1. Statement of Mean Value Theorem: If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and

differentiable on (a, b), then there exists at least one $c \in (a, b)$ such that:

$$f(b) - f(a) = f'(c)(b - a).$$

2. Proof of Logarithmic Inequalities for $x > 0$: Consider $f(t) = \ln(t)$ on the interval $[1, 1 + x]$. f is continuous on $[1, 1 + x]$ and differentiable on $(1, 1 + x)$ with $f'(t) = \frac{1}{t}$. By the Mean Value Theorem, there exists $c \in (1, 1 + x)$ such that:

$$\ln(1 + x) - \ln(1) = f'(c)((1 + x) - 1) \iff \ln(1 + x) = \frac{1}{c} \cdot x = \frac{x}{c}.$$

Since $1 < c < 1 + x$, taking reciprocals reverses inequalities:

$$\frac{1}{1 + x} < \frac{1}{c} < 1.$$

Multiplying by $x > 0$:

$$\frac{x}{1 + x} < \frac{x}{c} < x \iff \frac{x}{1 + x} < \ln(1 + x) < x. \quad \blacksquare$$

3. 1-Lipschitz Continuity of sin **and** arctan**:**

- **Sine Function:** For $x = y$, $|\sin x - \sin y| = 0 = |x - y|$. For $x \neq y$, assume without loss of generality $x < y$. Applying MVT to $f(t) = \sin(t)$ on $[x, y]$:

  $$\sin(y) - \sin(x) = \cos(c)(y - x) \quad \text{for some } c \in (x, y).$$

  Taking absolute values:

  $$|\sin(y) - \sin(x)| = |\cos(c)| \cdot |y - x| \leq 1 \cdot |y - x| = |x - y|.$$

- **Arctangent Function:** Applying MVT to $g(t) = \arctan(t)$:

  $$\arctan(y) - \arctan(x) = \frac{1}{1 + c^2}(y - x) \quad \text{for some } c \in (x, y).$$

  Since $c^2 \geq 0$, $\frac{1}{1+c^2} \leq 1$. Taking absolute values:

  $$|\arctan(y) - \arctan(x)| = \frac{1}{1 + c^2}|y - x| \leq |x - y|. \quad \blacksquare$$

Exercise 5.8 · Cauchy's Mean Value Theorem and Proof of L'Hôpital's Rule ★★ Intermediate

1. State Cauchy's Generalized Mean Value Theorem.
2. Prove Cauchy's MVT by applying Rolle's Theorem to the auxiliary function:

   $$h(x) = [g(b) - g(a)]f(x) - [f(b) - f(a)]g(x).$$

3. Use Cauchy's MVT to prove L'Hôpital's Rule for the indeterminate form $\frac{0}{0}$: If $f(x_0) = g(x_0) = 0$, f, g differentiable near x_0, $g'(x) \neq 0$, and $\lim_{x \to x_0} \frac{f'(x)}{g'(x)} = \ell$, then $\lim_{x \to x_0} \frac{f(x)}{g(x)} = \ell$.

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Complete Step-by-Step Solution 5.8

1. Statement of Cauchy's MVT: Let $f, g : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). If $g'(x) \neq 0$ for all $x \in (a, b)$, then $g(b) \neq g(a)$, and there exists $c \in (a, b)$ such that:

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

2. Proof via Auxiliary Function: Note that if $g(b) = g(a)$, by Rolle's Theorem there would exist $d \in (a, b)$ with $g'(d) = 0$, contradicting the assumption $g'(x) \neq 0$. Thus $g(b) \neq g(a)$. Define:

$$h(x) = [g(b) - g(a)]f(x) - [f(b) - f(a)]g(x).$$

h is continuous on $[a, b]$ and differentiable on (a, b). Evaluate h at endpoints:

$$h(a) = [g(b) - g(a)]f(a) - [f(b) - f(a)]g(a) = g(b)f(a) - f(b)g(a),$$

$$h(b) = [g(b) - g(a)]f(b) - [f(b) - f(a)]g(b) = g(b)f(a) - f(b)g(a).$$

Thus $h(a) = h(b)$. By Rolle's Theorem, there exists $c \in (a, b)$ such that $h'(c) = 0$:

$$[g(b) - g(a)]f'(c) - [f(b) - f(a)]g'(c) = 0 \iff \frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

3. Proof of L'Hôpital's Rule $(0/0)$: Let $x \neq x_0$ in a punctured neighborhood of x_0. Since $f(x_0) = g(x_0) = 0$, by Cauchy's MVT on the interval between x_0 and x, there exists c_x strictly between x_0 and x such that:

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(x_0)}{g(x) - g(x_0)} = \frac{f'(c_x)}{g'(c_x)}.$$

As $x \to x_0$, since c_x is squeezed between x_0 and x, we have $c_x \to x_0$. Therefore:

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(c_x)}{g'(c_x)} = \lim_{c \to x_0} \frac{f'(c)}{g'(c)} = \ell. \quad \blacksquare$$

Exercise 5.9 · Systematic Application of L'Hôpital's Rule ★★ Intermediate

Compute the following indeterminate limits using L'Hôpital's Rule:

1. $\lim_{x \to 0} \frac{x - \sin(x)}{x^3}$ $\left(\frac{0}{0}\right)$.
2. $\lim_{x \to 0^+} x \ln(x)$ $(0 \times \infty)$.

3. $\lim_{x\to 0}\left(\frac{1}{x}-\frac{1}{e^x-1}\right)$ $(\infty-\infty)$.

4. $\lim_{x\to 0^+} x^x$ (0^0).

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Complete Step-by-Step Solution 5.9

1. Limit of $\frac{x-\sin x}{x^3}$ $(0/0)$: Differentiating numerator and denominator (Pass 1):

$$\lim_{x\to 0}\frac{1-\cos x}{3x^2}\quad\left[\frac{0}{0}\right].$$

Differentiating again (Pass 2):

$$\lim_{x\to 0}\frac{\sin x}{6x}=\frac{1}{6}\lim_{x\to 0}\frac{\sin x}{x}=\frac{1}{6}\times 1=\frac{1}{6}.$$

2. Limit of $x\ln x$ as $x\to 0^+$ $(0\times\infty)$: Rewrite as $\frac{\ln x}{1/x}$ (∞/∞):

$$\lim_{x\to 0^+}\frac{\ln x}{1/x}\overset{\text{H}}{=}\lim_{x\to 0^+}\frac{1/x}{-1/x^2}=\lim_{x\to 0^+}(-x)=0.$$

3. Limit of $\frac{1}{x}-\frac{1}{e^x-1}$ $(\infty-\infty)$: Bring to a common denominator:

$$\frac{e^x-1-x}{x(e^x-1)}\quad\left[\frac{0}{0}\right].$$

Differentiate numerator and denominator (Pass 1):

$$\lim_{x\to 0}\frac{e^x-1}{(e^x-1)+xe^x}\quad\left[\frac{0}{0}\right].$$

Differentiate again (Pass 2):

$$\lim_{x\to 0}\frac{e^x}{e^x+e^x+xe^x}=\lim_{x\to 0}\frac{e^x}{2e^x+xe^x}=\frac{1}{2+0}=\frac{1}{2}.$$

4. Limit of x^x as $x\to 0^+$ (0^0): Let $y=x^x$. Then $\ln y=x\ln x$. From part 2, $\lim_{x\to 0^+}x\ln x=0$. By continuity of the exponential function:

$$\lim_{x\to 0^+}x^x=\exp\left(\lim_{x\to 0^+}x\ln x\right)=e^0=1.\quad\blacksquare$$

Exercise 5.10 · Darboux's Theorem: Intermediate Value Property of Derivatives ★★★ Advanced / Exam Level

Let $f : [a, b] \to \mathbb{R}$ be a differentiable function on $[a, b]$ (we do NOT assume f' is continuous).

1. Suppose $f'(a) < 0 < f'(b)$. Define x_0 as the point where f attains its minimum on $[a, b]$. Prove that $x_0 \in (a, b)$ and deduce that $f'(x_0) = 0$.
2. Prove Darboux's Theorem: If λ lies strictly between $f'(a)$ and $f'(b)$, then there exists $c \in (a, b)$ such that $f'(c) = \lambda$.
3. Deduce that a derivative f' can never have jump discontinuities.

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Complete Step-by-Step Solution 5.10

1. Minimum in the Interior (a, b): Since f is differentiable on $[a, b]$, f is continuous on the compact interval $[a, b]$. By the Extreme Value Theorem, f attains its minimum at some point $x_0 \in [a, b]$.

- Since $f'(a) = \lim_{x \to a^+} \frac{f(x)-f(a)}{x-a} < 0$, for $x > a$ sufficiently close to a, the quotient is negative:

$$\frac{f(x) - f(a)}{x - a} < 0 \implies f(x) < f(a).$$

  Thus a cannot be the global minimum point: $x_0 \neq a$.

- Since $f'(b) = \lim_{x \to b^-} \frac{f(x)-f(b)}{x-b} > 0$, for $x < b$ sufficiently close to b, the denominator is negative, so:

$$f(x) - f(b) < 0 \implies f(x) < f(b).$$

  Thus b cannot be the global minimum point: $x_0 \neq b$.

Therefore, $x_0 \in (a, b)$. Since x_0 is an interior extremum and f is differentiable at x_0, by Fermat's Theorem:

$$f'(x_0) = 0.$$

2. General Darboux Theorem: Let λ be between $f'(a)$ and $f'(b)$. Assume without loss of generality $f'(a) < \lambda < f'(b)$. Define the auxiliary function:

$$g(x) = f(x) - \lambda x.$$

Then g is differentiable on $[a, b]$, and:

$$g'(a) = f'(a) - \lambda < 0, \qquad g'(b) = f'(b) - \lambda > 0.$$

Applying the result of part 1 to g, there exists $c \in (a, b)$ such that:

$$g'(c) = 0 \iff f'(c) - \lambda = 0 \iff f'(c) = \lambda. \quad \blacksquare$$

3. Absence of Jump Discontinuities: A function h has a jump discontinuity at x_0 if both one-sided limits $\lim_{x \to x_0^+} h(x)$ and $\lim_{x \to x_0^-} h(x)$ exist but are unequal. If f' had a jump discontinuity, there would be an interval of values between the two limits that f' could never achieve in any neighborhood of x_0, directly violating Darboux's Intermediate Value Property. Therefore, derivatives cannot possess jump discontinuities. $\blacksquare$

Exercise 5.11 · Convexity, Jensen's Inequality, and Classical Means ★★★ Advanced / Exam Level

A function $f : I \to \mathbb{R}$ is convex if $\forall x, y \in I$ and $\forall t \in [0, 1]$, $f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$.

1. Prove that if $f \in C^2(I)$ and $f''(x) \geq 0$ for all $x \in I$, then f is convex.
2. State Jensen's Inequality for a convex function f:

$$f\left(\sum_{i=1}^{n} \lambda_i x_i\right) \leq \sum_{i=1}^{n} \lambda_i f(x_i) \quad \left(\lambda_i > 0, \sum \lambda_i = 1\right).$$

3. Applying Jensen's inequality to the concave function $f(x) = \ln(x)$ on $(0, +\infty)$ with weights $\lambda_i = 1/n$, prove the Arithmetic-Geometric Mean (AM-GM) inequality:

$$\sqrt[n]{x_1 x_2 \dots x_n} \leq \frac{x_1 + x_2 + \dots + x_n}{n}.$$

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Complete Step-by-Step Solution 5.11

1. $f''(x) \geq 0 \implies f$ is Convex: Let $x < y \in I$ and $t \in (0, 1)$. Let $z = tx + (1-t)y$. Then $x < z < y$. By Taylor-Lagrange with order 1 expanded around z:

$$f(x) = f(z) + f'(z)(x - z) + \frac{f''(\xi_1)}{2}(x - z)^2 \geq f(z) + f'(z)(x - z),$$

$$f(y) = f(z) + f'(z)(y - z) + \frac{f''(\xi_2)}{2}(y - z)^2 \geq f(z) + f'(z)(y - z),$$

since $f'' \geq 0$. Multiply the first inequality by t and the second by $1 - t$, and add them:

$$tf(x) + (1-t)f(y) \geq f(z) + f'(z)[t(x - z) + (1-t)(y - z)].$$

Notice that $t(x - z) + (1-t)(y - z) = [tx + (1-t)y] - z = z - z = 0$. Therefore:

$$tf(x) + (1-t)f(y) \geq f(z) = f(tx + (1-t)y).$$

This proves that f is convex.

2. Statement of Jensen's Inequality: If $f : I \to \mathbb{R}$ is convex, then for any $x_1, \ldots, x_n \in I$ and weights $\lambda_1, \ldots, \lambda_n \in [0, 1]$ with $\sum_{i=1}^{n} \lambda_i = 1$:

$$f\left(\sum_{i=1}^{n} \lambda_i x_i\right) \leq \sum_{i=1}^{n} \lambda_i f(x_i).$$

If f is concave (i.e., $-f$ is convex), the inequality reverses ($\geq$).

3. Proof of AM-GM Inequality: Let $f(x) = \ln(x)$ on $(0, +\infty)$. $f'(x) = 1/x$, $f''(x) = -1/x^2 < 0$. Since $f'' < 0$, $\ln(x)$ is strictly concave. Applying Jensen's inequality with equal weights $\lambda_i = 1/n$:

$$\ln\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) \geq \frac{1}{n}\sum_{i=1}^{n} \ln(x_i) = \sum_{i=1}^{n} \ln(x_i^{1/n}) = \ln\left(\prod_{i=1}^{n} x_i^{1/n}\right) = \ln\left(\sqrt[n]{x_1 x_2 \ldots x_n}\right).$$

Since the natural exponential function $x \mapsto e^x$ is strictly increasing, applying it to both sides preserves the inequality:

$$\frac{x_1 + x_2 + \cdots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \ldots x_n}. \quad \blacksquare$$

Exercise 5.12 · Synthesis Problem: Uniform Continuity and the Heine-Cantor Theorem ★★★

Advanced / Exam Level

1. Define uniform continuity of a function f on an interval I.
2. Prove that $f(x) = \frac{1}{x}$ is continuous on $(0, 1)$, but NOT uniformly continuous on $(0, 1)$.
3. State the Heine-Cantor Theorem: Every continuous function on a closed and bounded (compact) interval $[a, b]$ is uniformly continuous.
4. Prove that if $f : \mathbb{R} \to \mathbb{R}$ has a bounded derivative on $\mathbb{R}$ (i.e., $\exists M > 0, |f'(x)| \leq M$), then f is uniformly continuous on $\mathbb{R}$.

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Complete Step-by-Step Solution 5.12

1. Definition of Uniform Continuity: A function $f : I \to \mathbb{R}$ is uniformly continuous on I if:

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x, y \in I,\ (|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon).$$

Key distinction: δ depends solely on ε, not on the choice of point x.

2. $f(x) = 1/x$ **is Not Uniformly Continuous on** $(0, 1)$**:** Negation of uniform continuity:

$$\exists \varepsilon > 0,\ \forall \delta > 0,\ \exists x, y \in (0, 1),\ |x - y| < \delta \wedge |f(x) - f(y)| \geq \varepsilon.$$

Choose $\varepsilon_0 = 1$. Let $\delta > 0$ be arbitrary. Choose $n \in \mathbb{N}^*$ such that $1/n < \delta$. Define $x = \frac{1}{n}$ and $y = \frac{1}{n+1}$. Both $x, y \in (0, 1)$. The distance between them is:

$$|x - y| = \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)} < \frac{1}{n} < \delta.$$

However, the distance between function values is:

$$|f(x) - f(y)| = |n - (n+1)| = 1 \geq \varepsilon_0.$$

Thus f is not uniformly continuous on $(0, 1)$.

3. Heine-Cantor Theorem: Let $f : [a, b] \to \mathbb{R}$ be continuous on a compact interval $[a, b]$. Then f is uniformly continuous on $[a, b]$.

4. Bounded Derivative $\implies$ **Uniform Continuity:** Assume $|f'(t)| \leq M$ for all $t \in \mathbb{R}$. Let $x, y \in \mathbb{R}$ with $x < y$. Since f is differentiable, by the Mean Value Theorem, there exists $c \in (x, y)$ such that:

$$f(x) - f(y) = f'(c)(x - y).$$

Taking absolute values:

$$|f(x) - f(y)| = |f'(c)| \cdot |x - y| \leq M|x - y|.$$

Thus f is M-Lipschitz continuous on $\mathbb{R}$. Let $\varepsilon > 0$. Choose $\delta = \frac{\varepsilon}{M} > 0$. Then for all $x, y \in \mathbb{R}$:

$$|x - y| < \delta \implies |f(x) - f(y)| \leq M|x - y| < M\left(\frac{\varepsilon}{M}\right) = \varepsilon.$$

Since δ is completely independent of x and y, f is uniformly continuous on $\mathbb{R}$. ■

Chapter 6

Taylor Expansions, Asymptotic Developments & Riemann Integration

Essential Chapter Theorems & Formulas · Leibniz Formula, Taylor-Young, Riemann Sums & Integration Techniques

1. **Leibniz Formula for n-th Derivative:**

$$(u \cdot v)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} u^{(k)} v^{(n-k)}.$$

2. **Taylor-Young Formula at 0 (Maclaurin $\mathrm{DL}_n(0)$):**

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k + o(x^n) \quad \text{as } x \to 0.$$

3. **Reference Taylor Expansions at 0:**

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + o(x^n), \qquad \cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + o(x^{2n+1}).$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + o(x^{2n+2}), \qquad \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n-1}\frac{x^n}{n} + o(x^n).$$

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + o(x^n), \qquad (1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + \cdots + o(x^n).$$

4. **Riemann Sum Limit Theorem:** If f is continuous on $[a, b]$:

$$\lim_{n\to\infty} \frac{b-a}{n} \sum_{k=1}^{n} f\left(a + k\frac{b-a}{n}\right) = \int_a^b f(x)\,\mathrm{d}x.$$

5. **Fundamental Theorem of Calculus:** If $f \in C([a,b])$ and $F(x) = \int_a^x f(t)\,\mathrm{d}t$, then $F \in C^1([a,b])$ and $F'(x) = f(x)$.

6. **Integration by Parts (IPP) & Substitution:**

$$\int u(x)v'(x)\,\mathrm{d}x = u(x)v(x) - \int u'(x)v(x)\,\mathrm{d}x, \qquad \int_a^b f(\phi(t))\phi'(t)\,\mathrm{d}t = \int_{\phi(a)}^{\phi(b)} f(u)\,\mathrm{d}u.$$

Exercise 6.1 · The Leibniz Formula for the n-th Derivative ★ Foundational

1. State the Leibniz formula for the n-th derivative of a product of two n-times differentiable functions u and v.
2. Calculate the n-th derivative $f^{(n)}(x)$ for $f(x) = x^2 e^{3x}$ $(n \geq 2)$.

3. Calculate the n-th derivative $g^{(n)}(x)$ for $g(x) = (x^2+1)\cos(x)$.

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Complete Step-by-Step Solution 6.1

1. Leibniz Formula: If $u, v \in C^n(I)$:

$$(uv)^{(n)}(x) = \sum_{k=0}^{n} \binom{n}{k} u^{(k)}(x) v^{(n-k)}(x).$$

2. n-th Derivative of $f(x) = x^2 e^{3x}$: Let $u(x) = x^2$ and $v(x) = e^{3x}$. Compute the successive derivatives of $u(x)$:

$$u^{(0)}(x) = x^2, \quad u'(x) = 2x, \quad u''(x) = 2, \quad \text{and } u^{(k)}(x) = 0 \quad \text{for all } k \geq 3.$$

The derivatives of $v(x) = e^{3x}$ are:

$$v^{(m)}(x) = 3^m e^{3x}.$$

Applying the Leibniz formula (since all terms for $k \geq 3$ vanish):

$$\begin{aligned} f^{(n)}(x) &= \binom{n}{0} u(x) v^{(n)}(x) + \binom{n}{1} u'(x) v^{(n-1)}(x) + \binom{n}{2} u''(x) v^{(n-2)}(x) \\ &= 1 \cdot x^2 \cdot 3^n e^{3x} + n \cdot (2x) \cdot 3^{n-1} e^{3x} + \frac{n(n-1)}{2} \cdot 2 \cdot 3^{n-2} e^{3x} \\ &= 3^{n-2} e^{3x} \left[9x^2 + 6nx + n(n-1) \right]. \quad \blacksquare \end{aligned}$$

3. n-th Derivative of $g(x) = (x^2+1)\cos(x)$: Let $u(x) = x^2+1$ and $v(x) = \cos(x)$. Derivatives of u: $u'(x) = 2x$, $u''(x) = 2$, $u^{(k)}(x) = 0$ for $k \geq 3$. Derivatives of v: $v^{(m)}(x) = \cos\left(x + m\frac{\pi}{2}\right)$. Applying Leibniz:

$$\begin{aligned} g^{(n)}(x) &= \binom{n}{0} (x^2+1) \cos\left(x + \frac{n\pi}{2}\right) + \binom{n}{1} (2x) \cos\left(x + \frac{(n-1)\pi}{2}\right) \\ &\quad + \binom{n}{2} (2) \cos\left(x + \frac{(n-2)\pi}{2}\right) \\ &= (x^2+1) \cos\left(x + \frac{n\pi}{2}\right) + 2nx \sin\left(x + \frac{n\pi}{2}\right) - n(n-1) \cos\left(x + \frac{n\pi}{2}\right) \\ &= \left[x^2 + 1 - n(n-1) \right] \cos\left(x + \frac{n\pi}{2}\right) + 2nx \sin\left(x + \frac{n\pi}{2}\right). \quad \blacksquare \end{aligned}$$

Exercise 6.2 · Direct Product and Composition of Maclaurin Series ★ Foundational

Compute the Taylor-Young asymptotic expansion (development) at order 4 at $x_0 = 0$ for:

1. $f(x) = e^x \cos(x)$ at order 4.
2. $g(x) = \ln(1 + \sin(x))$ at order 3.
3. $h(x) = \frac{1}{\sqrt{1-x^2}}$ at order 4.

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Complete Step-by-Step Solution 6.2

1. Expansion of $f(x) = e^x \cos(x)$ to Order 4: The reference expansions at 0 are:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + o(x^4),$$

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4).$$

Multiplying and retaining terms of order ≤ 4:

$$\begin{aligned} e^x \cos(x) &= \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}\right)\left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) + o(x^4) \\ &= 1\left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) + x\left(1 - \frac{x^2}{2}\right) + \frac{x^2}{2}\left(1 - \frac{x^2}{2}\right) + \frac{x^3}{6}(1) + \frac{x^4}{24}(1) + o(x^4) \\ &= 1 - \frac{x^2}{2} + \frac{x^4}{24} + x - \frac{x^3}{2} + \frac{x^2}{2} - \frac{x^4}{4} + \frac{x^3}{6} + \frac{x^4}{24} + o(x^4) \\ &= 1 + x + \left(-\frac{1}{2} + \frac{1}{2}\right)x^2 + \left(-\frac{1}{2} + \frac{1}{6}\right)x^3 + \left(\frac{1}{24} - \frac{1}{4} + \frac{1}{24}\right)x^4 + o(x^4) \\ &= 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + o(x^4). \quad \blacksquare \end{aligned}$$

2. Expansion of $g(x) = \ln(1 + \sin x)$ to Order 3: We know:

$$u = \sin(x) = x - \frac{x^3}{6} + o(x^3) \xrightarrow{x \to 0} 0.$$

Expansion of $\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} + o(u^3)$: Compute powers of u:

$$u = x - \frac{x^3}{6} + o(x^3), \qquad u^2 = \left(x - \frac{x^3}{6}\right)^2 = x^2 + o(x^3), \qquad u^3 = x^3 + o(x^3).$$

Substitute:

$$\ln(1+\sin x) = \left(x - \frac{x^3}{6}\right) - \frac{1}{2}\left(x^2\right) + \frac{1}{3}\left(x^3\right) + o(x^3)$$
$$= x - \frac{x^2}{2} + \left(-\frac{1}{6} + \frac{1}{3}\right)x^3 + o(x^3) = x - \frac{x^2}{2} + \frac{x^3}{6} + o(x^3). \quad \blacksquare$$

3. Expansion of $h(x) = (1-x^2)^{-1/2}$ to Order 4: Using $(1+u)^\alpha = 1 + \alpha u + \frac{\alpha(\alpha-1)}{2}u^2 + o(u^2)$ with $\alpha = -1/2$ and $u = -x^2$:

$$\alpha = -\frac{1}{2}, \qquad \frac{\alpha(\alpha-1)}{2} = \frac{(-1/2)(-3/2)}{2} = \frac{3}{8}.$$

Substituting $u = -x^2$:

$$(1-x^2)^{-1/2} = 1 + \left(-\frac{1}{2}\right)(-x^2) + \frac{3}{8}(-x^2)^2 + o(x^4) = 1 + \frac{x^2}{2} + \frac{3x^4}{8} + o(x^4). \quad \blacksquare$$

Exercise 6.3 · Division and Inversion of Asymptotic Expansions ★ Foundational

1. Compute the asymptotic expansion of $\tan(x) = \frac{\sin(x)}{\cos(x)}$ to order 5 at $x_0 = 0$ using polynomial long division.
2. Compute the asymptotic expansion of $f(x) = \frac{1}{1+e^x}$ to order 3 at 0.

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Complete Step-by-Step Solution 6.3

1. Expansion of $\tan(x)$ to Order 5: We have $\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} + o(x^5)$ and $\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^5)$. Perform division according to increasing powers:

$$\begin{array}{r|l}
x - \frac{x^3}{6} + \frac{x^5}{120} & 1 - \frac{x^2}{2} + \frac{x^4}{24} \\
\cline{2-2}
-\left(x - \frac{x^3}{2} + \frac{x^5}{24}\right) & x + \frac{x^3}{3} + \frac{2x^5}{15} \\
\cline{1-1}
\frac{x^3}{3} - \frac{x^5}{30} & \\
-\left(\frac{x^3}{3} - \frac{x^5}{6}\right) & \\
\cline{1-1}
\frac{2x^5}{15} &
\end{array}$$

Thus:

$$\tan(x) = x + \frac{x^3}{3} + \frac{2x^5}{15} + o(x^5). \quad \blacksquare$$

2. Expansion of $f(x) = \frac{1}{1+e^x}$ to Order 3: Expand $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)$. Denominator:

$$1 + e^x = 2 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3) = 2\left[1 + \left(\frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{12}\right)\right].$$

Let $u = \frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{12} + o(x^3) \to 0$. Use $\frac{1}{1+u} = 1 - u + u^2 - u^3 + o(u^3)$: Compute powers of u:

$$u = \frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{12} + o(x^3), \qquad u^2 = \frac{x^2}{4} + \frac{x^3}{4} + o(x^3), \qquad u^3 = \frac{x^3}{8} + o(x^3).$$

Substitute:

$$\begin{aligned}\frac{1}{1+u} &= 1 - \left(\frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{12}\right) + \left(\frac{x^2}{4} + \frac{x^3}{4}\right) - \frac{x^3}{8} + o(x^3) \\ &= 1 - \frac{x}{2} + \left(-\frac{1}{4} + \frac{1}{4}\right)x^2 + \left(-\frac{1}{12} + \frac{1}{4} - \frac{1}{8}\right)x^3 + o(x^3) \\ &= 1 - \frac{x}{2} + 0x^2 + \left(\frac{-2+6-3}{24}\right)x^3 + o(x^3) = 1 - \frac{x}{2} + \frac{x^3}{24} + o(x^3).\end{aligned}$$

Multiply by $\frac{1}{2}$:

$$f(x) = \frac{1}{2} - \frac{x}{4} + \frac{x^3}{48} + o(x^3). \quad \blacksquare$$

Exercise 6.4 · Challenging Limits via Taylor Expansions ★★ Intermediate

Calculate the following indeterminate limits using asymptotic developments:

1. $L_1 = \lim_{x\to 0} \frac{\cos(x) - e^{-x^2/2}}{x^4}$.
2. $L_2 = \lim_{x\to 0} \frac{\tan(x) - \sin(x)}{x^3}$.
3. $L_3 = \lim_{x\to 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2(x)}\right)$.

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Complete Step-by-Step Solution 6.4

1. Computation of L_1: Expand to order 4:

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4).$$

For $e^{-x^2/2}$, let $u = -x^2/2$:

$$e^u = 1 + u + \frac{u^2}{2} + o(u^2) = 1 + \left(-\frac{x^2}{2}\right) + \frac{1}{2}\left(-\frac{x^2}{2}\right)^2 + o(x^4) = 1 - \frac{x^2}{2} + \frac{x^4}{8} + o(x^4).$$

Subtract the expansions:

$$\begin{aligned}\cos(x) - e^{-x^2/2} &= \left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) - \left(1 - \frac{x^2}{2} + \frac{x^4}{8}\right) + o(x^4) \\ &= \left(\frac{1}{24} - \frac{1}{8}\right)x^4 + o(x^4) = \left(\frac{1-3}{24}\right)x^4 + o(x^4) = -\frac{1}{12}x^4 + o(x^4).\end{aligned}$$

Divide by x^4:

$$L_1 = \lim_{x\to 0} \frac{-\frac{1}{12}x^4 + o(x^4)}{x^4} = -\frac{1}{12}. \quad \blacksquare$$

2. Computation of L_2: Expand $\tan(x)$ and $\sin(x)$ to order 3:

$$\tan(x) = x + \frac{x^3}{3} + o(x^3), \qquad \sin(x) = x - \frac{x^3}{6} + o(x^3).$$

Subtract:

$$\tan(x) - \sin(x) = \left(x + \frac{x^3}{3}\right) - \left(x - \frac{x^3}{6}\right) + o(x^3) = \left(\frac{1}{3} + \frac{1}{6}\right)x^3 + o(x^3) = \frac{x^3}{2} + o(x^3).$$

Divide by x^3:

$$L_2 = \lim_{x\to 0} \frac{x^3/2 + o(x^3)}{x^3} = \frac{1}{2}. \quad \blacksquare$$

3. Computation of L_3: Bring to a common denominator:

$$\frac{1}{x^2} - \frac{1}{\sin^2 x} = \frac{\sin^2 x - x^2}{x^2 \sin^2 x}.$$

Denominator: since $\sin x \sim x$, $x^2 \sin^2 x \sim x^4$. Numerator: expand $\sin(x) = x - \frac{x^3}{6} + o(x^3)$:

$$\sin^2(x) = \left(x - \frac{x^3}{6} + o(x^3)\right)^2 = x^2 - \frac{x^4}{3} + o(x^4).$$

Thus:

$$\sin^2(x) - x^2 = -\frac{x^4}{3} + o(x^4).$$

Divide by denominator equivalent:

$$L_3 = \lim_{x\to 0} \frac{-x^4/3 + o(x^4)}{x^4} = -\frac{1}{3}. \quad \blacksquare$$

Exercise 6.5 · Asymptotes and Relative Position of Curves ★★ Intermediate

Consider the function:

$$f(x) = \sqrt{x^2 + 2x + 3}.$$

1. Determine the asymptotic expansion of $f(x)$ as $x \to +\infty$ in powers of $\frac{1}{x}$ up to order $\frac{1}{x}$.

2. Deduce the equation of the oblique asymptote (Δ) to the curve $\mathcal{C}_f$ at $+\infty$.
3. Determine the position of the curve $\mathcal{C}_f$ relative to (Δ) in the neighborhood of $+\infty$.
4. Determine the asymptote and relative position as $x \to -\infty$.

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Complete Step-by-Step Solution 6.5

1. Expansion as $x \to +\infty$: For $x > 0$, factor $x = \sqrt{x^2}$:

$$f(x) = x\sqrt{1 + \frac{2}{x} + \frac{3}{x^2}}.$$

Let $u = \frac{2}{x} + \frac{3}{x^2} \to 0$ as $x \to +\infty$. Expand $(1+u)^{1/2} = 1 + \frac{1}{2}u - \frac{1}{8}u^2 + o(u^2)$:

$$u = \frac{2}{x} + \frac{3}{x^2}, \qquad u^2 = \left(\frac{2}{x} + \frac{3}{x^2}\right)^2 = \frac{4}{x^2} + o\left(\frac{1}{x^2}\right).$$

Substitute:

$$\begin{aligned}\sqrt{1+u} &= 1 + \frac{1}{2}\left(\frac{2}{x} + \frac{3}{x^2}\right) - \frac{1}{8}\left(\frac{4}{x^2}\right) + o\left(\frac{1}{x^2}\right) \\ &= 1 + \frac{1}{x} + \frac{3}{2x^2} - \frac{1}{2x^2} + o\left(\frac{1}{x^2}\right) = 1 + \frac{1}{x} + \frac{1}{x^2} + o\left(\frac{1}{x^2}\right).\end{aligned}$$

Multiply by x:

$$f(x) = x + 1 + \frac{1}{x} + o\left(\frac{1}{x}\right).$$

2. Oblique Asymptote at $+\infty$: The linear part gives the equation of the oblique asymptote:

$$(\Delta): \quad y = x + 1.$$

3. Relative Position at $+\infty$: Compute the vertical distance $f(x) - y_{(\Delta)}$:

$$f(x) - (x+1) = \frac{1}{x} + o\left(\frac{1}{x}\right).$$

As $x \to +\infty$, $\frac{1}{x} > 0$. Therefore, $f(x) - y_{(\Delta)} > 0$, meaning the curve $\mathcal{C}_f$ lies strictly **above** the asymptote (Δ).

4. Behavior as $x \to -\infty$: For $x < 0$, $\sqrt{x^2} = -x = |x|$.

$$f(x) = -x\sqrt{1 + \frac{2}{x} + \frac{3}{x^2}} = -x\left(1 + \frac{1}{x} + \frac{1}{x^2} + o(1/x^2)\right) = -x - 1 - \frac{1}{x} + o(1/x).$$

The asymptote at $-\infty$ is $(\Delta'): y = -x - 1$. The difference is $f(x) - (-x-1) = -\frac{1}{x} + o(1/x)$. Since $x < 0$, $-\frac{1}{x} > 0$, so $\mathcal{C}_f$ lies strictly **above** (Δ') at $-\infty$. ■

Exercise 6.6 · Limits of Riemann Sums ★★ Intermediate

Using the Riemann Sum Limit Theorem:

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right) = \int_0^1 f(x)\,\mathrm{d}x,$$

compute the exact limit of each of the following sequences:

1. $S_n = \sum_{k=1}^{n} \frac{n}{n^2+k^2}$.
2. $T_n = \sum_{k=1}^{n} \frac{1}{n+k}$.
3. $P_n = \frac{1}{n}\sum_{k=1}^{n} \sqrt{1-\left(\frac{k}{n}\right)^2}$.

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Complete Step-by-Step Solution 6.6

1. Limit of S_n: Factor $\frac{1}{n}$ from the sum:

$$S_n = \sum_{k=1}^{n} \frac{n}{n^2(1+(k/n)^2)} = \frac{1}{n}\sum_{k=1}^{n} \frac{1}{1+(k/n)^2}.$$

This is a standard Riemann sum for $f(x) = \frac{1}{1+x^2}$ on $[0,1]$. Since f is continuous on $[0,1]$:

$$\lim_{n\to\infty} S_n = \int_0^1 \frac{\mathrm{d}x}{1+x^2} = [\arctan(x)]_0^1 = \arctan(1) - \arctan(0) = \frac{\pi}{4}. \quad \blacksquare$$

2. Limit of T_n: Factor $\frac{1}{n}$:

$$T_n = \sum_{k=1}^{n} \frac{1}{n(1+k/n)} = \frac{1}{n}\sum_{k=1}^{n} \frac{1}{1+k/n}.$$

This is a Riemann sum for $f(x) = \frac{1}{1+x}$ on $[0,1]$.

$$\lim_{n\to\infty} T_n = \int_0^1 \frac{\mathrm{d}x}{1+x} = [\ln(1+x)]_0^1 = \ln(2) - \ln(1) = \ln(2). \quad \blacksquare$$

3. Limit of P_n: $P_n = \frac{1}{n}\sum_{k=1}^{n} \sqrt{1-(k/n)^2}$ is a Riemann sum for $f(x) = \sqrt{1-x^2}$ on $[0,1]$.

$$\lim_{n\to\infty} P_n = \int_0^1 \sqrt{1-x^2}\,\mathrm{d}x.$$

Geometrically, this integral represents the area of a quarter of the unit circle of radius 1:

$$\int_0^1 \sqrt{1-x^2}\,\mathrm{d}x = \frac{\pi(1)^2}{4} = \frac{\pi}{4}. \quad \blacksquare$$

Exercise 6.7 · Differentiating Integrals with Variable Bounds (Leibniz Integral Rule) ★★

Intermediate

Let $u, v : I \to \mathbb{R}$ be differentiable functions and $f : \mathbb{R} \to \mathbb{R}$ continuous. Define:

$$F(x) = \int_{u(x)}^{v(x)} f(t)\,\mathrm{d}t.$$

1. Prove using the Fundamental Theorem of Calculus and the Chain Rule that:
$$F'(x) = f(v(x))v'(x) - f(u(x))u'(x).$$
2. Calculate the derivative of $G(x) = \int_0^{x^2} e^{-t^2}\,\mathrm{d}t$.
3. Calculate the derivative of $H(x) = \int_x^{2x} \frac{\mathrm{d}t}{1+t^3}$ for $x > 0$.

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Complete Step-by-Step Solution 6.7

1. Proof of Leibniz Rule: Let f be continuous on $\mathbb{R}$. By the Fundamental Theorem of Calculus, f admits an antiderivative Φ such that $\Phi'(t) = f(t)$. By the evaluation formula:

$$F(x) = \int_{u(x)}^{v(x)} f(t)\,\mathrm{d}t = \Phi(v(x)) - \Phi(u(x)).$$

Since u and v are differentiable and Φ is differentiable, F is differentiable by the Chain Rule:

$$F'(x) = \frac{\mathrm{d}}{\mathrm{d}x}[\Phi(v(x))] - \frac{\mathrm{d}}{\mathrm{d}x}[\Phi(u(x))] = \Phi'(v(x))v'(x) - \Phi'(u(x))u'(x).$$

Since $\Phi' = f$:

$$F'(x) = f(v(x))v'(x) - f(u(x))u'(x). \quad \blacksquare$$

2. Derivative of $G(x) = \int_0^{x^2} e^{-t^2}\,\mathrm{d}t$: Here $u(x) = 0 \implies u'(x) = 0$, $v(x) = x^2 \implies v'(x) = 2x$, and $f(t) = e^{-t^2}$.

$$G'(x) = e^{-(x^2)^2} \cdot (2x) - 0 = 2xe^{-x^4}.$$

3. Derivative of $H(x) = \int_x^{2x} \frac{\mathrm{d}t}{1+t^3}$: Here $u(x) = x \implies u'(x) = 1$, $v(x) = 2x \implies v'(x) = 2$, and $f(t) = \frac{1}{1+t^3}$.

$$H'(x) = \frac{1}{1+(2x)^3} \cdot (2) - \frac{1}{1+x^3} \cdot (1) = \frac{2}{1+8x^3} - \frac{1}{1+x^3}. \quad \blacksquare$$

Exercise 6.8 · Systematic Integration by Parts (IPP) ★★ Intermediate

Calculate each of the following integrals using integration by parts:

1. $I_1 = \int x^2 e^{2x}\,\mathrm{d}x$.
2. $I_2 = \int_1^e x\ln(x)\,\mathrm{d}x$.
3. $I_3 = \int e^x \cos(x)\,\mathrm{d}x$ (circular / cyclic IPP).

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Complete Step-by-Step Solution 6.8

1. Calculation of $I_1 = \int x^2 e^{2x}\,\mathrm{d}x$ (Double IPP):

- First IPP: Set $u = x^2 \implies u' = 2x$, and $v' = e^{2x} \implies v = \frac{1}{2}e^{2x}$.

$$I_1 = \frac{1}{2}x^2 e^{2x} - \int x e^{2x}\,\mathrm{d}x.$$

- Second IPP for $\int x e^{2x}\,\mathrm{d}x$: Set $u = x \implies u' = 1$, and $v' = e^{2x} \implies v = \frac{1}{2}e^{2x}$.

$$\int x e^{2x}\,\mathrm{d}x = \frac{1}{2}x e^{2x} - \frac{1}{2}\int e^{2x}\,\mathrm{d}x = \frac{1}{2}x e^{2x} - \frac{1}{4}e^{2x}.$$

- Combining:

$$I_1 = \frac{1}{2}x^2 e^{2x} - \frac{1}{2}x e^{2x} + \frac{1}{4}e^{2x} + C = \frac{1}{4}e^{2x}(2x^2 - 2x + 1) + C. \quad \blacksquare$$

2. Calculation of $I_2 = \int_1^e x\ln(x)\,\mathrm{d}x$: Set $u = \ln(x) \implies u' = \frac{1}{x}$, and $v' = x \implies v = \frac{x^2}{2}$.

$$I_2 = \left[\frac{x^2}{2}\ln(x)\right]_1^e - \int_1^e \frac{x^2}{2}\cdot\frac{1}{x}\,\mathrm{d}x = \left(\frac{e^2}{2}\ln(e) - 0\right) - \frac{1}{2}\int_1^e x\,\mathrm{d}x$$
$$= \frac{e^2}{2} - \frac{1}{2}\left[\frac{x^2}{2}\right]_1^e = \frac{e^2}{2} - \frac{1}{4}(e^2 - 1) = \frac{e^2+1}{4}. \quad \blacksquare$$

3. Calculation of $I_3 = \int e^x \cos(x)\,\mathrm{d}x$ (Cyclic IPP): Set $u = \cos x \implies u' = -\sin x$, and $v' = e^x \implies v = e^x$.

$$I_3 = e^x\cos x - \int e^x(-\sin x)\,\mathrm{d}x = e^x\cos x + \int e^x \sin x\,\mathrm{d}x.$$

Apply IPP to $\int e^x \sin x\,\mathrm{d}x$: Set $u = \sin x \implies u' = \cos x$, and $v' = e^x \implies v = e^x$.

$$\int e^x \sin x\,\mathrm{d}x = e^x\sin x - \int e^x\cos x\,\mathrm{d}x = e^x\sin x - I_3.$$

Substituting back into I_3:

$$I_3 = e^x \cos x + e^x \sin x - I_3 \iff 2I_3 = e^x(\cos x + \sin x).$$

Therefore:

$$I_3 = \frac{e^x}{2}(\sin x + \cos x) + C. \quad \blacksquare$$

Exercise 6.9 · Integration by Substitution (Change of Variables) ★★ Intermediate

Calculate the following integrals using an appropriate substitution:

1. $J_1 = \int_0^1 \frac{x}{\sqrt{1+x^2}}\,\mathrm{d}x$ (substitute $u = 1 + x^2$).
2. $J_2 = \int_0^{\pi/2} \frac{\cos(x)}{1+\sin^2(x)}\,\mathrm{d}x$ (substitute $u = \sin x$).
3. $J_3 = \int_1^4 \frac{\mathrm{d}x}{x(1+\sqrt{x})}$ (substitute $u = \sqrt{x}$).

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Complete Step-by-Step Solution 6.9

1. Calculation of J_1: Let $u = 1 + x^2 \implies \mathrm{d}u = 2x\,\mathrm{d}x \iff x\,\mathrm{d}x = \frac{\mathrm{d}u}{2}$. Bounds: when $x = 0$, $u = 1$; when $x = 1$, $u = 2$.

$$J_1 = \int_1^2 \frac{\mathrm{d}u/2}{\sqrt{u}} = \frac{1}{2}\int_1^2 u^{-1/2}\,\mathrm{d}u = \frac{1}{2}\left[2\sqrt{u}\right]_1^2 = \left[\sqrt{u}\right]_1^2 = \sqrt{2} - 1. \quad \blacksquare$$

2. Calculation of J_2: Let $u = \sin(x) \implies \mathrm{d}u = \cos(x)\,\mathrm{d}x$. Bounds: when $x = 0$, $u = \sin 0 = 0$; when $x = \pi/2$, $u = \sin(\pi/2) = 1$.

$$J_2 = \int_0^1 \frac{\mathrm{d}u}{1+u^2} = \left[\arctan(u)\right]_0^1 = \arctan(1) - \arctan(0) = \frac{\pi}{4}. \quad \blacksquare$$

3. Calculation of J_3: Let $u = \sqrt{x} \iff x = u^2 \implies \mathrm{d}x = 2u\,\mathrm{d}u$. Bounds: when $x = 1$, $u = 1$; when $x = 4$, $u = 2$. Substitute:

$$J_3 = \int_1^2 \frac{2u\,\mathrm{d}u}{u^2(1+u)} = 2\int_1^2 \frac{\mathrm{d}u}{u(1+u)}.$$

Decompose into partial fractions:

$$\frac{1}{u(1+u)} = \frac{1}{u} - \frac{1}{1+u}.$$

Integrate:

$$J_3 = 2\int_1^2 \left(\frac{1}{u} - \frac{1}{1+u}\right) du = 2[\ln|u| - \ln|1+u|]_1^2 = 2\left[\ln\left(\frac{u}{1+u}\right)\right]_1^2$$
$$= 2\left(\ln(2/3) - \ln(1/2)\right) = 2\ln\left(\frac{2/3}{1/2}\right) = 2\ln\left(\frac{4}{3}\right). \quad \blacksquare$$

Exercise 6.10 · **Partial Fraction Decomposition of Rational Functions** ★★★ Advanced / Exam Level

Consider the rational fraction:

$$f(x) = \frac{2x^2 - 3x + 5}{(x-1)(x^2+1)}.$$

1. Determine the partial fraction decomposition of $f(x)$ in $\mathbb{R}(X)$:

$$\frac{2x^2 - 3x + 5}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx + C}{x^2+1}.$$

2. Determine constants A, B, C by identifying coefficients or evaluating at special values.
3. Calculate the indefinite integral:

$$I = \int \frac{2x^2 - 3x + 5}{(x-1)(x^2+1)}\, dx.$$

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Complete Step-by-Step Solution 6.10

1. Decomposition Form: The denominator has a real linear factor $(x-1)$ and an irreducible quadratic factor (x^2+1) with $\Delta = -4 < 0$. The degree of the numerator (2) is strictly less than the degree of the denominator (3). The decomposition form over $\mathbb{R}$ is:

$$\frac{2x^2 - 3x + 5}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx + C}{x^2+1}.$$

2. Determination of Coefficients A, B, C:

- **Finding A:** Multiply both sides by $(x-1)$ and set $x = 1$:

$$A = \left.\frac{2x^2 - 3x + 5}{x^2+1}\right|_{x=1} = \frac{2(1)^2 - 3(1) + 5}{1^2+1} = \frac{4}{2} = 2.$$

- **Finding C:** Set $x = 0$ in the identity:

$$\frac{5}{(-1)(1)} = \frac{A}{-1} + \frac{C}{1} \implies -5 = -2 + C \implies C = -3.$$

- **Finding B:** Multiply both sides by x and take the limit as $x \to \infty$:

$$\lim_{x\to\infty} \frac{2x^3 - 3x^2 + 5x}{(x-1)(x^2+1)} = 2 = A + B \implies 2 = 2 + B \implies B = 0.$$

Thus $A = 2$, $B = 0$, $C = -3$. The partial fraction decomposition is:

$$\frac{2x^2 - 3x + 5}{(x-1)(x^2+1)} = \frac{2}{x-1} - \frac{3}{x^2+1}.$$

3. Integration:

$$\begin{aligned} I &= \int \left(\frac{2}{x-1} - \frac{3}{x^2+1} \right) \mathrm{d}x \\ &= 2\int \frac{\mathrm{d}x}{x-1} - 3\int \frac{\mathrm{d}x}{x^2+1} \\ &= 2\ln|x-1| - 3\arctan(x) + C. \quad \blacksquare \end{aligned}$$

Exercise 6.11 · The Wallis Integrals and Asymptotic Formula for π ★★★ Advanced / Exam Level

Define the sequence of Wallis integrals for $n \in \mathbb{N}$ by:

$$W_n = \int_0^{\pi/2} \sin^n(x)\,\mathrm{d}x.$$

1. Compute W_0 and W_1.
2. Using integration by parts on $\sin^n(x) = \sin(x)\sin^{n-1}(x)$, establish the recurrence relation:

$$W_n = \frac{n-1}{n} W_{n-2} \quad \text{for all } n \geq 2.$$

3. Deduce explicit formulas for W_{2p} and W_{2p+1} in terms of double factorials:

$$W_{2p} = \frac{(2p)!}{2^{2p}(p!)^2}\frac{\pi}{2}, \qquad W_{2p+1} = \frac{2^{2p}(p!)^2}{(2p+1)!}.$$

4. Prove that $(n+1)W_{n+1}W_n = \frac{\pi}{2}$ for all $n \geq 0$, and deduce the Wallis product formula for π:

$$\lim_{p\to\infty} \frac{1}{2p}\left(\frac{2\cdot 4\cdot 6\ldots(2p)}{1\cdot 3\cdot 5\ldots(2p-1)}\right)^2 = \frac{\pi}{2}.$$

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Complete Step-by-Step Solution 6.11

1. Initial Terms:

$$W_0 = \int_0^{\pi/2} 1\,\mathrm{d}x = \frac{\pi}{2}, \qquad W_1 = \int_0^{\pi/2} \sin(x)\,\mathrm{d}x = [-\cos x]_0^{\pi/2} = 0 - (-1) = 1.$$

2. Recurrence Relation via IPP: For $n \geq 2$, write $W_n = \int_0^{\pi/2} \sin(x)\sin^{n-1}(x)\,\mathrm{d}x$. Set $u = \sin^{n-1}(x) \implies u' = (n-1)\cos(x)\sin^{n-2}(x)$, and $v' = \sin(x) \implies v = -\cos(x)$.

$$\begin{aligned} W_n &= [-\cos(x)\sin^{n-1}(x)]_0^{\pi/2} + (n-1)\int_0^{\pi/2} \cos^2(x)\sin^{n-2}(x)\,\mathrm{d}x \\ &= 0 + (n-1)\int_0^{\pi/2} (1-\sin^2 x)\sin^{n-2}(x)\,\mathrm{d}x \\ &= (n-1)W_{n-2} - (n-1)W_n. \end{aligned}$$

Rearranging:

$$W_n + (n-1)W_n = (n-1)W_{n-2} \iff nW_n = (n-1)W_{n-2} \iff W_n = \frac{n-1}{n}W_{n-2}.$$

3. Closed-Form Expressions:

- For $n = 2p$:
$$W_{2p} = \frac{2p-1}{2p}\cdot\frac{2p-3}{2p-2}\cdots\frac{1}{2}W_0 = \frac{(2p-1)!!}{(2p)!!}\frac{\pi}{2} = \frac{(2p)!}{2^{2p}(p!)^2}\frac{\pi}{2}.$$
- For $n = 2p+1$:
$$W_{2p+1} = \frac{2p}{2p+1}\cdot\frac{2p-2}{2p-1}\cdots\frac{2}{3}W_1 = \frac{(2p)!!}{(2p+1)!!} = \frac{2^{2p}(p!)^2}{(2p+1)!}.$$

4. Product Identity and Wallis Formula: Compute the product $(n+1)W_{n+1}W_n$: For $n = 2p$:

$$(2p+1)W_{2p+1}W_{2p} = (2p+1)\cdot\frac{2^{2p}(p!)^2}{(2p+1)!}\cdot\frac{(2p)!}{2^{2p}(p!)^2}\frac{\pi}{2} = \frac{\pi}{2}.$$

Since $0 < \sin x < 1$ on $(0, \pi/2)$, $\sin^{n+1} x < \sin^n x < \sin^{n-1} x$, which implies $W_{n+1} < W_n < W_{n-1}$. Dividing by W_{n+1}:

$$1 < \frac{W_n}{W_{n+1}} < \frac{W_{n-1}}{W_{n+1}} = \frac{n+1}{n} \xrightarrow{n\to\infty} 1.$$

By the Squeeze Theorem, $W_n \sim W_{n+1}$. Thus $\frac{\pi}{2} = (n+1)W_{n+1}W_n \sim nW_n^2 \implies W_n \sim \sqrt{\frac{\pi}{2n}}$. Substituting $n = 2p$:

$$\frac{W_{2p}}{W_{2p+1}} = \left(\frac{(2p)!!}{(2p-1)!!}\right)^2 \frac{1}{2p+1}\frac{\pi}{2} \xrightarrow{p\to\infty} 1.$$

Rearranging yields the historic Wallis product formula for π:

$$\frac{\pi}{2} = \lim_{p\to\infty}\prod_{k=1}^{p}\frac{4k^2}{4k^2-1} = \lim_{p\to\infty}\left(\frac{2\cdot 4\cdot 6\ldots(2p)}{1\cdot 3\cdot 5\ldots(2p-1)}\right)^2\frac{1}{2p+1}. \quad \blacksquare$$

Exercise 6.12 · Synthesis Problem: Error Bounds in Numerical Quadrature (Trapezoidal Rule)
★★★ Advanced / Exam Level

Let $f \in C^2([a,b])$ with $M_2 = \max_{x\in[a,b]} |f''(x)|$. Approximate the integral $I = \int_a^b f(x)\,\mathrm{d}x$ using the composite Trapezoidal Rule on n subintervals of length $h = \frac{b-a}{n}$:

$$T_n = \frac{h}{2}\left[f(a) + 2\sum_{k=1}^{n-1} f(a+kh) + f(b)\right].$$

1. On an elementary interval $[x_k, x_{k+1}]$ with midpoint $m_k = \frac{x_k+x_{k+1}}{2}$, express $f(x)$ via Taylor's formula around m_k.
2. Prove that the elementary error on $[x_k, x_{k+1}]$ satisfies:
$$\left|\int_{x_k}^{x_{k+1}} f(x)\,\mathrm{d}x - \frac{h}{2}[f(x_k) + f(x_{k+1})]\right| \le \frac{M_2 h^3}{12}.$$
3. Sum over all n subintervals to establish the global error bound:
$$|I - T_n| \le \frac{M_2(b-a)^3}{12n^2}.$$
4. Application: How many subintervals n are required to approximate $\int_0^1 e^{-x^2}\,\mathrm{d}x$ using the Trapezoidal Rule with an absolute error strictly less than 10^{-4}?

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Complete Step-by-Step Solution 6.12

1. Taylor Expansion around Midpoint m_k: Let $h = x_{k+1} - x_k$. The midpoint is $m_k = x_k + h/2$. For $x \in [x_k, x_{k+1}]$, let $t = x - m_k \in [-h/2, h/2]$. By Taylor's formula with integral remainder:
$$f(m_k + t) = f(m_k) + tf'(m_k) + \int_0^t (t-s)f''(m_k+s)\,\mathrm{d}s.$$

2. Elementary Trapezoidal Error: Integrating $f(m_k+t)$ over $t \in [-h/2, h/2]$:
$$\int_{x_k}^{x_{k+1}} f(x)\,\mathrm{d}x = \int_{-h/2}^{h/2} f(m_k+t)\,\mathrm{d}t = hf(m_k) + 0 + \int_{-h/2}^{h/2}\left(\int_0^t (t-s)f''(m_k+s)\,\mathrm{d}s\right)\mathrm{d}t.$$

The trapezoidal approximation is:
$$\frac{h}{2}[f(x_k) + f(x_{k+1})] = \frac{h}{2}\left[f(m_k - h/2) + f(m_k + h/2)\right] = hf(m_k) + \frac{h^3}{8}f''(m_k) + O(h^5).$$

Subtracting and bounding with $|f''(x)| \le M_2$: Standard integration yields:

$$\left| \int_{x_k}^{x_{k+1}} f(x)\,\mathrm{d}x - \frac{h}{2}[f(x_k) + f(x_{k+1})] \right| \le \frac{M_2 h^3}{12}.$$

3. Global Error Bound: Summing over all n subintervals $k = 0, 1, \ldots, n-1$:

$$|I - T_n| \le \sum_{k=0}^{n-1} \frac{M_2 h^3}{12} = n \cdot \frac{M_2 h^3}{12}.$$

Substitute $h = \frac{b-a}{n}$:

$$|I - T_n| \le n \cdot \frac{M_2}{12} \left(\frac{b-a}{n} \right)^3 = \frac{M_2 (b-a)^3}{12n^2}. \quad \blacksquare$$

4. Application to $\int_0^1 e^{-x^2}\,\mathrm{d}x$: Here $[a, b] = [0, 1]$, so $b - a = 1$. Compute derivatives of $f(x) = e^{-x^2}$:

$$f'(x) = -2xe^{-x^2}, \qquad f''(x) = (4x^2 - 2)e^{-x^2}.$$

On $[0, 1]$:

- At $x = 0$: $|f''(0)| = |-2| = 2$.
- At $x = 1$: $|f''(1)| = |2e^{-1}| = \frac{2}{e} \approx 0.736$.
- Derivative of f'': $f'''(x) = (12x - 8x^3)e^{-x^2} = 4x(3 - 2x^2)e^{-x^2} = 0 \implies x = \sqrt{3/2} \approx 1.22 > 1$.

Thus the maximum of $|f''(x)|$ on $[0, 1]$ occurs at $x = 0$:

$$M_2 = \max_{x \in [0,1]} |f''(x)| = 2.$$

The error bound gives:

$$|I - T_n| \le \frac{2 \times (1)^3}{12n^2} = \frac{1}{6n^2}.$$

We require:

$$\frac{1}{6n^2} < 10^{-4} \iff 6n^2 > 10^4 = 10000 \iff n^2 > \frac{10000}{6} \approx 1666.67.$$

Taking square roots:

$$n > \sqrt{1666.67} \approx 40.82.$$

Therefore, choosing $n = 41$ subintervals guarantees an error strictly less than $10^{-4} = 0.0001$. $\blacksquare$